SUPPLEMENTAL MATERIAL

Supplemental Methods (Data S1)

1. Search Strategy for the association between chronic hypertension and adverse perinatal outcomes

A) Search strategy for Web of Science

1. Pregnancy outcome* or obstetric outcome* or birth outcome* or pregnancy complication* or gestational complication*/or obstetric complication/ or birth complication* or labor complication* or uterine complication* or normal birth* or live birth* or pre-term deliver* or preterm deliver* or pre-term birth*or preterm birth* or preterm labor* or pre-term labor* or premature deliver* or premature birth* or premature labor* or prematurity or cesarean* or cesarean section* or csection* or c section* or spontaneous abortion* or miscarriage* or miscarry or stillbirth* or still birth* or intrauterine death* or intra-uterine death* or fetal death* or fetal mortality or neonatal mortalit* or neonatal death* or neo-natal death or neo-natal mortalit* or newborn death* or newborn mortalit* or new born death* or new born mortalit* or perinatal mortalit* or perinatal death* or infant mortalit* or infant death* or postneonatal mortalit* or post-neonatal mortalit* or postneonatal death* or post-neonatal death* or preeclampsia or pre eclampsia or preeclamptic or pre eclamptic or preeclamptic or pre-eclamptic toxemia* or preeclamptic toxemia* or PET or pregnancy toxemia* or toxemia* or EPH toxemia* or EPH gestosis or hypertension-edema-proteinuria gestosis or hypertension edema proteinuria gestosis/ or proteinuria-edemahypertension gestosis/ or proteinuria edema hypertension gestosis or edema proteinuria hypertension gestosis or edema-proteinuria-hypertension gestosis/ or eclampsia/ or eclamptic or HELLP or hemolysis elevated liver enzymes low platelet count/ or hemolysis-elevated liver enzymes-low platelet count or antenatal hemorrage* or antepartum hemorrage* or postnatal hemorrage* or postpartum hemorrage* or postpartum complication* or postnatal complication* or post birth complication* or post labor complication* or maternal outcome* or maternal complication* or fetal outcome* or fetal complication* or neonate complication* or neonatal complication*or newborn complication*/ or gestational age/ or special care baby unit admission* or SCBU admission* or NICU admission* or neonatal intensive care unit admission* or small for gestational age* or SGA* or IUGR* or intrauterine growth restriction* or LBW*or low birth weight* or VLBW* or very low birth weight* or neonatal intraventricular hemorrhage*or Intraventricular hemorrhage of newborn*

2. Essential hypertensi* or chronic hypertensi* or chronic hypertension in pregnanc*

3. Search 1 and 2 were then combined (1 ‘AND’ 2)

B) Search strategy for Embase (via OVID)

1. ('pregnancy outcome' OR 'pregnancy complications' OR 'high risk pregnancy' OR 'labor' OR 'delivery' OR 'fetus outcome' OR 'live birth' OR 'premature labor' OR 'immature and premature labor' OR 'prematurity' OR 'cesarean section' OR 'spontaneous abortion' OR 'stillbirth' OR 'fetus death' OR 'infant mortality' OR 'fetus mortality' OR 'maternal mortality' OR 'perinatal mortality' OR 'preeclampsia' OR 'eclampsia' OR 'eclampsia and preeclampsia' OR 'hellp' OR 'disseminated intravascular coagulation' OR 'magnesium sulfate' OR 'newborn death' OR 'gestational age' OR 'infant low birth weight' OR 'low birth weight' OR 'very low birth weight' OR 'extremely low birth weight' OR 'apgar score' OR 'newborn intensive care' OR 'small for date infant' OR 'pregnancy outcome*' OR 'maternal outcome*' OR 'pregnancy complication*' OR 'obstetric

outcome*' OR 'obstetric complication*' OR 'normal birth*' OR 'live birth*' OR 'premature birth*' OR 'preterm birth*' OR 'preterm deliver*' OR 'born preterm' OR 'cesarean*' OR 'csection*' OR 'miscarriage*' OR 'stillbirth*' OR 'intrauterine death*' OR 'antenatal hemorrhage*' OR 'antenatal haemorrhage*' OR 'antepartum haemorrhage*' OR 'antepartum hemorrhage*' OR 'postpartum hemorrhage*' OR 'postpartum haemorrhage*' OR 'postnatal haemorrhage*' OR 'postnatal hemorrhage*' OR 'postpartum complication*' OR 'postnatal complication*' OR 'special care baby unit admission*' OR 'scbu admission*' OR 'neonatal intensive care unit admission*' OR 'nicu admission*' OR 'sga' OR 'iugr' OR 'neonatal intraventricular haemorrhage' OR 'neonatal intraventricular hemorrhage' OR 'lbw' OR 'vlbw' OR 'fetus outcome*' OR 'high risk pregnan*')

2. Essential hypertensi*/ or chronic hypertensi*/ or chronic hypertension in pregnan*

3. Search 1 and 2 were then combined (1 'AND' 2)

C) Search strategy for Medline (via OVID platform)

1. pregnancy outcome* or pregnancy complication* obstetric complication* or obstetric labor complications/ or pregnancy, high-risk/ or delivery, obstetric or labor or live birth or obstetric labor, premature or premature birth or premature birth* or preterm birth* preterm deliver* or born preterm or caesarean section cesarean* or c-section or caesarean* or abortion, spontaneous or stillbirth or fetal death/ or infant mortality or maternal mortality or perinatal mortality or pre-eclampsia or eclampsia/ or HELLP/ or gestational age/ or infant, low birth weight/ or pregnancy/or or obstetric outcome* or normal birth* or live birth* or miscarriage* or stillbirth* or intrauterine death* or neonatal death*or antenatal haemorrhage* or postpartum haemorrhage* or antenatal hemorrhage* or postpartum hemorrage* or antepartum hemorrage* or antepartum haemorrhage* or postpartum complication*/ or special care baby unit admission* or SCBU admission* or neonatal intensive care unit admission* or NICU admission* or small for gestationa age or SGA or intrauterine growth restriction or IUGR or neonatal intraventricular hemorrage* or neonatal intraventricular haemorrhage*
2. Essential hypertensi* or chronic hypertensi* or chronic hypertension in pregnan*
   Searches 1 and 2 were then combined (1 'AND' 2)

2. Selection process and eligibility criteria

Two of us (SAK and PB) independently reviewed the titles and abstracts of all articles, studies that clearly not meeting the predefined criteria excluded, then the full text of potentially eligible studies obtained. Following that, both investigators independently screened the full text articles to define the included studies and a third investigator (ASK) resolved any inconsistencies about included or excluded articles between the two investigators. In addition, PRISMA flow chart used to illustrate the process for selecting the included studies with the number of references at each phase of this review.

We included cohort and case control studies in which chronic hypertension reported as an exposure (compared to normotensive women); and the outcomes of interest were adverse maternal, fetal, or neonatal outcomes. We excluded study designs other than observational studies, along with case reports, case series, editorials, reviews, conference abstracts, book chapters and animal/in vitro studies. When two or more studies included the same cohort, we included the one with largest population or with more relevant outcomes. Also, studies with additional comorbidities were also excluded. Moreover, as, pre-eclampsia associated with higher risk of adverse outcomes and this review focusing on the impact of chronic hypertension; we included women with chronic hypertension without superimposed pre-eclampsia, when possible. Additionally, when studies reported results for both groups separately, we used the group without pre-eclampsia as it is one of the outcomes of interest.

3. Data collection process and quality assessment

An electronic standardised data extraction form was developed, and pilot tested on seven studies. One of us (SAK extracted the data for all included studies, and two obstetricians (FMC and DF) extracted the data independently for more than 50% of the included studies to ensure the validity of extracted data. The following details extracted from each study: source of data, authors' name, study design, study period, population characteristics (sample size, appropriateness of case and control recruitment, inclusion/exclusion criteria, and definition of exposure, measures, and definitions of outcomes (maternal, fetal and neonatal), appropriateness of analyses, confounders adjusted for (if any), crude and adjusted estimates. Any disagreement about the extracted data was resolved by the third investigator (ASK).

We used the Newcastle-Ottawa Scale (NOS) to assess the study quality. The NOS has a version for case-control studies and one for cohort studies, and it uses a 'star system', in which stars are assigned to show higher quality based on three criteria: selection of the study groups; comparability of the groups; and the ascertainment of the exposure and/or outcome of interest. The quality assessment was assessed independently by two investigators (SAK) and (LP). Any inconsistency between the two investigators resolved by a third investigator (ASK)

Supplemental Methods (Data S2)

1. Search Strategy for the association between antihypertensive treatment and adverse perinatal outcomes

A) Search strategy for Web of Science

1. Anti hypertensive agent*/ or / Antihypertensive agent* Anti-hypertensive agent* / or /Cardiovascular Agent*/ or /anti-hypertensive drug*/ or /antihypertensive drug*/ or /antihypertensive/ or /anti hypertensive drug*/ or /antihypertensive agents therap*/ or /anti hypertensive agents therap*/ or/ anti-hypertensive agents therap*/ or /antihypertensive agents pharmacol*/ or /anti hypertensive agents pharmacol*/ or / anti-hypertensive agents pharmacol*/ or / calcium channel blocker/ OR /beta blocker/ OR /ACEi/ OR /adrenergic receptor antagonist/ or / adrenergic receptor blocker/ OR /alpha-2 adrenergic receptor agonist/ OR /angiotensin ii receptor/ OR /hydralazine/ OR /sodium nitroprusside/ OR /clonidine hydrochloride/ OR /moxonidine/ OR /renin inhibitor/ OR /thiazide/ OR /loop diuretic/ OR /potassium sparing diuretic
2. hypertensi*
3. Pregnancy outcome* or obstetric outcome* or birth outcome* or pregnancy complication* or gestational complication*/or obstetric complication/ or birth complication* or labor complication* or uterine complication* or normal birth* or live birth* or pre-term deliver* or preterm deliver* or pre-term birth*or preterm birth* or preterm labor* or pre-term labor* or premature deliver* or premature birth* or premature labor* or prematurity or cesarean* or cesarean section* or csection* or c section* or spontaneous abortion* or miscarriage* or miscarry or stillbirth* or still birth* or intrauterine death* or intra-uterine death* or fetal death* or fetal mortality or neonatal mortalit* or neonatal death* or neo-natal death or neo-natal mortalit* or newborn death* or newborn mortalit* or new born death* or new born mortalit* or perinatal mortalit* or perinatal death* or infant mortalit* or infant death* or postneonatal mortalit* or post-neonatal mortalit* or postneonatal death* or post-neonatal death* or preeclampsia or pre eclampsia or preeclamptic or pre eclamptic or preeclamptic or pre-eclamptic toxemia* or preeclamptic toxemia* or PET or pregnancy toxemia* or toxemia* or EPH toxemia* or EPH gestosis or hypertension-edema-proteinuria gestosis or hypertension edema proteinuria gestosis/ or proteinuria-edemahypertension gestosis/ or proteinuria edema hypertension gestosis or edema proteinuria hypertension gestosis or edema-proteinuria-hypertension gestosis/ or eclampsia/ or eclamptic or HELLP or hemolysis elevated liver enzymes low platelet count/ or hemolysis-elevated liver enzymes-low platelet count or antenatal hemorrage* or antepartum hemorrage* or postnatal hemorrage* or postpartum hemorrage* or postpartum complication* or postnatal complication* or post birth complication* or post labor complication* or maternal outcome* or maternal complication* or fetal outcome* or fetal complication* or neonate complication* or neonatal complication*or newborn complication*/ or gestational age/ or special care baby unit admission* or SCBU admission* or NICU admission* or neonatal intensive care unit admission* or small for gestational age* or SGA* or IUGR* or intrauterine growth restriction* or LBW*or low birth weight* or VLBW* or very low birth weight* or neonatal intraventricular hemorrhage*or Intraventricular hemorrhage of newborn*
4. Search 1 AND 2 AND 3

B) Search strategy for Embase (via OVID platform)

1. ('pregnancy outcome' OR 'pregnancy complications' OR 'high risk pregnancy' OR 'labor' OR 'delivery' OR 'fetus outcome' OR 'live birth' OR 'premature labor' OR 'immature and premature labor' OR 'prematurity' OR 'cesarean section' OR 'spontaneous abortion' OR 'stillbirth' OR 'fetus death' OR 'infant mortality' OR 'fetus mortality' OR 'maternal mortality' OR 'perinatal mortality' OR 'preeclampsia' OR 'eclampsia' OR 'eclampsia and preeclampsia' OR 'hellp' OR 'disseminated intravascular coagulation' OR 'magnesium sulfate' OR 'newborn death' OR 'gestational age' OR 'infant low

birth weight' OR 'low birth weight' OR 'very low birth weight' OR 'extremely low birth weight' OR 'apgar score' OR 'newborn intensive care' OR 'small for date infant' OR 'pregnancy outcome*' OR 'maternal outcome*' OR 'pregnancy complication*' OR 'obstetric outcome*' OR 'obstetric complication*' OR 'normal birth*' OR 'live birth*' OR 'premature birth*' OR 'preterm birth*' OR 'preterm deliver*' OR 'born preterm' OR 'cesarean*' OR 'csection*' OR 'miscarriage*' OR 'stillbirth*' OR 'intrauterine death*' OR 'antenatal hemorrhage*' OR 'antenatal haemorrhage*' OR 'antepartum haemorrhage*' OR 'antepartum hemorrhage*' OR 'postpartum hemorrhage*' OR 'postpartum haemorrhage*' OR 'postnatal haemorrhage*' OR 'postnatal hemorrhage*' OR 'postpartum complication*' OR 'postnatal complication*' OR 'special care baby unit admission*' OR 'scbu admission*' OR 'neonatal intensive care unit admission*' OR 'nicu admission*' OR 'sga' OR 'iugr' OR 'neonatal intraventricular haemorrhage' OR 'neonatal intraventricular hemorrhage' OR 'lbw' OR 'vlbw' OR 'fetus outcome*' OR 'high risk pregnan*')

2. ('calcium channel blocker' OR 'beta blocker' OR 'acei' OR 'adrenergic receptor antagonist/blocker' OR 'alpha-2 adrenergic receptor agonist' OR 'angiotensin ii receptor' OR 'antihypertensive' OR 'hydralazine' OR 'sodium nitroprusside' OR 'clonidine hydrochloride' OR 'moxonidine' OR 'renin inhibitor' OR 'thiazide' OR 'loop diuretic' OR 'potassium sparing diuretic')
3. Hypertension
4. 1 AND 2 AND 3

C) Search strategy for Medline (via OVID platform)

1. Anti hypertensive agent*/ or / Antihypertensive agent* Anti-hypertensive agent* / or /Cardiovascular Agent*/ or /anti-hypertensive drug*/ or /antihypertensive drug*/ or /antihypertensive/ or /anti hypertensive drug*/ or /antihypertensive agents therap*/ or /anti hypertensive agents therap*/ or/ anti-hypertensive agents therap*/ or /antihypertensive agents pharmacol*/ or /anti hypertensive agents pharmacol*/ or / anti-hypertensive agents pharmacol*/ or / calcium channel blocker/ OR /beta blocker/ OR /ACEi/ OR /adrenergic receptor antagonist/ or / adrenergic receptor blocker/ OR /alpha-2 adrenergic receptor agonist/ OR /angiotensin ii receptor/ OR /hydralazine/ OR / nitroprusside/ OR /clonidine / OR /thiazide/ OR /loop diuretic/ OR /potassium sparing diuretic
2. Hyperten*
3. pregnancy outcome* or pregnancy complication* obstetric complication* or obstetric labor complications/ or pregnancy, high-risk/ or delivery, obstetric or labor or live birth or obstetric labor, premature or premature birth or premature birth* or preterm birth* preterm deliver* or born preterm or caesarean section cesarean* or c-section or caesarean* or abortion, spontaneous or stillbirth or fetal death/ or infant mortality or maternal mortality or perinatal mortality or pre-eclampsia or eclampsia/ or HELLP/ or gestational age/ or infant, low birth weight/ or pregnancy/or or obstetric outcome* or normal birth* or live birth* or miscarriage* or stillbirth* or intrauterine death* or neonatal death*or antenatal haemorrhage* or postpartum haemorrhage* or antenatal hemorrhage* or postpartum hemorrage* or antepartum hemorrage* or antepartum haemorrhage* or postpartum complication*/ or special care baby unit admission* or SCBU admission* or neonatal intensive care unit admission* or NICU admission* or small for gestationa age or SGA or intrauterine growth restriction or IUGR or neonatal intraventricular hemorrage* or neonatal intraventricular haemorrhage*
4. Search 1 AND 2 AND 3

We used similar to the previously mentioned method, but the exposure is antihypertensive treatment, while the comparative groups divided into 1) untreated normotensive women and 2) untreated women with chronic hypertension.

2. Selection process and eligibility criteria

One of us (SAK) reviewed the titles and abstracts of all articles, studies not meeting the including criteria excluded. Then out of 254 articles, two reviewers (SAK and PB) independently reviewed these studies to decide about including studies.

3. Data collection process and quality assessment

Similarly, one investigator (SAK) extracted the data for all included studies (n=16), and second investigator (DF) extracted the data independently for more than 50% of the included studies to ensure the validity of extracted data. The following details extracted from each study: source of data, authors' name, study design, study period, population characteristics (sample size, appropriateness of case and control recruitment, inclusion/exclusion criteria, and definition of exposure (type of antihypertensive agent / dose/ initiation of treatment), types of hypertension being treated, definitions of outcomes (maternal/fetal and neonatal), appropriateness of analyses, confounders adjusted for (if any), crude and adjusted estimates. Moreover, Newcastle-Ottawa Scale (NOS) was used to assess the study quality, and that was done independently by two investigators (SAK) and (LP). Any inconsistency between the two investigators resolved by a third investigator (ASK)

Table S1: Characteristics of studies for pre-eclampsia (PE) outcome

Author, year published	*Country (ies)*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle-Ottawa grade*
Rey, 1994[26]	Canada	Cohort (hospital-based)	20,375 women	1987 to 1991	Blood pressure (BP) >140/90 mm Hg with a diagnosed before pregnancy or hypertension in at least two measurements<20 weeks of pregnancy and/or 6 weeks after delivery	Superimposed PE: increase in SBP or DPB compared with the first trimester of 30 mm Hg and 15 mm Hg, respectively, with the appearance of protein excretion of >300 mg/day in a 24-hour collection, de novo thrombocytopenia, liver enzymes elevation, or coagulation disorders	6
Conde-Agudelo, 2000a[27]	Latin America and the Caribbean*	Cohort (multicountry)	878,680 pregnancies	1985 to 1997	Chronic hypertension diagnosed using ICD-10 codes	DBP≥90 mmHg on two or more consecutive occasions 24 hours apart or a DBP≥10 mmHg on any one occasion plus proteinuria (one 24-hour urine collection with a total protein excretion of > 300 mg or ≥1+ on a urine dipstick) (ICD-10 codes)	7
Samadi, 2001[28]		Case control (national-based)	182687 women	1988 to 1996	Chronic hypertension diagnosed using ICD-9 codes	ICD-9 codes	7
Mostello, 2002[29]	USA	Case control (population-based)	4702 women	1989 to 1997	History of chronic hypertension	NR (outcome includes PE and eclampsia)	7
Zetterstrom, 20053[0]¥	Sweden	Cohort (population-based)	681,515 women	1992 to 1998	BP≥140/90 mmHg diagnosed before pregnancy, or before 20 weeks' gestation using (ICD-9 and ICD-10 codes)	Preeclampsia: DBP≥90 combined with proteinuria > 300 mg/day or ≥1+ on a urine dipstick.	8
Catov, 2007[31]	Denmark	Cohort (national-based)	69007 women	1997 to 2003	Hypertension reported before pregnancy or at the first interview (16weeks), and/or	BP >140/90mmHg on at least three occasions) in combination with proteinuria of ≥300 mg/l.	8

					reported taking antihypertensive medication		
Gilbert, 2007[32]	USA	Cohort (population-based)	4,324,902 pregnancies	1991 to 2001	Chronic hypertension diagnosed using ICD-9 codes	ICD-9 codes	8
Kiondo, 2012[33]	Uganda	Case control (hospital-based)	559 women	2008 to 2009	History of chronic hypertension before pregnancy or at booking	BP≥160/110 mmHg or two measurements of ≥140/90 mmHg with one 24-hour urine collection with a total protein excretion of > 300 mg or ≥1+ on a urine dipstick	6
Bateman, 2012[3]	USA	Cohort (population-based)	12,947,000	2007 to 2008	Chronic hypertension was defined as hypertension without comorbidities using ICD-9 codes	PE diagnosed using ICD-9 codes	8
Yanit, 2012[34]	USA	Cohort (population-based)	527,937 mother-infant pairs	2006	Chronic hypertension diagnosed using ICD-9 codes	PE diagnosed using ICD-9 codes	8
Orbach, 2013[35‡]	Israel	Cohort (population-based)	100,029 births	1998 to 2008	Chronic hypertension diagnosed using ICD-9 codes	PE diagnosed using ICD-9 codes	8
Lisonkova, 2013[36]	USA	Cohort (population-based)	456,668 mother-infant pairs	2003 to 2008	Chronic hypertension diagnosed using ICD-9 codes	PE diagnosed using ICD-9 codes	8
Guerrier, 2013[37]	Nigeria	Case control (hospital-based)	1676 women	2010 to 2011	History of chronic hypertension	Preeclampsia was defined as new hypertension with BP≥160/110 after 20 weeks of gestation in a woman who was normotensive before 20 weeks gestation, associated with proteinuria (≥2+ on a urine dipstick). Eclampsia was defined as occurrence of seizure and/or altered level of consciousness not caused by epilepsy or other convulsive disorders, with signs of severe preeclampsia.	6

Aksornphusitaphong, 2013[38]	Thailand	Case control (hospital-based)	898 women	2005 to 2010	History of chronic hypertension	Superimposed preeclampsia was defined as a new onset of proteinuria of at least 300 mg/24 h in hypertensive women but no proteinuria prior to 20 weeks' gestation, or a sudden increase in proteinuria or blood pressure in women with hypertension and proteinuria before 20 weeks' gestation.	6
Block-Abraham, 2014[39]	USA	Cohort (hospital-based)	614 women	2007 to 2010	hypertension as SBP>140 mm Hg or DBP>90 mm Hg or both	new-onset or worsening proteinuria and maternal systolic blood pressure 140 mm Hg or greater or diastolic blood pressure 90 mm Hg or greater on two separate occasions, 6 or more hours apart, after 20 weeks of gestation. Preeclampsia superimposed on chronic hypertension was defined as worsening blood pressure and increasing proteinuria after 20 weeks of gestation	6
Bilano, 2014[40]	23 countries**	Cohort (International, multicountry)	276,103 women	(2004 to 2005 in Africa and Latin America), and (between 2007 to 2008 in Asia)	Hypertension diagnosed before pregnancy, or before 20 weeks' gestation	high blood pressure (≥140 mmHg systolic or ≥90 mmHg diastolic or increases of 30 mmHg systolic or 15 mmHg diastolic from the baseline on at least two occasions six or more hours apart) that develops from the 20th, and proteinuria.	7
Abalos, 2014[41]	29 countries***	Cohort (International, multicountry)	312,115 pregnancies	2004 to 2008	BP >140/90 mmHg diagnosed before pregnancy, or before 20 weeks' gestation	presence of hypertension (blood pressure >140/90 mmHg) associated with proteinuria	6
Broekhuijsen, 2015[42]	Netherlands	Cohort (population-based)	988,389 mother-infant pairs	2002 to 2007	BP ≥140/90 mmHg diagnosed before pregnancy, or before 20 weeks' gestation	Preeclampsia is defined as at least one diastolic blood pressure measurement of at least 90mmHg combined with proteinuria of at least 300 mg/day	8

						or 1þ on a urine dipstick. Mild preeclampsia is defined as a diastolic blood pressure from 90 to 109mmHg combined with proteinuria of <500mg/day or 1þ or 2þ on a urine dipstick. Severe preeclampsia is defined as preeclampsia with either a diastolic blood pressure of at least 110mmHg or albuminuria of at least 500mg/day or both	
Panaitescu, 2017[43]	UK	Cohort (hospital-based)	109,932 pregnancies	March 2006 and July 2015/ February 2007 and November 2015	History of chronic hypertension	Superimposed on PE was diagnosed according to the guidelines of the ISSHP: as development of significant proteinuria after 20 weeks' gestation in a previously non-proteinuric woman	8
You S-H, 2018[44]	Taiwan	Cohort (population-based)	2,884,347	2001 to 2014	Chronic hypertension was diagnosed using (ICD-9 codes)	Two occasions of ≥ 140/90 mmHg after 20 weeks of gestation accompanied by proteinuria > 300 mg/day or ≥1+ on dipstick based (using ICD-9 codes)	8
Youngstrom, 2018[45‡]	USA	Cohort (hospital-based)	1,306 mother-infant pairs	2000 to 2014	History of hypertension **or** the use of antihypertensive medication before pregnancy	BP ≥140/90 with either proteinuria (protein excretion ≥300 mg in 24 hours or protein-to-creatinine ratio≥0.3), thrombocytopenia (less than 100,000/mL), transaminases (aspartate aminotransferase > twice the upper limit of normal), or elevated creatinine≥1.2 mg/dL	8

* Uruguay, Argentina, Peru, Colombia, Honduras, Paraguay, Salvador, Chile , Bolivia, Costa Rica, Panama, Dominican Republic, Nicaragua, Brazil, Ecuador, Mexico, Bahamas, and Venezuela

**23 countries; Algeria, Angola, Democratic Republic of Congo, Niger, Nigeria, Kenya, and Uganda from Africa; Argentina, Brazil, Cuba, Ecuador, Mexico, Nicaragua, Paraguay, and Peru from Latin America; and Cambodia, China, India, Nepal, Philippines, Sri Lanka, Thailand, and Vietnam from Asia

***29 countries, African Region (Angola, DR Congo, Kenya, Niger, Nigeria and Uganda); Region of the Americas (Argentina, Brazil, Ecuador, Mexico, Nicaragua, Paraguay and Peru); Eastern Mediterranean Region (Afghanistan, Jordan, Lebanon, occupied Palestinian territory, Palestine, Pakistan and Qatar); South-East Asia Region (India, Nepal, Sri Lanka and Thailand); Western Pacific Region (Cambodia, China, Japan, Mongolia, Philippines and Vietnam)

¥ We combined the odds ratios for mild and severe pre-eclampsia

‡ The odds ratios (ORs) from these studies were combined for treated and untreated women with chronic hypertension.

Table S2: Characteristics of studies for hemolysis, elevated liver enzyme levels, and low platelet levels (HEELP) syndrome outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle-Ottawa grade*
Gilbert, 2007[32]	USA	Cohort (population-based)	4,324,902 pregnancies	1991 to 2001	Chronic hypertension diagnosed using ICD-9 codes	HELLP syndrome diagnosed using ICD-9 codes	8
Fitzpatrick, 2014[46]	UK	Case-control (national-based	605 women	2011-2012	NR	Elevated liver enzymes: (Serum aspartate aminotransferase ≥70 international units/L OR Gammaglutamy l transferase ≥70 international units/L OR Alanine aminotransferase ≥70 international units/L or greater) AND Low platelets, defined as platelet count< 1003109/L AND Hemolysis, defined by abnormal (fragmented or contracted red cells) peripheral blood smear or serum lactate dehydrogenase levels≥600 international units/L or total bilirubin ≥20.5 micromole/L OR BP≥140/90 mm Hg OR Proteinuria (≥1+ 0.3 g/L on dipstick testing, a protein:creatinine ratio ≥30 mg/mmol on a random sample, or a urine protein excretion ≥300 mg /24 h")	7
Malmstrom, 2018[47]	Norway	Cohort (population-based)	418,897 pregnancies	1999 to 2014	Clinical definitions based on Norwegian Society of Obstetricians and Gynaecologists	HELLP syndrome diagnosed based on Norwegian Society of Obstetricians and Gynaecologists	8

Table S3: Characteristics of studies for cesarean section (CS) outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle -Ottawa grade*
Sass, 1990[70]	Brazil	Case control (hospital-based)	337 births	1985 to 1986	DBP ≥90mmHg before pregnancy or up to 20 weeks of pregnancy or hypertension at 10 weeks postpartum (for those whose antenatal care was not at that hospital)	CS (not specified)	4
Hjertberg, 1992[92]	Sweden	Cohort (hospital-based)	2593 births	1986 to 1987	Chronic hypertension before pregnancy before 24weeks' gestation	CS (not specified)	6
Rey, 1994[26]	Canada	Cohort (hospital-based)	20,375 mothers	1987 to 1991	BP >140/90 mm Hg with a diagnosed before pregnancy or hypertension in at least two measurements<20 weeks of pregnancy and/or 6 weeks after delivery	CS (not specified)	6
Jain, 1997[50]	USA	Cohort (hospital-based)	109,428 mother-infant pairs	1982 to 1987	SBP≥ 140 mmHg or a DBP≥90 mmHg before pregnancy or before 20 weeks' gestation.	CS (not specified)	7
Lydakis, 1998[85]	UK	Cohort (hospital-based)	3,729 births	1980 to 1997	Hypertension (DBP ≥110 mmHg OR DBP >90 mmHg on two or more occasions ≥4 h apart) at the first booking visit before the 20 weeks' gestation in the absence of trophoblastic disease, OR hypertension at any stage of pregnancy that continued for more than 6 weeks after delivery	Emergency CS	6

Hartikainen, 1998[51]	Finland	Cohort (hospital-based)	8,050 mother-infant pairs	1985 to 1986	Hypertension diagnosed before pregnancy and/or DBP > 90 mm Hg and/or antihypertensive medication, each<20 weeks' gestation	CS (not specified)	6
Lydakis, 2001[54]	UK	Cohort (hospital-based)	238 births	1980 to 1997	Hypertension (DBP ≥110 mmHg OR DBP >90 mmHg on two or more occasions ≥4 h apart) at the first booking visit before the 20 weeks' gestation in the absence of trophoblastic disease, OR hypertension at any stage of pregnancy that continued for more than 6 weeks after delivery	Elective CS	6
Vanek, 2004[73]	Israel	Cohort (hospital-based)	114,963 mother-infant pairs	1988 to 1999	BP≥140/90 mmHg that measured at least twice at least 4 h apart that preceded pregnancy, or hypertension present before 20 weeks' gestation or that persists longer than the usual postpartum period (12 weeks post-delivery)	CS (not specified)	6
Roberts, 2005[55]	Australia	Cohort (population-based)	227,067 women and 231,811 infants	2000 to 2002	Chronic hypertension diagnosed ICD-10 codes	Emergency and elective CS	7
Gilbert, 2007[32]	USA	Cohort (population-based)	4,324,902 pregnancies	1991 to 2001	Chronic hypertension diagnosed using ICD-9 codes	CS (not specified)	8
Cruz, 2011[78]	USA	Cohort (multicentre hospital-based)	17,752 births	2002 and 2008	Hypertension reported before pregnancy, or SBP≥140 mm Hg or DBP ≥90 mm Hg before 20 weeks' gestation	CS (not specified)	7

Tuuli, 2011[59]	USA	Cohort (hospital-based)	58,135 mother-infant pairs	1990 to 2008	SBP≥ 140 mmHg or a DBP≥90 mmHg before pregnancy or before 20 weeks' gestation	CS (not specified)	8
Madi, 2012[80]	Brazil	Cohort (hospital-based)	3,689 mother-infant pairs	1998 to 2009.	BP≥140/90 mmHg that measured at least twice, before 20 weeks' gestation, or persisting for ≥12 weeks post-delivery	CS (not specified)	7
Bateman, 2012[3]	USA	Cohort (population-based)	12,947,000 pregnancies	2007 to 2008	Chronic hypertension was defined as hypertension without comorbidities using ICD-9 codes	CS (not specified)	8
Ye, 2014[62]	China	Cohort (population-based)	108,550 Pregnancies from 106, 869 women	2011 to 2011	BP≥140/90 mmHg before pregnancy or diagnosed before 20 weeks' gestation or developed hypertension after 20 weeks of gestation and continued for 12 weeks of postpartum	CS (not specified)	7
Janoudi, 2015[96]	Canada	Cohort (hospital-based)	134,088 births	2011to 2012	NR	CS (not specified)	8
Broekhuijsen, 2015[42]	Netherlands	Cohort (population-based)	988,389 mother-infant pairs	2002 to 2007	BP ≥140/90 mmHg diagnosed before pregnancy, or before 20 weeks' gestation	Elective CS	8
Panaitescu, 2017[44]	UK	Cohort (hospital-based)	109,932 pregnancies	March 2006 and July 2015/ February 2007 and November 2015	History of chronic hypertension	Emergency and elective CS	8

Table S4: Characteristics of studies for post-partum hemorrhage (PPH) outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle-Ottawa grade*
Vanek, 2004[73]	Israel	Cohort (hospital-based)	114,963 mother-infant pairs	1988 to 1999	BP≥140/90 mmHg that measured at least twice, before 20 weeks' gestation, or persisting for ≥12 weeks post-delivery	Postpartum haemorrhage	6
Roberts, 2005[55]	Australia	Cohort (population-based)	227,067 women and 231,811 infants	2000 to 2002	ICD-10 codes	Postpartum haemorrhage	7
Gilbert, 2007[32]	USA	Cohort (population-based)	4,324,902 pregnancies	1991 to 2001	Chronic hypertension diagnosed using ICD-9 codes	Diagnosed using ICD-9 codes	8
Cruz, 2011[78]	USA	Cohort (multicentre hospital-based)	17,752	2002 and 2008	Hypertension reported before pregnancy, or SBP≥140 mm Hg or DBP ≥90 mm Hg before 20 weeks' gestation	Postpartum haemorrhage	7
Ye, 2014[62]	China	Cohort (population-based)	108,550 Pregnancies from 106, 869 women	2011 to 2011	BP≥140/90 mmHg before pregnancy or diagnosed before 20 weeks' gestation or developed hypertension after 20 weeks of gestation and continued for 12 weeks of postpartum	Postpartum haemorrhage	7
Broekhuijsen, 2015[42]	Netherlands	Cohort (population-based)	988,389 mother-infant pairs	2002 to 2007	BP ≥140/90 mmHg diagnosed before pregnancy, or before 20 weeks' gestation	Postpartum haemorrhage	8

Table S5: Characteristics of studies for maternal mortality outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle -Ottawa grade*
Gilbert, 2007[32]	USA	Cohort (population-based)	4,324,902 pregnancies	1991 to 2001	Chronic hypertension diagnosed using ICD-9 codes	Maternal in hospital death	8
Kuklina, 2009[97]	USA	Cohort (population-based)	34,321,769 births	1998 to 2006	Chronic hypertension diagnosed using ICD-9 codes	Maternal death	8
Bateman, 2012[3]	USA	Cohort (population-based)	12,947,000 pregnancies	2007 to 2008	Chronic hypertension diagnosed using ICD-9 codes	Maternal death	8
Campbell, 2013[98]	USA	Cohort (population-based)	1,078,553	1995 to 2003	Chronic hypertension diagnosed using ICD-9 codes	Maternal death considered present if it was identified in the discharge diagnosis codes, if it was noted on the birth certificate, or if it was present on both when possible	8

Table S6: Characteristics of studies for stillbirth outcome

Author, year published	*Country (ies)*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle -Ottawa grade*
Copper, 1994[99]	USA	Cohort (population-based)	34,350 births	1982-1986	NR	Birth of a fetus at 20 weeks' gestation or later with an APGAR score of 0 at 1 and 5 minutes	7
Ananth, 1995a[93]	USA	Cohort (population-based)	371,123 mother-infant pairs	1988 and 1991	NR	Fetal death (early≤20 weeks and late>28 weeks) and further defined into antepartum and intrapartum death	8
Jain, 1997[50]	USA	Cohort (hospital-based)	109,428 mother-infant pairs	1982 to 1987	SBP≥ 140 mmHg or a DBP≥90 mmHg before pregnancy or before 20 weeks' gestation	***NR***	7
Lydakis, 1998[85]	UK	Cohort (hospital-based)	3,729 births	1980 to 1997	Hypertension (DBP ≥110 mmHg OR DBP >90 mmHg on two or more occasions ≥4 h apart) at the first booking visit before the 20 weeks' gestation in the absence of trophoblastic disease, OR hypertension at any stage of pregnancy that continued for more than 6 weeks after delivery	Intrauterine fetal death at >24 weeks' gestation with an Apgar score 0 at 1 min	6
Conde-Agudelo, 2000b[100]	Latin America and the Caribbean*	Cohort (multicountry)	837,232 births	1985 to 1997	History of chronic hypertension using ICD-10 codes	Fetal death at≥20 weeks' gestation	7
Allen, 2004[72]	Canada	Cohort (population-based)	123,160 births	1988 to 2000	Hypertension reported before pregnancy, or prior to 20 weeks' gestation	Fetal death at≥20 weeks' gestation and birthweight ≥ 500 grams	9

Canterino, 2004[94]	USA	Cohort (population-based)	21,610,873 mother-infant pairs	1995 to 2000	Hypertension before pregnancy or before 20 weeks' gestation	Fetal death at≥24 weeks' gestation	7
Roberts, 2005[55]	Australia	Cohort (population-based)	227,067 women and 231,811 infants	2000 to 2002	Chronic hypertension diagnosed using ICD-10 codes	Fetal death at≥20 weeks' gestation	7
Aagaard-Tillery, 2006[101]	USA	Case control (population-based)	4306	1992 to 2002	NR	Fetal death at≥20 weeks' gestation or weighing ≥400g	6
Gilbert, 2007[32]	USA	Cohort (population-based)	4,324,902 pregnancies	1991 to 2001	Chronic hypertension diagnosed using ICD-9 codes	ICD-9 –CM codes	8
Zetterstrom, 2008[90]	Sweden	Cohort (population-based)	866,188 mother-infant pairs	1992 to 2004	BP≥140/90 mmHg diagnosed before pregnancy, or before 20 weeks' gestation using (ICD-9 and ICD-10 codes)	Intrauterine fetal death at ≥28 weeks' gestation	8
Reddy, 2010[102]	USA	Cohort (population-based)	160,954	2002 to 2008	Chronic hypertension diagnosed using ICD-9 codes	Antepartum stillbirth defined as fetus having no signs of life before labor with an Apgar scores of 0 and 0 at 1 minute and 5 minutes	7
Tuuli, 2011[59]	USA	Cohort (hospital-based)	58,135 mother-infant pairs	1990 to 2008	SBP≥ 140 mmHg or a DBP≥90 mmHg before pregnancy or before 20 weeks' gestation	Fetal death at≥20 weeks' gestation	8
Yanit, 2012[34]	USA	Cohort (population-based)	527,937 mother-infant pairs	2006	Chronic hypertension diagnosed using ICD-9 codes	NR	8
Bateman, 2012[3]	USA	Cohort (population-based)	12,947,000 pregnancies	2007 to 2008	Chronic hypertension diagnosed using ICD-9 codes	ICD-9 codes	8

Ahmad, 2012[91]	Norway	Cohort (population-based)	2,027,042 mother-infant pairs	1967 to 2006	SBP≥ 140 mmHg or a DBP≥90 mmHg before 20 weeks' gestation (ICD-8 and ICD-10 codes)	Fetal death ≥20 weeks' gestation	7
Madi, 2012[80]	Brazil	Cohort (hospital-based)	3,689 mother-infant pairs	1998 to 2009.	BP≥140/90 mmHg that measured at least twice, before 20 weeks' gestation, or persisting for ≥12 weeks post-delivery	Fetal death	7
Yerlikaya, 2016[103]	UK	Cohort (hospital-based)	113,415 mother-infant pairs	2006 to 2015	History of chronic hypertension	Fetal death ≥24 weeks' gestation	8
Panaitescu, 2017[44]	UK	Cohort (hospital-based)	109,932 pregnancies	March 2006 and July 2015/ February 2007 and November 2015	History of chronic hypertension	Fatal death ≥24 weeks	8
Xiong, 2018[104]	China	Cohort (population-based)	6,970,032 births	2012 to 2016	SBP≥ 140 mmHg or a DBP≥90 mmHg before pregnancy or before 20 weeks' gestation.	Stillbirth defined according to WHO of 3rd trimester stillbirth definition	8
Youngstrom, 2018[45]‡	USA	Cohort (hospital-based)	1,306 mother-infant pairs	2000 to 2014	History of hypertension **or** the use of antihypertensive medication before pregnancy	NR	8

*Uruguay, Argentina, Peru, Colombia, Honduras, Paraguay, Salvador, Chile , Bolivia, Costa Rica, Panama, Dominican Republic, Nicaragua, Brazil, Ecuador, Mexico, Bahamas, and Venezuela. ‡ Odds ratios from this study was combined for treated and untreated women with chronic hypertension.

Table S7: Characteristics of studies for very preterm birth (VPTB) outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle -Ottawa grade*
Ananth, 1995b[9]	USA	Cohort (population-based)	276,876	1988 and 1990	Chronic hypertension obtained from medical records; no further details provided	<33 weeks' gestation	8
McCowan, 1996[49]	New Zealand	Cohort (hospital-based)	20,224 mother-infant pairs	1991 to 1993	DBP>90 mmHg before 20 weeks gestation, pre-existing history of essential hypertension and/or on antihypertensive medication before the pregnancy	<32 weeks' gestation	6
Roberts, 2005[55]	Australia	Cohort (population-based)	227,067 women and 231,811 infants	2000 to 2002	chronic hypertension diagnosed using ICD-10 codes	28-32 weeks' gestation	*7*
Tuuli, 2011[59]	USA	Cohort (hospital-based)	58,135 mother-infant pairs	1990 to 2008	SBP≥ 140 mmHg or a DBP≥90 mmHg before pregnancy or before 20 weeks' gestation	<32 weeks' gestation	*8*
Ferrazzani, 2011[58]	Italy	Cohort (hospital-based)	1,154 Women	1986 and 1995	Hypertension diagnosed before pregnancy and/or DBP ≥90 mm Hg and/or on antihypertensive medication, each before 20 weeks' gestation uncomplicated by de novo proteinuria	<32 weeks' gestation	6
Yanit, 2012[34]	USA	Cohort (population-based)	527,937 mother-infant pairs	2006	Chronic hypertension diagnosed using ICD-9 codes	<32 weeks' gestation	8
Broekhuijsen, 2015[42]	Netherlands	Cohort (population-based)	988,389 mother-infant pairs	2002 to 2007	BP ≥140/90 mmHg diagnosed before pregnancy, or before 20 weeks' gestation	<32 weeks' gestation	8
Yang, 2015[66]	China	Cohort (national-based)	344,929	2010 and 2013	SBP≥ 140 mmHg or a DBP≥90 mmHg on a single occasion after participants rested for ≥10 minutes	<32 weeks' gestation, but after 28 weeks	6

Panaitescu, 2017[44]	UK	Cohort (hospital-based)	109,932 pregnancies	March 2006 and July 2015/ February 2007 and November 2015	History of chronic hypertension	<34 weeks' gestation (medically indicated and spontaneous) *	8

**we combined the estimates for both medically indicated and spontaneous as it was the only study reported that for VPTB*

Table S8: Characteristics of studies for preterm birth (PTB) outcome

Author, year published	*Country (ies)*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle -Ottawa grade*
Acien,1990[48]	Spain	Cohort (hospital-based)	236 pregnancies and 238 newborns	1979 and 1986	NR	33-36 weeks' gestation	4
Rey, 1994[26]	Canada	Cohort (hospital-based)	20,375 mothers	1987 to 1991	BP>140/90 mm Hg with a diagnosed before pregnancy or hypertension in at least two measurements<20 weeks of pregnancy and/or 6 weeks after delivery	NR	6
Velentgas, 1994[84]	USA	Cohort (population-based)	14,562 mother-infant pairs	1984-1990	NR	<37 weeks' gestation	6
Ananth, 1995b[9]	USA	Cohort (population-based)	276,876	1988 and 1990	Chronic hypertension obtained from medical records; no further details provided	<37 weeks' gestation	8
McCowan, 1996[49]	New Zealand	Cohort (hospital-based)	20,224 mother-infant pairs	1991 to 1993	DBP>90 mmHg before 20 weeks gestation, pre-existing history of essential hypertension and/or on antihypertensive medication before the pregnancy	<37 weeks' gestation	6
Jain, 1997[50]	USA	Cohort (hospital-based)	109,428 mother-infant pairs	1982 to 1987	SBP≥ 140 mmHg or a DBP≥90 mmHg before pregnancy or before 20 weeks' gestation.	NR	7
Hartikainen, 1998[51]	Finland	Cohort (hospital-based)	8,050 mother-infant pairs	1985 to 1986	Hypertension diagnosed before pregnancy and/or DBP >90 mm Hg and/ or antihypertensive medication, each<20 weeks' gestation	<37 weeks' gestation	6
Meis, 1998[52]	USA	Cohort (multi-centres)	2929 Women	1992 to 1994	BP≥140/90 mmHg before 20 weeks' gestation	<37 weeks' gestation (<259 days') gestation, (medically indicated)	7

Samadi, 1998[53]	USA	Case control (hospital-based)	25,060	1988 and 1993	Chronic hypertension diagnosed using ICD-9 codes	<37 weeks' gestation (spontaneous)	7
Lydakis, 2001[54]	UK	Cohort (hospital-based)	238 births	1980 to 1997	Hypertension (DBP ≥110 mmHg OR DBP >90 mmHg on two or more occasions ≥4 h apart) at the first booking visit before the 20 weeks' gestation in the absence of trophoblastic disease, OR hypertension at any stage of pregnancy that continued for more than 6 weeks after delivery	<37 weeks' gestation	6
Roberts, 2005[55]	Australia	Cohort (population-based)	227,067 women and 231,811 infants	2000 to 2002	Chronic hypertension diagnosed using ICD-10 codes	33-36 weeks' gestation	7
Graham, 2007[56]	USA	Cohort (population-based)	202,931 mother-infant pairs	1999 to 2003	Hypertension diagnosed before pregnancy	<37 weeks' gestation	7
Gilbert, 2007[32]	USA	Cohort (population-based)	4,324,902 pregnancies	1991 to 2001	Chronic hypertension diagnosed using ICD-9 codes	<37 weeks' gestation	8
Carter, 2011[57]	USA	Cohort (population-based)	259,576 Births	2000 to 2008	NR	<37 weeks' gestation	6
Ferrazzani, 2011[58]	Italy	Cohort (hospital-based)	1,154 Women	1986 and 1995	Hypertension diagnosed before pregnancy and/or DBP ≥90 mm Hg and/or on antihypertensive medication, each before 20 weeks' gestation uncomplicated by de novo proteinuria	<37 weeks' gestation	6
Tuuli, 2011[59]	USA	Cohort (hospital-based)	58,135 mother-infant pairs	1990 to 2008	SBP≥ 140 mmHg or a DBP≥90 mmHg before pregnancy or before 20 weeks' gestation	<37 weeks' gestation	8

Bateman, 2012[3]	USA	Cohort (population-based)	12,947,000 pregnancies	2007 to 2008	Chronic hypertension diagnosed using ICD-9 codes	<37 weeks' gestation (spontaneous)	8
Yanit, 2012[34]	USA	Cohort (population-based)	527,937 mother-infant pairs	2006	Chronic hypertension diagnosed using ICD-9 codes	<37 weeks' gestation	8
Orbach, 2013[35‡]	Israel	Cohort (population-based)	100,029 births	1998 to 2008	Chronic hypertension diagnosed using ICD-9 codes	<37 weeks' gestation	8
Su, 2013[60]	Taiwan	Cohort (population-based)	10,908 mother-infant pairs	2005 to 2005	Chronic hypertension diagnosed using ICD-9-CM codes	<37 weeks' gestation	8
Morisaki, 2014[61]	29 countries*	Cohort (international, multicountry)	299,878 women	2010 to 2011	NR	<37 weeks' gestation	7
Ye, 2014[62]	China	Cohort (population-based)	108,550 Pregnancies from 106, 869 women	2011 to 2011	BP≥140/90 mmHg before pregnancy or diagnosed before 20 weeks' gestation or developed hypertension after 20 weeks of gestation and continued for 12 weeks of postpartum	<37 weeks' gestation (after 20 weeks and before 37)	7
Arora, 2015[63]	5 countries**	Cohort (multicountry)	37 661 births	2007 to 2009	History of chronic hypertension	<37 weeks' gestation	6
Broekhuijsen, 2015[42]	Netherlands	Cohort (population-based)	988,389 mother-infant pairs	2002 to 2007	BP ≥140/90 mmHg diagnosed before pregnancy, or before 20 weeks' gestation	<37 weeks' gestation	8
Derakhshi, 2014[64]	Iran	Case control (hospital-based)	600 births	2012	NR	<37 weeks' gestation	5

Tucker, 2015[65]	USA	Cohort (Medicaid patients)	15,428 women	2011 to 2012	NR	24–36 weeks' gestation	7
Yang, 2015[66]	China	Cohort (national-based)	344,929	2010 and 2013	SBP≥ 140 mmHg or a DBP≥90 mmHg on a single occasion after participants rested for ≥10 minutes	<37 weeks' gestation	6
Premkumar, 2016[67]	USA	Cohort (hospital-based)	23,425 women	2002 to 2015	SBP> 140 mm Hg or DBP> 90 mm Hg recorded on at least 2 separate occasions >6 hours apart before 20 weeks' gestation	<37 weeks' gestation< 37 weeks' gestation (medically indicated and spontaneous)	9
Souza, 2016[68]	Brazil	Case control (multicentre)	2614 Births	2011 to 2012	NR	<37 weeks' gestation	6
Panaitescu, 2017[44]	UK	Cohort (hospital-based)	109,932 pregnancies	March 2006 and July 2015/ February 2007 and Novembe r 2015	History of chronic hypertension	<37 weeks' gestation (medically indicated and spontaneous)	8
Campbell, 2018[69]	Canada	Cohort (population-based)	26,654 live births	2009 to 2014	NR	<37 weeks' gestation (live births)	6
Youngstrom, 2018[45‡]	USA	Cohort (hospital-based)	1,306 mother-infant pairs	2000 to 2014	History of hypertension or the use of antihypertensive medication before pregnancy	<37 weeks' gestation	8

*29 countries: African Region (Angola, DR Congo, Kenya, Niger, Nigeria and Uganda); Region of the Americas (Argentina, Brazil, Ecuador, Mexico, Nicaragua, Paraguay and Peru); Eastern Mediterranean Region (Afghanistan, Jordan, Lebanon, occupied Palestinian territory, Palestine, Pakistan and Qatar); South-East Asia Region (India, Nepal, Sri Lanka and Thailand); Western Pacific Region (Cambodia, China, Japan, Mongolia, Philippines and Vietnam)

** Czech Republic, Hungary, Romania, Slovakia, and Ukraine

‡ The ORs from these studies were combined for treated and untreated women with chronic hypertension.

Table S9: Characteristics of studies for small for gestational age (SGA) outcome

Author, year published	*Country (ies)*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle -Ottawa grade*
Sass, 1990[70]	Brazil	Case control (hospital-based)	337 Births	1985 to 1986	DBP ≥90mmHg before pregnancy or up to 20 weeks' gestation or hypertension at 10 weeks postpartum (for those whose antenatal care was not at that hospital).	birthweight <10th percentile for gestational age	4
Acien,1990[48]	Spain	Cohort (hospital-based)	236 pregnancies and 238 newborns	1979 and 1986	NR	birthweight <10th percentile for gestational age	4
Rey, 1994[26]	Canada	Cohort (hospital-based)	20,375 mothers	1987 to 1991	BP>140/90 mm Hg with a diagnosed before pregnancy or hypertension in at least two measurements<20 weeks of pregnancy and/or 6 weeks after delivery	birthweight <10th percentile for gestational age	6
Ananth, 1995b[9]	USA	Cohort (population-based)	276,876	1988 and 1990	Chronic hypertension obtained from medical records; no further details provided	birthweight <10th percentile for gestational age	8
McCowan, 1996[49]	New Zealand	Cohort (hospital-based)	20,224 mother-infant pairs	1991 to 1993	DBP>90 mmHg before 20 weeks gestation, pre-existing history of essential hypertension and/or on antihypertensive medication before the pregnancy	birthweight <5th percentile for gestational age	6
Haelterman, 1997[71]	France	Cohort (multicentre)	1938 mother-infant pairs	1991 to 1993	DBP ≥90 mmHg on two or more consecutive occasions at least 4 h apart or SBP>160 mmHg before 21 weeks' gestation OR women with a prior history of hypertension and were on antihypertensive treatment.	birthweight <5th percentile for gestational age	6

Jain, 1997[50]	USA	Cohort (hospital-based)	109,428 mother-infant pairs	1982 to 1987	SBP≥ 140 mmHg or a DBP≥90 mmHg before pregnancy or before 20 weeks' gestation.	Intrauterine growth restriction (NR)	7
Hartikainen, 1998[51]	Finland	Cohort (hospital-based)	8,050 mother-infant pairs	1985 to 1986	Hypertension diagnosed before pregnancy and/or DBP > 90 mm Hg and/ or antihypertensive medication, each<20th week	birthweight <10th percentile for gestational age	6
Lydakis, 2001[54]	UK	Cohort (hospital-based)	238 births	1980 to 1997	Hypertension (DBP ≥110 mmHg OR DBP >90 mmHg on two or more occasions ≥4 h apart) at the first booking visit before the 20th week of pregnancy in the absence of trophoblastic disease, OR hypertension at any stage of pregnancy that continued for more than 6 weeks after delivery.	birthweight <5th percentile for gestational age	6
Allen, 2004[72]	Canada	Cohort (population-based)	123,160 births	1988 to 2000	Hypertension reported before pregnancy, or before to 20 weeks' gestation	birthweight <5th percentile for gestational age	9
Vanek, 2004[73]	Israel	Cohort (hospital-based)	114,963 mother-infant pairs	1988 to 1999	BP≥140/90 mmHg that measured at least twice, before 20 weeks' gestation, or persisting for ≥12 weeks post-delivery	NR	6
Roberts, 2005[55]	Australia	Cohort (population-based)	227,067 women and 231,811 infants	2000 to 2002	chronic hypertension diagnosed using ICD-10 codes	birthweight <10th percentile for gestational age	7
Odibo, 2006[74]	USA	Case control (hospital-based)	2472	1997 to 2004	NR	birthweight <10th percentile for gestational age	7

Rasmussen, 2006[75]	Norway	Cohort (population-based)	404,400 mother-infant pairs	1999 to 2002	NR	birthweight <10th percentile for gestational age	6
Zetterstrom, 2006[76]	Sweden	Cohort (population-based)	560,188 mother-infant pairs	1992 to 1998	BP≥140/90 mmHg diagnosed before pregnancy, or before 20 weeks' gestation using (ICD-9 and ICD-10 codes)	<2 standard deviations below the mean birth weight adjusted for gestational age and sex	8
Gilbert, 2007[32]	USA	Cohort (population-based)	4,324,902 pregnancies	1991 to 2001	Chronic hypertension diagnosed using ICD-9 codes	ICD-9 codes	8
Catov, 2008[77]	Denmark	Cohort (national-based)	81,008 mother-infant pairs	1996 to 2002	Hypertension reported before pregnancy or at the first interview (16 weeks), and/or reported taking antihypertensive medication	<2 standard deviations below the mean birth weight	8
Cruz, 2011[78]	USA	Cohort (multicentre hospital-based)	17,752	2002 and 2008	Hypertension reported before pregnancy, or SBP≥140 mm Hg or DBP ≥90 mm Hg before 20 weeks' gestation	birthweight <10th percentile for gestational age	7
Poon, 2011[79]	UK	Cohort (hospital-based)	33,602 pregnancies	2006 to 2009	History of chronic hypertension	birthweight <5th percentile for gestational age	6
Tuuli, 2011[59]	USA	Cohort (hospital-based)	58,135 mother-infant pairs	1990 to 2008	SBP≥ 140 mmHg or a DBP≥90 mmHg before pregnancy or before 20 weeks' gestation	birthweight <10th percentile for gestational age	8
Bateman, 2012[3]	USA	Cohort (population-based)	12,947,000 pregnancies	2007 to 2008	Chronic hypertension diagnosed using ICD-9 codes	ICD-9 codes	8
Madi, 2012[80]	Brazil	Cohort (hospital-based)	3,689 mother-infant pairs	1998 to 2009.	BP≥140/90 mmHg that measured at least twice, before 20 weeks' gestation, or persisting for ≥12 weeks post-delivery	NR	7

Yanit, 2012[35]	USA	Cohort (population-based)	527,937 mother-infant pairs	2006	Chronic hypertension diagnosed using ICD-9 codes	birthweight <10th percentile for gestational age	8
Anderson, 2013[81]	New Zealand	Cohort (population-based)	24,434 mother-infant pairs	2006 to 2009	Chronic hypertension diagnosed based on the International Society for the Study of Hypertension in Pregnancy guidelines	birthweight <10th percentile for gestational age	8
Orbach, 2013[35‡]	Israel	Cohort (population-based)	100,029 births	1998 to 2008	Chronic hypertension diagnosed using ICD-9 codes	ICD-9 codes	8
Su, 2013[60]	Taiwan	Cohort (population-based)	10,908 mother-infant pairs	2005 to 2005	Chronic hypertension diagnosed using ICD-9 CM codes	birthweight <10th percentile for gestational age	8
Ota, 2014[82]	29 countries*	Cohort (international, multicountry)	245,773 mother-infant pairs	2010 to 2011	NR	birthweight <10th percentile for gestational age	7
Xaverius, 2014[83]	USA	Cohort (population-based)	142,017 mother-infant pairs	2000-2009	History of chronic hypertension	birthweight <10th percentile for gestational age	7
Broekhuijsen, 2015[42]	Netherlands	Cohort (population-based)	988,389 mother-infant pairs	2002 to 2007	BP ≥140/90 mmHg diagnosed before pregnancy, or before 20 weeks' gestation	birthweight <10th percentile for gestational age	8
Panaitescu, 2017[44]	UK	Cohort (hospital-based)	109,932 pregnancies	March 2006 and July 2015/ February 2007 and November 2015	History of chronic hypertension	birthweight <10th percentile for gestational age	8

Campbell, 2018[69]	Canada	Cohort (population-based)	26,654 live births	2009 to 2014	NR	birthweight <10th percentile for gestational age	6
Youngstrom, 2018[45]‡	USA	Cohort (hospital-based)	1,306 mother-infant pairs	2000 to 2014	History of hypertension or the use of antihypertensive medication before pregnancy	birthweight <10th percentile for gestational age	8

*29 countries: African Region (Angola, DR Congo, Kenya, Niger, Nigeria and Uganda); Region of the Americas (Argentina, Brazil, Ecuador, Mexico, Nicaragua, Paraguay and Peru); Eastern Mediterranean Region (Afghanistan, Jordan, Lebanon, occupied Palestinian territory, Palestine, Pakistan and Qatar); South-East Asia Region (India, Nepal, Sri Lanka and Thailand); Western Pacific Region (Cambodia, China, Japan, Mongolia, Philippines and Vietnam)

‡ The ORs from these studies were combined for treated and untreated women with chronic hypertension.

Table S10: Characteristics of studies for low birth weight (LBW) outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle -Ottawa grade*
Velentgas, 1994[84]	USA	Cohort (population-based)	14,562 mother-infant pairs	1984-1990	NR	birth weight of <2500 grams	6
Ananth, 1995b[9]	USA	Cohort (population-based)	276,876	1988 and 1990	Chronic hypertension obtained from medical records; no further details provided	birth weight of <2500 grams‡	8
Hartikainen, 1998[51]	Finland	Cohort (hospital-based)	8,050 mother-infant pairs	1985 to 1986	Hypertension diagnosed before pregnancy and/or DBP > 90 mm Hg and/or antihypertensive medication, each<20 weeks' gestation	birth weight of <2500 grams	6
Lydakis, 1998[85]	UK	Cohort (hospital-based)	3,729 births	1980 to 1997	Hypertension (DBP ≥110 mmHg OR DBP >90 mmHg on two or more occasions ≥4 h apart) at the first booking visit before the 20 weeks' gestation in the absence of trophoblastic disease, OR hypertension at any stage of pregnancy that continued for more than 6 weeks after delivery	birth weight of <2000 grams	6
Gilbert, 2007[32]	USA	Cohort (population-based)	4,324,902 pregnancies	1991 to 2001	Chronic hypertension diagnosed using ICD-9 codes	birth weight of ≤2500 grams	8
Graham, 2007[56]	USA	Population-based	202,931 mother-infant pairs	1999 to 2003	Hypertension diagnosed before pregnancy	birth weight of <2500 grams	7
Odell, 2006[86]	USA	Population-based (Black women only)	16,578	1996 to 2000	History of chronic hypertension diagnosed using ICD 9 codes	birth weight of <2500 grams	7

Vahdaninia, 2008[87]	Iran	Cohort (hospital-based)	3733 mother-infant pairs	2005	NR	birth weight of <2500 grams	7
Madi, 2012[80]	Brazil	Cohort (hospital-based)	3,689 mother-infant pairs	1998 to 2009.	BP≥140/90 mmHg that measured at least twice, before 20 weeks' gestation, or persisting for ≥12 weeks post-delivery	birth weight of ≤2500 grams	7
Su, 2013[60]	Taiwan	Cohort (population-based)	10,908 mother-infant pairs	2005 to 2005	Chronic hypertension diagnosed using ICD-9 CM codes	birth weight of <2500 grams	8
Ye, 2014[62]	China	Cohort (population-based)	108,550 Pregnancies from 106, 869 women	2011 to 2011	BP≥140/90 mmHg before pregnancy or diagnosed before 20 weeks' gestation or developed hypertension after 20 weeks of gestation and continued for 12 weeks of postpartum	birth weight of <2500 grams	7
Harvey, 2017[88]	USA	Case control (hospital-based)	862 births	2010 to 2011	NR	birth weight of <2500 grams	8
Campbell, 2018[69]	Canada	Cohort (population-based)	26,654 live births	2009 to 2014	NR	birth weight of <2500 grams	6
Hailu,2018[89]	Ethiopia	Case control (hospital-based)	441 births	2016	NR	birth weight of <2500 grams	6

‡ We combined the ORs for birthweight < 1,499 g and birthweight (1,500-2,499 g).

Table S11: Characteristics of studies for neonatal death outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle -Ottawa grade*
Jain, 1997[50]	USA	Cohort (hospital-based)	109,428 mother-infant pairs	1982 to 1987	SBP≥ 140 mmHg or a DBP≥90 mmHg before pregnancy or before 20 weeks' gestation.	NR	7
Gilbert, 2007[32]	USA	Cohort (population-based)	4,324,902 pregnancies	1991 to 2001	Chronic hypertension diagnosed using ICD-9 codes	ICD-9 –CM codes	8
Zetterstrom, 2008[90]	Sweden	Cohort (population-based)	866,188 mother-infant pairs	1992 to 2004	BP≥140/90 mmHg diagnosed before pregnancy, or before 20 weeks' gestation using (ICD-9 and ICD-10 codes)	Neonatal death of an infant born live at any week of gestation, within 27 days of birth	8
Madi, 2012[80]	Brazil	Cohort (hospital-based)	3,689 mother-infant pairs	1998 to 2009.	BP≥140/90 mmHg that measured at least twice, before 20 weeks' gestation, or persisting for ≥12 weeks post-delivery	Neonatal mortality	7
Youngstrom, 2018[45‡]	USA	Cohort (hospital-based)	1,306 mother-infant pairs	2000 to 2014	History of hypertension or the use of antihypertensive medication before pregnancy	NR	8

‡ The ORs from this study was combined for treated and untreated women with chronic hypertension.

Table S12: Characteristics of studies for perinatal death outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle -Ottawa grade*
Acien,1990[48]	Spain	Cohort (hospital-based)	236 pregnancies and 238 newborns	1979 and 1986	NR	Late fetal death with a birth weight≥1000 g, and early neonatal death<less 7 days	4
Sass, 1990[70]	Brazil	Case control (hospital-based)	337 Births	1985 to 1986	DBP ≥90mmHg before pregnancy or up to 20 weeks of pregnancy or hypertension at 10w postpartum (for those whose antenatal care was NOT at that hospital).	NR	4
Rey, 1994[26]	Canada	Cohort (hospital-based)	20,375 mothers	1987 to 1991	BP>140/90 mm Hg with a diagnosed before pregnancy or hypertension in at least two measurements<20 weeks of pregnancy and/or 6 weeks after delivery	NR	6
McCowan, 1996[49]	New Zealand	Cohort (hospital-based)	20,224 mother-infant pairs	1991 to 1993	DBP>90 mmHg before 20 weeks gestation, pre-existing history of essential hypertension and/or on antihypertensive medication before the pregnancy	Fetal deaths after 20 weeks, early and late neonatal deaths per 1000 total births	6
Jain, 1997[50]	USA	Cohort (hospital-based)	109,428 mother-infant pairs	1982 to 1987	SBP≥ 140 mmHg or a DBP≥90 mmHg before pregnancy or before 20 weeks' gestation.	NR	7

Hartikainen, 1998[51]	Finland	Cohort (hospital-based)	8,050 mother-infant pairs	1985 to 1986	Hypertension diagnosed before pregnancy and/or DBP > 90 mm Hg and/or antihypertensive medication, each<20 weeks' gestation	Fetal death with a birth weight≥ 500 g, and/or gestational age≥ 24 completed weeks, and neonatal death<less 7 days	6
Vanek, 2004[73]	Israel	Cohort (hospital-based)	114,963 mother-infant pairs	1988 to 1999	BP≥140/90 mmHg that measured at least twice, before 20 weeks' gestation, or persisting for ≥12 weeks post-delivery	NR	6
Ahmad, 2012[91]	Norway	Cohort (population-based)	2,027,042 mother-infant pairs	1967 to 2006	SBP≥ 140 mmHg or a DBP≥90 mmHg before 20 weeks' gestation using (ICD-8 and ICD-10 codes)	Fetal death with a birth weight≥ 500 g, and/or gestational age≥22 weeks (154 days) and neonatal death<less 7 days	7
Ye, 2014[62]	China	Cohort (population-based	108,550 Pregnancies from 106, 869 women	2011 to 2011	BP≥140/90 mmHg before pregnancy or diagnosed before 20 weeks' gestation or developed hypertension after 20 weeks of gestation and continued for 12 weeks of postpartum	Fetuses and neonates who were born dead, or died in the first 28 days after delivery	7
Broekhuijsen, 2015[42]	Netherlands	Cohort (population-based)	988,389 mother-infant pairs	2002 to 2007	BP ≥140/90 mmHg diagnosed before pregnancy, or before 20 weeks' gestation	Fetal death or neonatal death within 7 days after birth	8
Youngstrom, 2018[45‡]	USA	Cohort (hospital-based)	1,306 mother-infant pairs	2000 to 2014	History of hypertension or the use of antihypertensive medication before pregnancy	NR	8

‡ The ORs from this study was combined for treated and untreated women with chronic hypertension.

Table S13: Characteristics of studies for neonatal intensive care unit (NICU) admission outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle-Ottawa grade*
Hjertberg, 1992[92]	Sweden	Cohort (hospital-based)	2593	1986 to 1987	Chronic hypertension before pregnancy and/or before 24 weeks' gestation	Admission to NICU	***6***
McCowan, 1996[49]	New Zealand	Cohort (hospital-based)	20,224 mother-infant pairs	1991 to 1993	DBP>90 mmHg before 20 weeks gestation, pre-existing history of essential hypertension and/or on antihypertensive medication before the pregnancy	Admission to NICU	6
Hartikainen, 1998[51]	Finland	Cohort (hospital-based)	8,050 mother-infant pairs	1985 to 1986	Hypertension diagnosed before pregnancy and/or DBP > 90 mm Hg and/or antihypertensive medication, each<20 weeks' gestation	Admission to NICU	6
Roberts, 2005[55]	Australia	Cohort (population-based)	227,067 women and 231,811 infants	2000 to 2002	Chronic hypertension diagnosed using ICD-10 codes	Admission to NICU	***7***
Tuuli, 2011[59]	USA	Cohort (hospital-based)	58,135 mother-infant pairs	1990 to 2008	SBP≥ 140 mmHg or a DBP≥90 mmHg before pregnancy or before 20 weeks' gestation	Admission to NICU	***8***
Cruz, 2011[78]	USA	Cohort (Multicentre hospital- based)	17,752	2002 and 2008	Hypertension reported before pregnancy, or SBP≥140 mm Hg or DBP ≥90 mm Hg before 20 weeks' gestation	Admission to NICU	***7***
Madi, 2012[80]	Brazil	Cohort (hospital-based)	3,689 mother-infant pairs	1998 to 2009.	BP≥140/90 mmHg that measured at least twice, before 20 weeks' gestation, or persisting for ≥12 weeks post-delivery	Admission to NICU	***7***

Table S14: Characteristics of studies for miscarriage outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle -Ottawa grade*
Sass, 1990[70]	Brazil	Case control (hospital-based)	337 Births	1985 to 1986	DBP ≥90mmHg before pregnancy or up to 20 weeks of pregnancy or hypertension at 10w postpartum (for those whose antenatal care was NOT at that hospital).	NR	4
Panaitescu, 2017[44]	UK	Cohort (hospital-based)	109,932 pregnancies	March 2006 and July 2015/ February 2007 and November 2015	History of chronic hypertension	Late miscarriage included spontaneous delivery or fetal death at 16 + 0 to 23 + 6 weeks' gestation	8

Table S15: Characteristics of studies for stillbirth stratified by maternal ethnicity

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle -Ottawa grade*
Ananth, 1995a[93]	USA	Cohort (population-based)	Black:14,417 White:15,819	1988 and 1990	Chronic hypertension obtained from medical records; no further details provided	Fetal death at≥20 weeks' gestation	8
Canterino, 2004[94]	USA	Cohort (population-based)	Black:3,269,211 White:18,341,662	1995 to 2000	Hypertension before pregnancy or before 20 weeks' gestation	Fetal death at≥24 weeks' gestation	7

Table S16: Characteristics of studies for PTB stratified by maternal ethnicity

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle -Ottawa grade*
Ananth, 1995b[9]	USA	Cohort (population-based)	Black:14,417 White:15,819	1988 and 1990	Chronic hypertension obtained from medical records; no further details provided	<37 weeks' gestation	8
Rey, 1997[95]	Canada	Cohort (hospital-based)	Black:2,450 White:17,854	1987 to 1991	BP>140/90 mm Hg with a diagnosed before pregnancy or hypertension in at least two measurements<20 weeks of pregnancy and/or 6 weeks after delivery	Prematurity/NR	6
Samadi, 1998[53]	USA	Case control (hospital-based)	Black:25,060 White: none	1988 and 1993	Chronic hypertension diagnosed using ICD-9 codes	<37 weeks' gestation	7
Graham, 2007[56]	USA	Cohort (population-based)	Black:91,718 White:111,213	1999 to 2003	Hypertension diagnosed before pregnancy	<37 weeks' gestation	7
Premkumar, 2016[67]	USA	Cohort (hospital-based)	Black:2,272 White:10,843	2002 to 2015	SBP> 140 mm Hg or DBP> 90 mm Hg recorded on at least 2 separate occasions >6 hours apart before 20 weeks' gestation	<37 weeks' gestation	9

Table S17: Characteristics of studies for SGA stratified by maternal ethnicity

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle -Ottawa grade*
Rey, 1997[95]	Canada	Cohort (hospital-based)	Black:2,450 White:17,854	1987 to 1991	BP>140/90 mm Hg with a diagnosed before pregnancy or hypertension in at least two measurements<20 weeks of pregnancy and/or 6 weeks after delivery	Prematurity/NR	6
Ananth, 1995b[9]	USA	Cohort (population-based)	Black:14,417 White:15,819	1988 and 1990	Chronic hypertension obtained from medical records; no further details provided	<37 weeks' gestation	8

Table S18: Characteristics of studies for LBW stratified by maternal ethnicity

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of chronic hypertension*	*Definition of outcome*	*Newcastle -Ottawa grade*
Velentgas, 1994[84]	USA	Cohort (population-based)	Black:285 White:3,144	1984-1990	NR	birth weight of <2500 grams	6
Ananth, 1995b[9]	USA	Cohort (population-based)	Black:14,417 White:15,819	1988 and 1990	Chronic hypertension obtained from medical records, no further details provided	birth weight of <2500 grams‡	8
Lydakis, 1998[85]	UK	Cohort (hospital-based)	Black:79 White:64	1980 to 1997	Hypertension (DBP ≥110 mmHg OR DBP >90 mmHg on two or more occasions ≥4 h apart) at the first booking visit before the 20 weeks' gestation in the absence of trophoblastic disease, OR hypertension at any stage of pregnancy that continued for more than 6 weeks after delivery	birth weight of <2000 grams	6
Graham, 2007[56]	USA	Cohort (population-based)	Black:91,718 White:111,213	1999 to 2003	Hypertension diagnosed before pregnancy	birth weight of <2500 grams	7
Odell, 2006[86]	USA	Population-based (Black women only)	Black:12,258 White: None	1996 to 2000	History of chronic hypertension diagnosed using ICD 9 codes	birth weight of <2500 grams	7

‡ We combined the ORs for birthweight < 1,499 g and birthweight (1,500-2,499 g).

Table S19: Characteristics of studies for pre-eclampsia outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle-Ottawa grade*
Mabie, 1986[112]†	USA	Cohort (hospital-based)	169 pregnancies in 156 women	1980 to 1984	Women with CH who were on antihypertensive treatment	Untreated women with CH	Superimposed preeclampsia: worsening hypertension (30 mmHg SBP or 15 mmHg DBP) plus either nondependent edema or proteinuria of +1 or greater by dipstick.	6
Bayliss, 2002[109]‡	UK	Cohort (hospital-based)	491 pregnancies	1980 to 1999	Women with CH who were on antihypertensive treatment	Untreated women with CH	The onset of significant levels of proteinuria after 20 weeks gestation	6
Orbach, 2013[35]	Israel	Cohort (population- based)	98,253 births	1998 to 2008	Women with CH who were on antihypertensive treatment (methyldopa or atenolol) dispensed during the 1st trimester	Untreated normotensive women	Proteinuria diagnosed using ICD-9 codes	8
Rezk, 2016[110]	Egypt	Cohort (hospital-based)	222 women	2012 to 2016	Women with CH who were taking Methyldopa	Untreated women with CH	Superimposed preeclampsia: a new onset proteinuria with 0.3 g of protein or more in a 24-h urine specimen after 20 weeks' gestation), eclampsia (generalized convulsions)	4
Nzelu, 2018[111]	UK	Cohort (hospital-based)	419	2011 to 2016	women with CH on antihypertensive treatment	Untreated women with CH	Superimposed pre-eclampsia: hypertension, with at least 1 of the following problems: renal involvement (proteinuria 300 mg/24 h and/or creatinine 90 mmol/L or 1 mg/dL), liver impairment (transaminases >70 IU/L), neurologic complications (eg, eclampsia), thrombocytopenia (platelet count <150,000/mL).	7

Youngstrom, 2018[45]	USA	Cohort (hospital-based)	1,094 mother-infant pairs	2000 to 2014	women with CH on antihypertensive treatment	Untreated normotensive women	BP ≥140/90 with either proteinuria (protein excretion ≥300 mg in 24 hours or protein-to-creatinine ratio≥0.3), thrombocytopenia (less than 100,000/mL), transaminases (aspartate aminotransferase > twice the upper limit of normal), or elevated creatinine≥1.2 mg/dL	8
Mito, 2019[114§]	Japan	Cohort (population-based)	231 women	2008 to 2016	women with CH on antihypertensive treatment at the first trimester	Untreated women with CH	Superimposed pre-eclampsia diagnosed if a woman developed new-onset proteinuria in the with a rise in blood pressure or a sudden increase in pre-existing proteinuria.	7

CH: Chronic hypertension; BP: blood pressure. †The raw data for this study include women on methyldopa, other monotherapy, or multiple drug treatment vs. untreated women. ‡The odds ratios of this study were combined for different agents. §The raw data for this study include women on amlodipine or other antihypertensive drug treatment vs. untreated women

Table S20: Characteristics of studies for cesarean section (CS) outcome

Author, year published	*Country (ies)*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle-Ottawa grade*
Rezk, 2016[110]	Egypt	Cohort (hospital-based)	222 women	2012 to 2016	Women with CH who were taking Methyldopa	Untreated women with CH	Caesarean section (not specified)	4
Hoeltzenbei, 2017[107‡]	Germany	Cohort	787 pregnancies	2000 to 2014	pregnancies with methyldopa exposure at least during the first trimester, but no longer than 20 weeks' gestation	Untreated normotensive women	Caesarean section (not specified)	5

CH: Chronic hypertension

Table S21: Characteristics of studies for miscarriage outcome

Author, year published	*Country (ies)*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle -Ottawa grade*
Bayliss, 2002[109]	UK	Cohort (hospital-based)	491 pregnancies	1980 to 1999	Women with CH who were on antihypertensive treatment	Untreated women with CH	miscarriages	6
Hoeltzenbei, 2017[107]	Germany	Cohort	787 pregnancies	2000 to 2014	pregnancies with methyldopa exposure at least during the first trimester, but no longer than 20 weeks' gestation	Untreated normotensive women	Spontaneous pregnancy loss of a fetus <500 g or in case of unknown weight <24 completed weeks after last menstrual period	5

CH: Chronic hypertension

Table S22: Characteristics of studies for stillbirth outcome

Author, year published	*Country (ies)*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle -Ottawa grade*
Bayliss, 2002[109]	UK	Cohort (hospital-based)	491 pregnancies	1980 to 1999	Women with CH who were on antihypertensive treatment	Untreated women with CH	Stillbirth	6
Hoeltzenbei, 2017[107]	Germany	Cohort	787 pregnancies	2000 to 2014	pregnancies with methyldopa exposure at least during the first trimester, but no longer than 20 weeks' gestation	Untreated normotensive women	Stillbirth	5
Rezk, 2016[110]	Egypt	Cohort (hospital-based)	222 women	2012 to 2016	Women with CH who were taking Methyldopa	Untreated women with CH	Intrauterine fetal death	4
Youngstrom, 2018[45]	USA	Cohort (hospital-based)	1,094 mother-infant pairs	2000 to 2014	women with CH on antihypertensive treatment	Untreated normotensive women	Stillbirth	8

CH: Chronic hypertension

Table S23a: Characteristics of studies for Preterm birth outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle -Ottawa grade*
Ray, 2001[108]†	Canada	Cohort (hospital-based)	583 mother-infant pairs	1986 to 1995	women with CH on antihypertensive treatment	Untreated women with CH	<37 weeks' gestation	6
Lennestål, 2009[105]	Sweden	Cohort (population-based)	1,032,094 pregnancies	1995 to 2006	Women who reported using antihypertensive treatment in early pregnancy, and with a delivery diagnosis of CH for women who were on beta-blocking drug	Untreated normotensive women	<37 weeks' gestation	8
Banhidy, 2010[106]	Hungary	Case control (population based)	36,155 births	1980 to 1996	Women with CH on antihypertensive treatment	Untreated normotensive women	<37 weeks' gestation	6
Su, 2013[60]§	Taiwan	Cohort (population-based)	2,727 mother-infant pairs	2005	women with CH on antihypertensive treatment	Untreated women with CH	<37 weeks' gestation	8
Orbach, 2013[35]	Israel	Cohort (population-based)	98,253 births	1998 to 2008	Women with CH who were on antihypertensive treatment (methyldopa or atenolol) dispensed during the 1st trimester	Untreated normotensive women	<37 weeks' gestation	8
Rezk, 2016[110]	Egypt	Cohort (hospital-based)	222 women	2012 to 2016	Women with CH who were taking methyldopa	Untreated women with CH	<37 weeks' gestation	4
Hoeltzenbei, 2017[107]	Germany	Cohort	787 pregnancies	2000 to 2014	pregnancies with methyldopa exposure at least during the first trimester, but no longer than 20 weeks' gestation	Untreated normotensive women	<37 weeks' gestation	5

Youngstrom, 2018[45]	USA	Cohort (hospital-based)	1,094 mother-infant pairs	2000 to 2014	women with CH on antihypertensive treatment	Untreated normotensive women	<37 weeks' gestation	8
Mito, 2019[114]‡	Japan	Cohort (population-based)	231 women	2008 to 2016	women with CH on antihypertensive treatment at the first trimester	Untreated women with CH	<37 weeks' gestation	7

CH: Chronic hypertension.

†We combined odds ratios from this study for women using beta-blockers or other agents.

§The odds ratios of these studies were combined for different agents.

‡The raw data for this study include women on amlodipine or other antihypertensive drug treatment vs. untreated women

Table S23b: Characteristics for preterm birth outcome: (exposed to beta-blockers vs. untreated)

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle -Ottawa grade*
Ray, 2001[108]	Canada	Cohort (hospital-based)	391mother-infant pairs	1986 to 1995	women with CH on β-blocker treatment	Untreated women with CH	<37 weeks' gestation	6
Su, 2013[60]	Taiwan	Cohort (population-based)	1,420 mother-infant pairs	2005	Pregnant women with CH who were exposed to β-blockers	Untreated women with CH	<37 weeks' gestation	8
Orbach, 2013[35]	Israel	Cohort (population-based)	97,927 births	1998 to 2008	Pregnant women with CH who were exposed to atenolol	Untreated normotensive women	<37 weeks' gestation	8

Table S23c: Characteristics for Preterm birth outcome: (exposed to RAS vs. other agents)

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle -Ottawa grade*
Ahmed, 2018[116]	Australia	Cohort (population-based)	456 pregnancies	2005 and 2012	Pregnancies with CH on ARBs exposure at first trimester	Pregnant women on methyldopa	<37 weeks' gestation	7

CH: Chronic hypertension; ARB: angiotensin-receptor blockers

Table S24a: Characteristics of studies for small for gestational age (SGA) outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle -Ottawa grade*
Mabie, 1986[112]†	USA	Cohort (hospital-based)	169 pregnancies in 156 women	1980 to 1984	Women with CH who were on antihypertensive treatment	Untreated women with CH	Birthweight <10th percentile	6
Ray, 2001[108]*	Canada	Cohort (hospital-based)	583 mother-infant pairs	1986 to 1995	women with CH on antihypertensive treatment	Untreated women with CH	Birthweight <10th percentile	6
Bayliss, 2002[109]*	UK	Cohort (hospital-based)	491 pregnancies	1980 to 1999	Women with CH who were on antihypertensive treatment	Untreated women with CH	Birthweight <10th percentile	6
Lennestål, 2009[105]	Sweden	Cohort (population-based)	1,032,094 pregnancies	1995 to 2006	Women who reported using antihypertensive treatment in early pregnancy, and with a delivery diagnosis of chronic hypertension for women who were on beta-blocking drug	Untreated normotensive women	Birthweight < -2SD	8
Su, 2013[60]*	Taiwan	Cohort (population-based)	2,727 mother-infant pairs	2005	women with CH on antihypertensive treatment	Untreated women with CH	Birthweight <10th percentile	8
Orbach, 2013[35]	Israel	Cohort (population-based)	98,253 births	1998 to 2008	Women with CH who were on antihypertensive treatment (methyldopa or atenolol) dispensed during the 1st trimester	Untreated normotensive women	SGA diagnosed using ICD-9	8
Rezk, 2016[110]	Egypt	Cohort (hospital-based)	222 women	2012 to 2016	Women with CH who were taking Methyldopa	Untreated women with CH	Birthweight <10th percentile	4
Nzelu, 2018[111]	UK	Cohort (hospital-based)	419	2011 to 2016	women with CH on antihypertensive treatment	Untreated women with CH	Birthweight <5th percentile	7

Fisher, 2018[117]	USA	Cohort (population-based)	4,282 mother-infant pairs	2006 to 2011	Women with CH on antihypertensive treatment	Untreated normotensive women	Birthweight <10th percentile	6
Youngstrom, 2018[45]	USA	Cohort (hospital-based)	1,094 mother-infant pairs	2000 to 2014	women with CH on antihypertensive treatment	Untreated normotensive women	Birthweight <10th percentile	8

CH: Chronic hypertension . * We combined odds ratios from these studies were combined for different agents. †The raw data for this study include women on methyldopa, other monotherapy, or multiple drug treatment vs. untreated women.

Table S24b: Characteristics of studies for SGA outcome: (beta blockers vs. untreated women)

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle -Ottawa grade*
Ray, 2001[108]	Canada	Cohort (hospital-based)	391 mother-infant pairs	1986 to 1995	Pregnant women with CH who were exposed to β-blockers	Untreated women with CH	Birth weight<10th percentile	6
Bayliss, 2002[109]	UK	Cohort (hospital-based)	229 pregnancies	1980 to 1999	Pregnant women with CH who were exposed to atenolol<15 weeks' gestation	Untreated women with CH	Birth weight<10th percentile	6
Su, 2013[60]	Taiwan	Cohort (population-based)	1,420 mother-infant pairs	2005	Pregnant women with CH who were exposed to β-blockers	Untreated women with CH	Birth weight<10th percentile	8
Orbach, 2013[35]	Israel	Cohort (population-based)	97,927 births	1998 to 2008	Pregnant women with CH who were exposed to atenolol	Untreated normotensive women	SGA diagnosed using ICD-9	8

CH: Chronic hypertension; β-blockers: beta blockers

Table S24c: Characteristics of studies for SGA outcome: (single agent/multiple agents vs. untreated women)

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle -Ottawa grade*
Ray, 2001[108]	Canada	Cohort (hospital-based)	500 mother-infant pairs (untreated. n=247, sing. n=144 multi. n=109)	1986 to 1995	women with CH who were on single agent (βB), or on multiple agents	Untreated women with CH	Birthweight<10th percentile	6
Bayliss, 2002[109]	UK	Cohort (hospital-based)	268 pregnancies (sing. n=229, multi. n=243)	1980 to 1999	Women with CH who were on single agent (atenolol), or on multiple agents	Untreated women with CH	Birthweight<10th percentile	6
Su, 2013[60]	Taiwan	Cohort (population-based)	1,543 mother-infant pairs (untreated n=1,006, sing. n=144, multi. n=123)	2005	women with CH who were on single agent (βB), or on multiple agents	Untreated women with CH	Birthweight <10th percentile	8

CH: Chronic hypertension; βB: beta-blockers

Table S24d: Characteristics of studies for SGA outcome: (centrally acting antiadrenergic vs. untreated women)

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle -Ottawa grade*
Mabie, 1986[112]	USA	Cohort (hospital-based)	136 pregnancies	1980 to 1984	Pregnant women who were exposed to methyldopa	Untreated women with CH	Birthweight<10th percentile	6
Su, 2013[60]	Taiwan	Cohort (population-based)	1,187 mother-infant pairs	2005	Pregnant women who were exposed to centrally acting antiadrenergic	Untreated women with CH	Birthweight<10th percentile	8
Orbach, 2013[35]	Israel	Cohort (population-based)	98,160 births	1998 to 2008	Pregnant women who were exposed to methyldopa	Untreated normotensive women	SGA diagnosed using ICD-9	8
Rezk, 2016[110]	Egypt	Cohort (hospital-based)	222 women	2012 to 2016	Pregnant women who were exposed to methyldopa	Untreated women with CH	Birthweight<10th percentile	4

CH: Chronic hypertension.

Table S24e: Characteristics of studies for SGA outcome: (methyldopa vs other agents)

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle -Ottawa grade*
Mabie, 1986[112]	USA	Cohort (hospital-based)	64 pregnancies	1980 to 1984	Pregnant women with CH who were exposed to methyldopa	Pregnant women with CH on hydrochlorothiazide	Birthweight<10th percentile	6
Su, 2013[60]	Taiwan	Cohort (population-based)	595 mother-infant pairs	2005	Pregnant women who were exposed to centrally acting antiadrenergic	Pregnant women with CH on β-blockers	Birthweight<10th percentile	8
Orbach, 2013[35]	Israel	Cohort (population-based)	447 births	1998 to 2008	Pregnant women with CH who were exposed to methyldopa	Pregnant women with CH who were exposed to atenolol	SGA diagnosed using ICD-9	8
Xie,2014[115]	Canada	Cohort (population-based)	628 Women with CH	1980 to 1987 or 1990 to 2005	Pregnant women with CH who were exposed to methyldopa	Pregnant women with CH on β-blockers	Birthweight<10th percentile	7
Ahmed, 2018[116]	Australia	Cohort (population-based)	456 pregnancies	2005 and 2012	Pregnant women with CH who were exposed to methyldopa	Pregnancies with CH on ARBs exposure at first trimester	Birthweight<10th percentile	7

CH: Chronic hypertension; ARB: angiotensin-receptor blockers

Table S24f: Characteristics of studies for SGA outcome: (beta blockers vs. methyldopa)

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle -Ottawa grade*
Su, 2013[60]	Taiwan	Cohort (population-based)	595 mother-infant pairs	2005	Pregnant women with CH on β-blockers	Pregnant women who were exposed to centrally acting antiadrenergic	Birthweight<10th percentile	8
Orbach, 2013[35]	Israel	Cohort (population-based)	447 births	1998 to 2008	Pregnant women with CH who were exposed to atenolol	Pregnant women with CH who were exposed to methyldopa	SGA diagnosed using ICD-9	8
Xie,2014[115]	Canada	Cohort (population-based)	628 Women with CH	1980 to 1987 or 1990 to 2005	Pregnant women with CH on β-blockers	Pregnant women with CH who were exposed to methyldopa	Birthweight<10th percentile	7

CH: Chronic hypertension

Table S25: Characteristics of studies for low birth weight (LBW) outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle-Ottawa grade*
Lennestål, 2009[105]	Sweden	Cohort (population-based)	1,032,094 pregnancies	1995 to 2006	Women who reported using antihypertensive treatment in early pregnancy, and with a delivery diagnosis of chronic hypertension for women who were on beta-blocking drug	Untreated normotensive women	Birthweight < 2,500g	8
Banhidy, 2010[106]	Hungary	Case control (population based)	36,155 births	1980 to 1996	Women with CH on antihypertensive treatment	Untreated normotensive women	Birthweight < 2,500g	6
Su, 2013[60]§	Taiwan	Cohort (population-based)	2,727 mother-infant pairs	2005	women with CH on antihypertensive treatment	Untreated women with CH	Birthweight < 2,500g	8
Orbach, 2013[35]	Israel	Cohort (population-based)	98,253 births	1998 to 2008	Women with CH who were on antihypertensive treatment (methyldopa or atenolol) dispensed during the 1st trimester	Untreated normotensive women	Birthweight < 2,500g	8
Mito, 2019[114]†	Japan	Cohort (population-based)	231 women	2008 to 2016	women with CH on antihypertensive treatment at the first trimester	Untreated women with CH	Birthweight < 2,500g	7

CH: Chronic hypertension.

§The ORs of these studies were combined for different agents.†The raw data for this study include women on amlodipine or other antihypertensive drug treatment vs. untreated women.

Table S26: Characteristics of studies for neonatal intensive care unit (NICU) admission outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle -Ottawa grade*
Rezk, 2016[110]	Egypt	Cohort (hospital-based)	222 women	2012 to 2016	Women with CH who were taking Methyldopa	Untreated women with CH	Admission to NICU	4
Helou, 2017[113]	Australia	Cohort (hospital-based)	49 women	2010	Women with CH on antihypertensive treatment	Untreated women with CH	Admission to NICU or SCN	6

CH: Chronic hypertension; SCN: Special care nursery.

Table S27: Characteristics of studies for perinatal death outcome

Author, year published	*Country*	*Design*	*Sample size*	*Study duration*	*Definition of exposure*	*Comparative group*	*Definition of outcome*	*Newcastle -Ottawa grade*
Mabie, 1986[112]†	USA	Cohort (hospital-based)	169 pregnancies in 156 women	1980 to 1984	Women with CH who were on antihypertensive treatment	Untreated women with CH	Not reported	6
Ray, 2001[108]*	Canada	Cohort (hospital-based)	1,948 mother-infant pairs	1986 to 1995	women with CH or other HDP on antihypertensive treatment	Untreated women with CH or HDP	Perinatal death after 20 weeks gestation and up to 30 days after birth	6
Orbach, 2013[35]	Israel	Cohort (population- based)	98,253 births	1998 to 2008	Women with CH who were on antihypertensive treatment (methyldopa or atenolol) dispensed during the first trimester	Untreated normotensive women	Not reported	8
Youngstrom, 2018[45]	USA	Cohort (hospital-based)	1,094 mother-infant pairs	2000 to 2014	women with CH on antihypertensive treatment	Untreated normotensive women	Stillbirth and neonatal death	8

CH: chronic hypertension.

†The raw data for this study include women on methyldopa, other monotherapy, or multiple drug treatment vs. untreated women.

*We combined odds ratios from this study for women using beta-blockers or other agents

Figure S1a: Forest plot of crude estimates of the association between chronic hypertension and PE

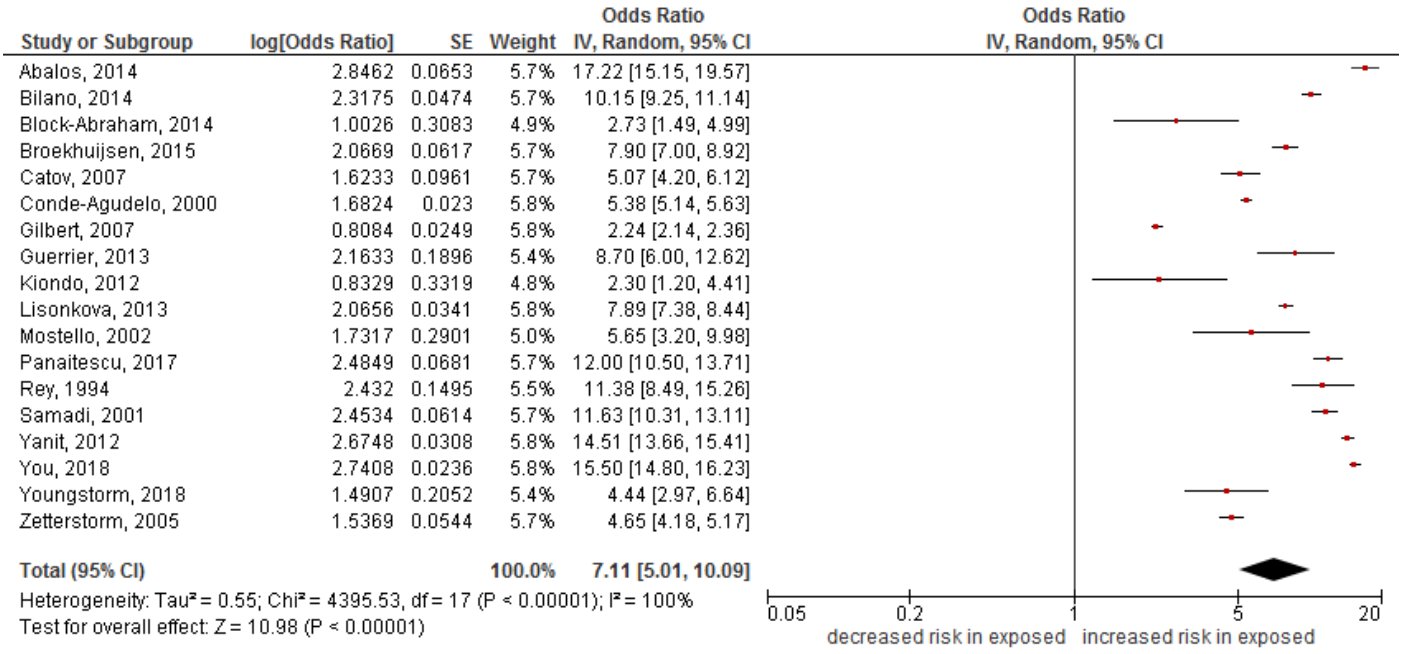


Figure S1b: Funnel plot for adjusted estimates of PE (chronic hypertension versus normotensive)

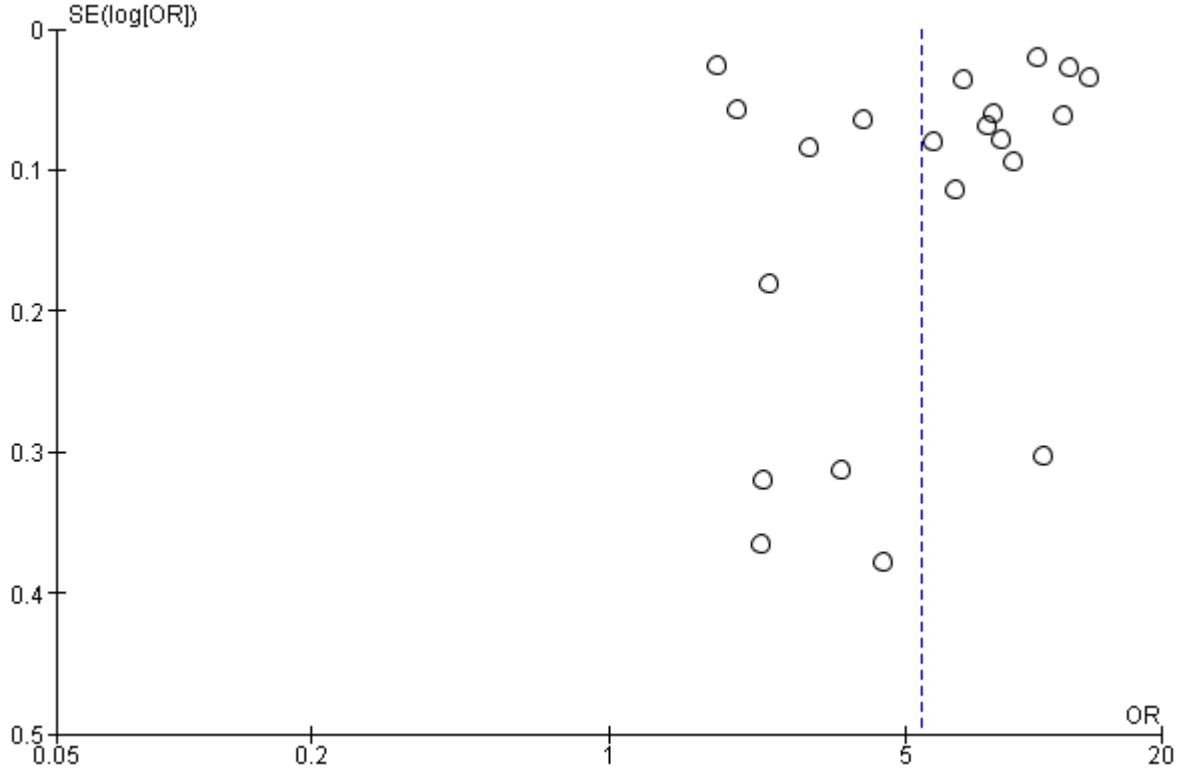


Figure S2a: Forest plot of crude estimates of the association between chronic hypertension and HEELP syndrome

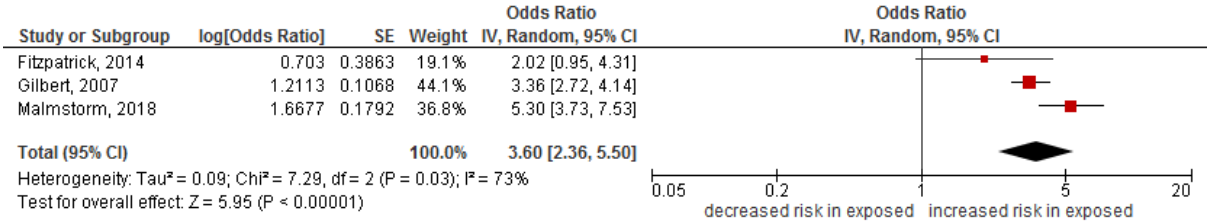


Figure S2b: Forest plot of adjusted estimates of the association between chronic hypertension and HEELP syndrome

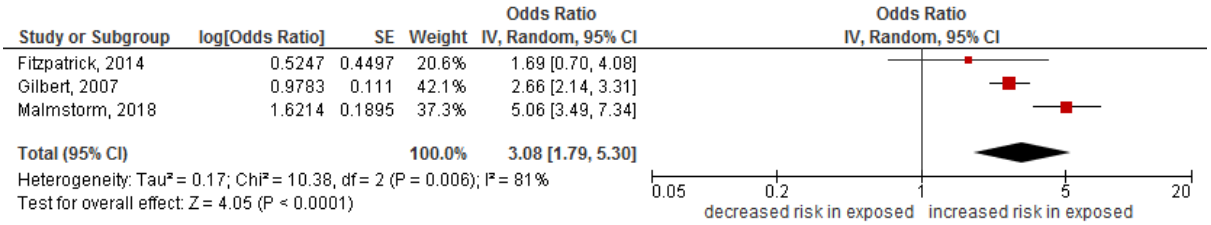


Figure S3a: Forest plot of crude estimates of the association between chronic hypertension and CS

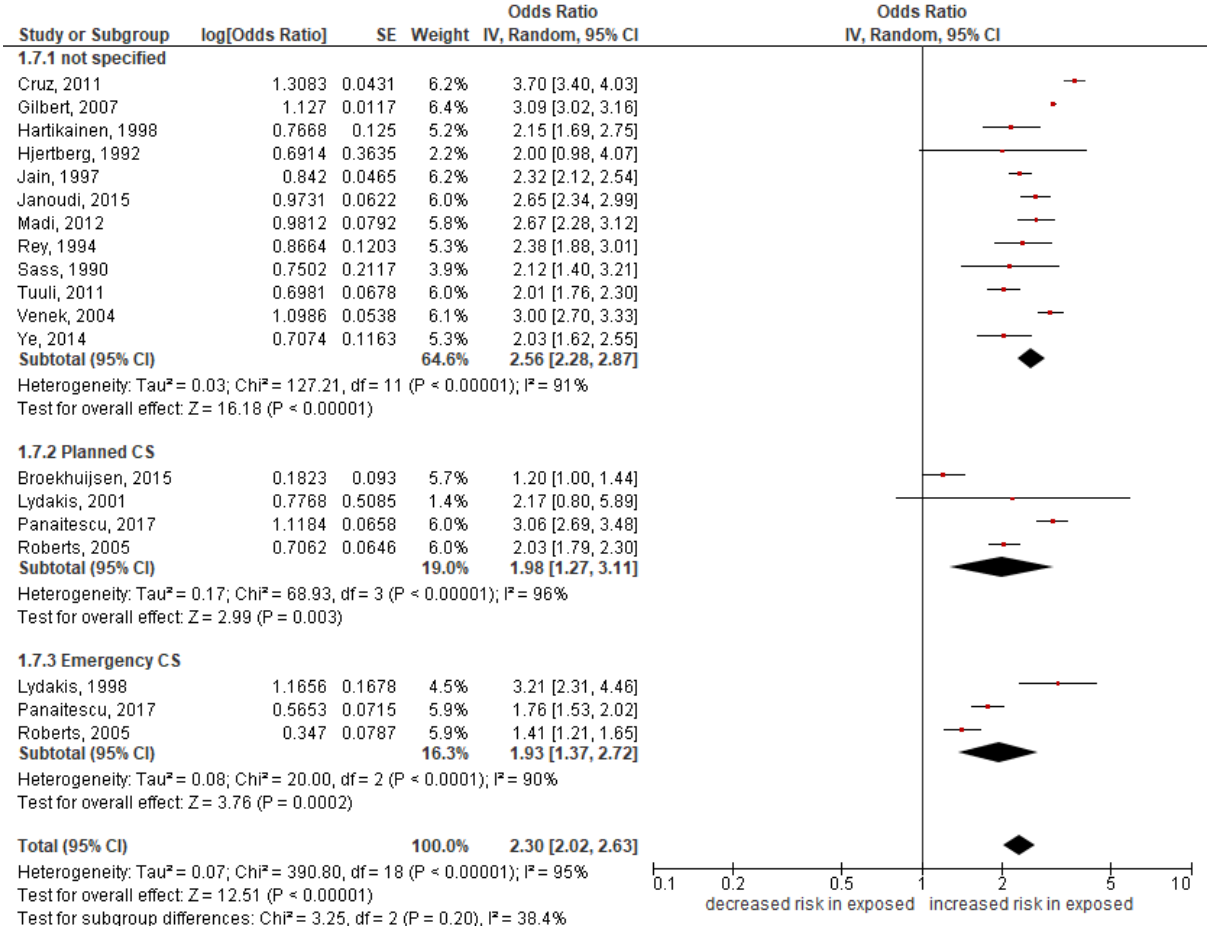


Figure S3b: Forest plot of adjusted estimates of the association between chronic hypertension and CS

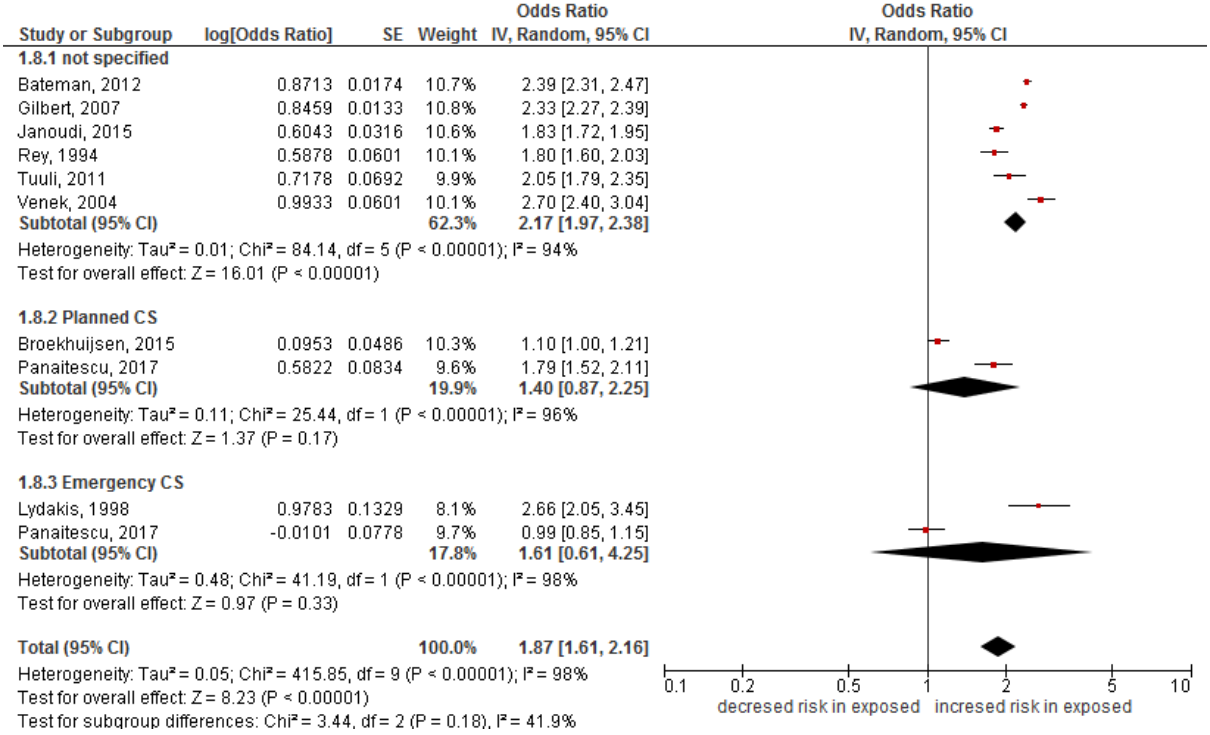


Figure S3c: Funnel plot for adjusted estimates of CS (chronic hypertension versus normotensive)

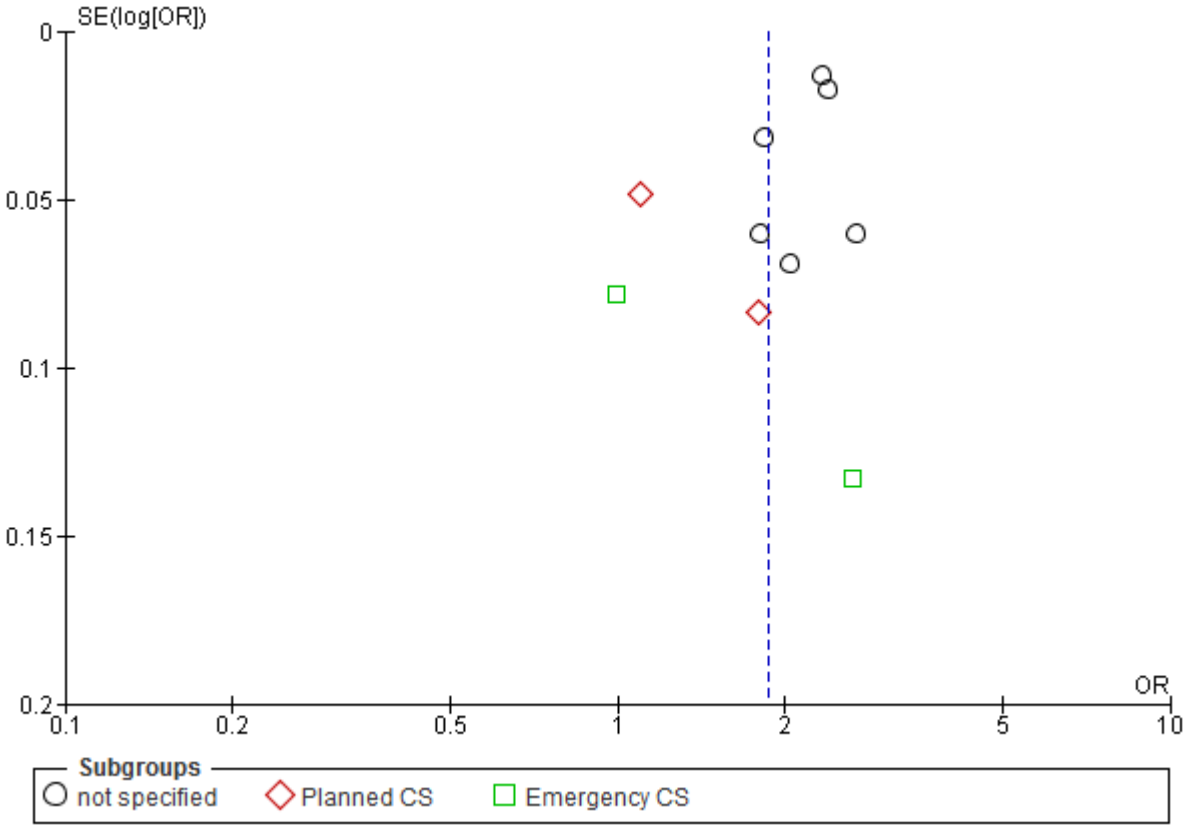


Figure S4a: Forest plot of crude estimates of the association between chronic hypertension and PPH

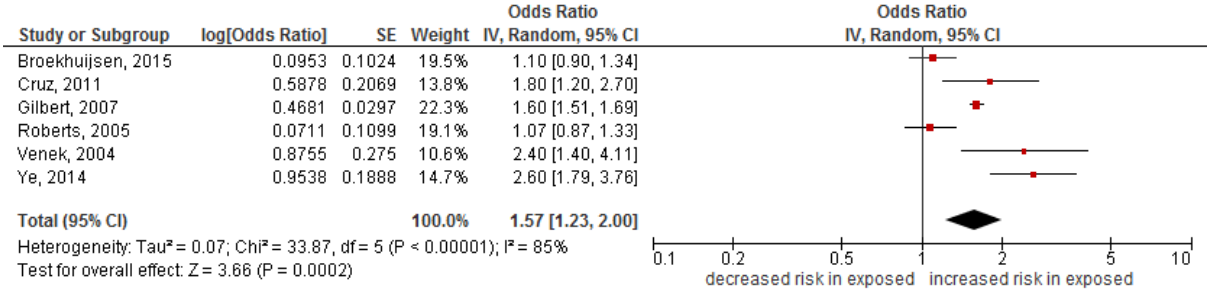


Figure S4b: Forest plot of adjusted estimates of the association between chronic hypertension and PPH

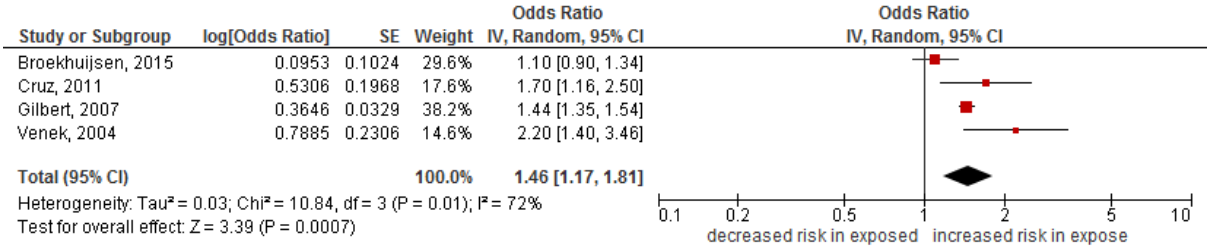


Figure S5: Forest plot of crude estimates of the association between chronic hypertension and maternal mortality

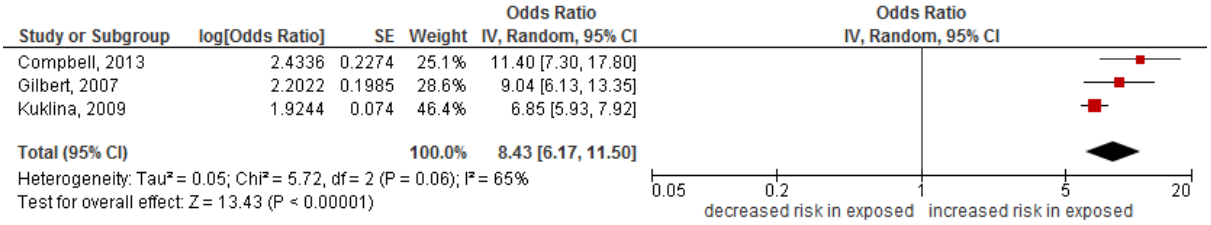


Figure S6a: Forest plot of crude estimates of the association between chronic hypertension and stillbirth

Study or Subgroup	log[Odds Ratio]	SE	Weight	Odds Ratio IV, Random, 95% CI
Aagord-Tittery, 2006	1.2442	0.3551	2.0%	3.47 [1.73, 6.96]
Ahmed, 2012	0.8074	0.111	7.6%	2.24 [1.80, 2.79]
Allen, 2004	1.1632	0.266	3.1%	3.20 [1.90, 5.39]
Ananth, 1995	1.1741	0.12	7.2%	3.24 [2.56, 4.09]
Canterino, 2004	1.2981	0.031	10.6%	3.66 [3.45, 3.89]
Conde-Agudelo, 2000b	0.9859	0.0448	10.2%	2.68 [2.45, 2.93]
Gilbert, 2007	1.1878	0.0354	10.5%	3.28 [3.06, 3.52]
Lydakis, 1998	1.5602	0.3645	1.9%	4.76 [2.33, 9.72]
Madi, 2012	1.8358	0.3198	2.4%	6.27 [3.35, 11.74]
Panaitescu, 2017	1.3987	0.2203	4.0%	4.05 [2.63, 6.24]
Reddy, 2010	0.9858	0.1519	6.0%	2.68 [1.99, 3.61]
Roberts, 2005	0.8116	0.2325	3.7%	2.25 [1.43, 3.55]
Tuuli, 2011	0.8796	0.2154	4.1%	2.41 [1.58, 3.68]
Xiong, 2018	0.9026	0.0565	9.8%	2.47 [2.21, 2.75]
Yanit, 2012	0.9977	0.1518	6.0%	2.71 [2.01, 3.65]
Yerlikaya, 2016	1.5173	0.2205	4.0%	4.56 [2.96, 7.02]
Youngstorm, 2018	0.7575	0.4642	1.3%	2.13 [0.86, 5.30]
Zetterstorm, 2006	0.9959	0.1641	5.6%	2.71 [1.96, 3.73]
Total (95% CI)			**100.0%**	**3.00 [2.69, 3.35]**

Heterogeneity: Tau² = 0.03; Chi² = 83.53, df = 17 (P < 0.00001); I² = 80%
Test for overall effect: Z = 19.82 (P < 0.00001)

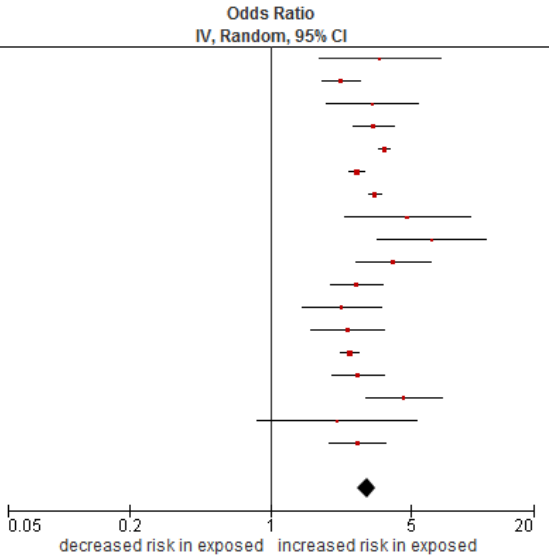


Figure S6b: Forest plot of adjusted estimates of the association between chronic hypertension and stillbirth

Study or Subgroup	log[Odds Ratio]	SE	Weight	Odds Ratio IV, Random, 95% CI
Aagord-Tittery, 2006	1.2413	0.5665	0.2%	3.46 [1.14, 10.50]
Ahmed, 2012	0.7514	0.1097	4.0%	2.12 [1.71, 2.63]
Allen, 2004	1.1632	0.266	0.7%	3.20 [1.90, 5.39]
Ananth, 1995	1.0367	0.1221	3.2%	2.82 [2.22, 3.58]
Bateman, 2012	0.802	0.0505	18.9%	2.23 [2.02, 2.46]
Canterino, 2004	0.8587	0.0475	21.4%	2.36 [2.15, 2.59]
Conde-Agudelo, 2000b	0.8109	0.0678	10.5%	2.25 [1.97, 2.57]
Copper, 1994	0.5306	0.3189	0.5%	1.70 [0.91, 3.18]
Gilbert, 2007	0.8587	0.0428	26.4%	2.36 [2.17, 2.57]
Jain, 1997	1.0986	0.1582	1.9%	3.00 [2.20, 4.09]
Panaitescu, 2017	0.8671	0.2321	0.9%	2.38 [1.51, 3.75]
Reddy, 2010	0.6931	0.1468	2.2%	2.00 [1.50, 2.67]
Tuuli, 2011	0.6259	0.2221	1.0%	1.87 [1.21, 2.89]
Xiong, 2018	0.8416	0.11	4.0%	2.32 [1.87, 2.88]
Yanit, 2012	0.9163	0.1968	1.2%	2.50 [1.70, 3.68]
Yerlikaya, 2016	0.9632	0.2328	0.9%	2.62 [1.66, 4.13]
Youngstorm, 2018	0.6575	0.4007	0.3%	1.93 [0.88, 4.23]
Zetterstorm, 2006	0.7129	0.1637	1.8%	2.04 [1.48, 2.81]
Total (95% CI)			**100.0%**	**2.32 [2.22, 2.42]**

Heterogeneity: Tau² = 0.00; Chi² = 13.11, df = 17 (P = 0.73); I² = 0%
Test for overall effect: Z = 38.32 (P < 0.00001)

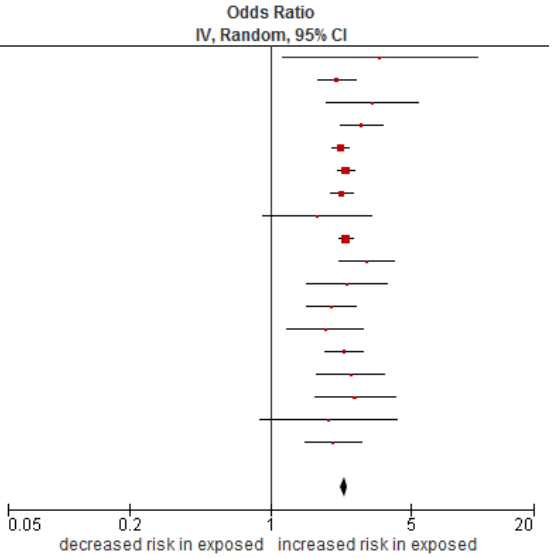


Figure S6c: Funnel plot for adjusted estimates of stillbirth (chronic hypertension versus normotensive)

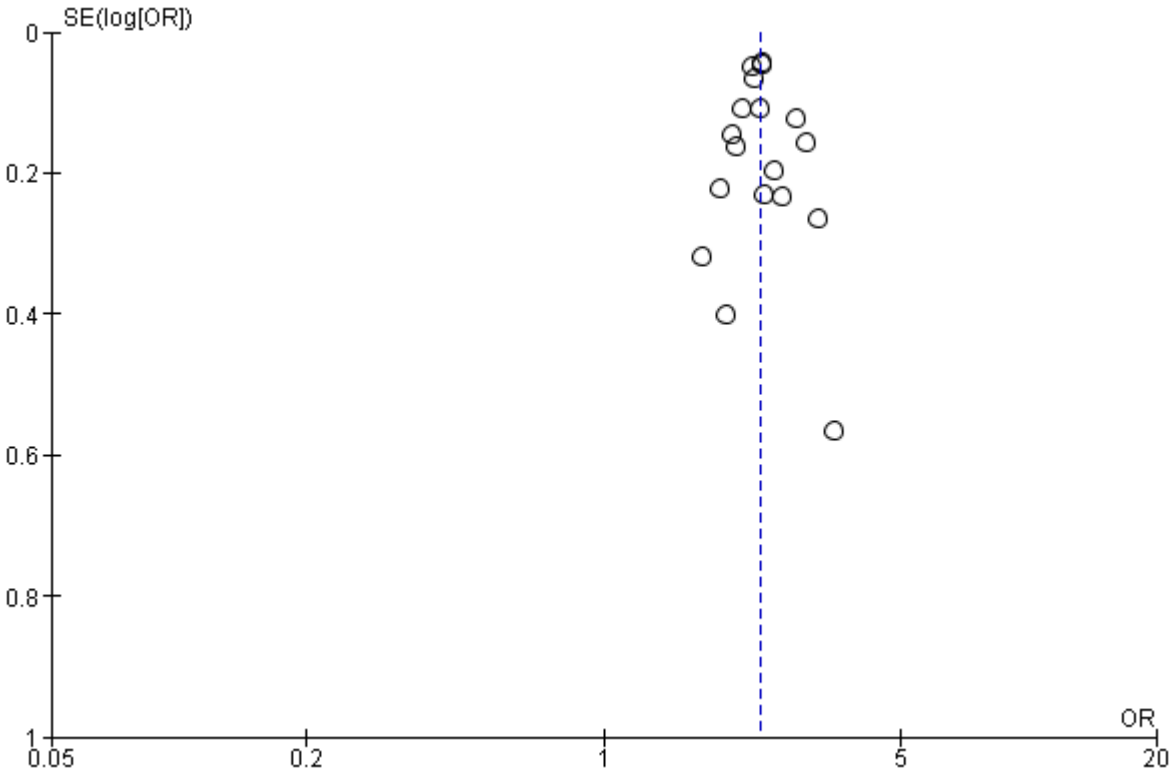


Figure S7a: Forest plot of crude estimates of the association between chronic hypertension and VPTB

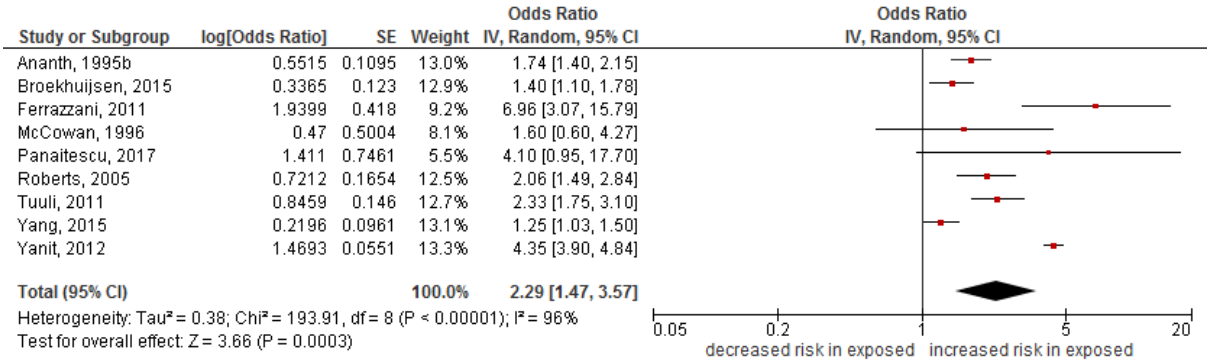


Figure S7b: Forest plot of adjusted estimates of the association between chronic hypertension and VPTB

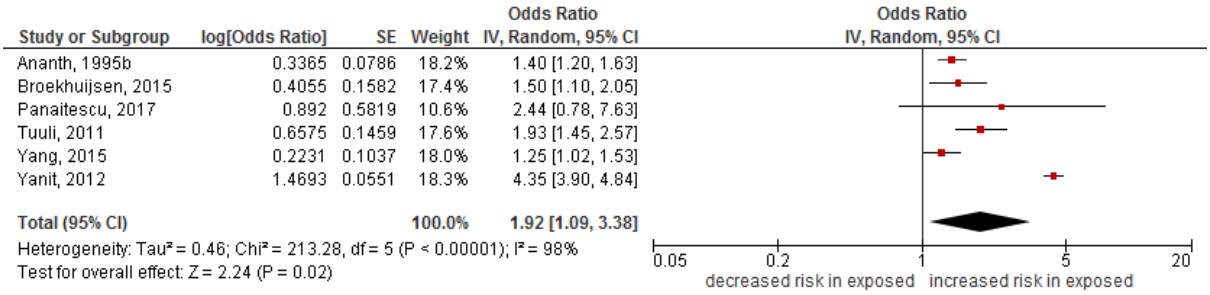


Figure S8a: Forest plot of crude estimates of the association between chronic hypertension and PTB

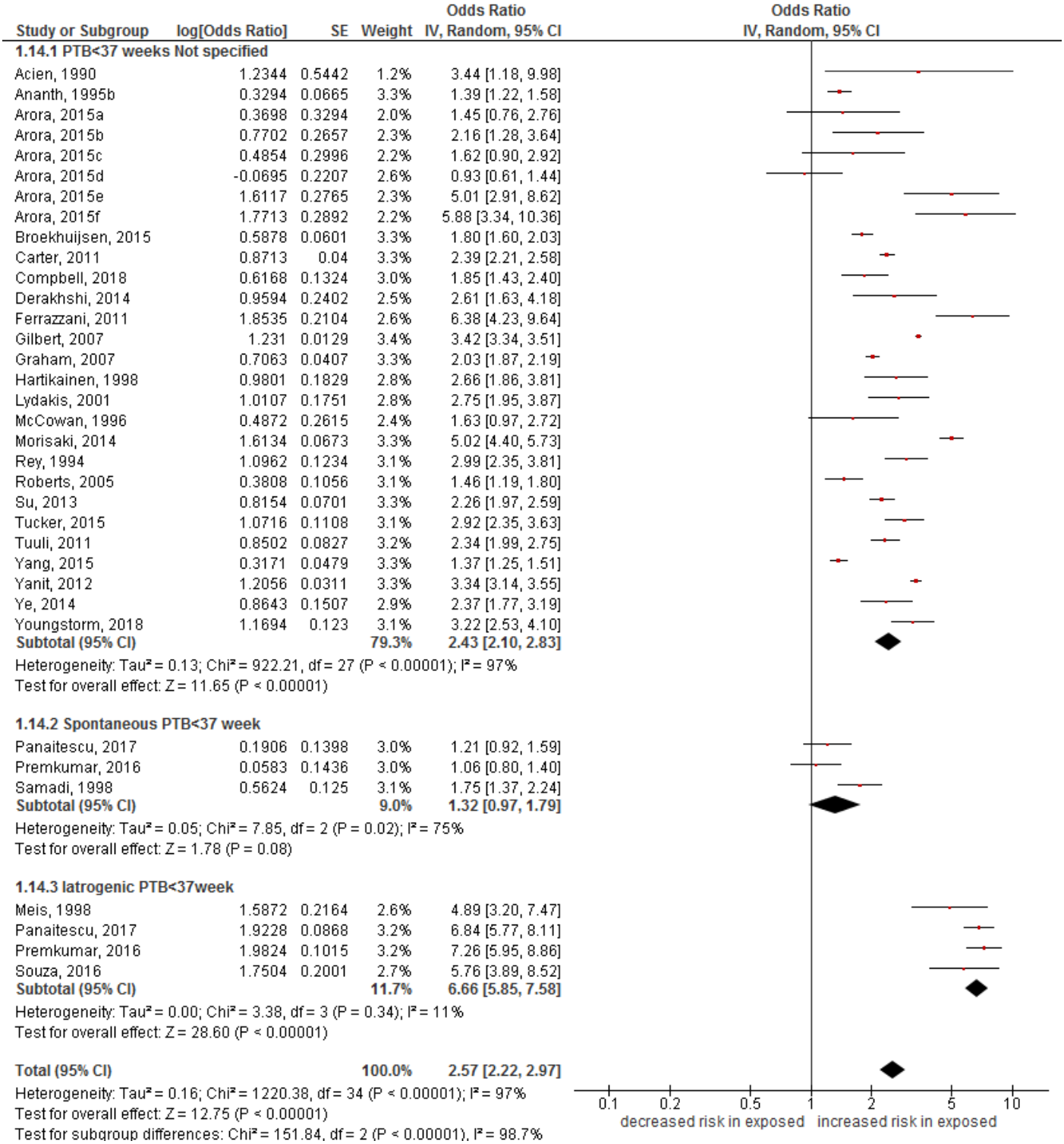


Figure S8b: Forest plot of adjusted estimates of the association between chronic hypertension and PTB

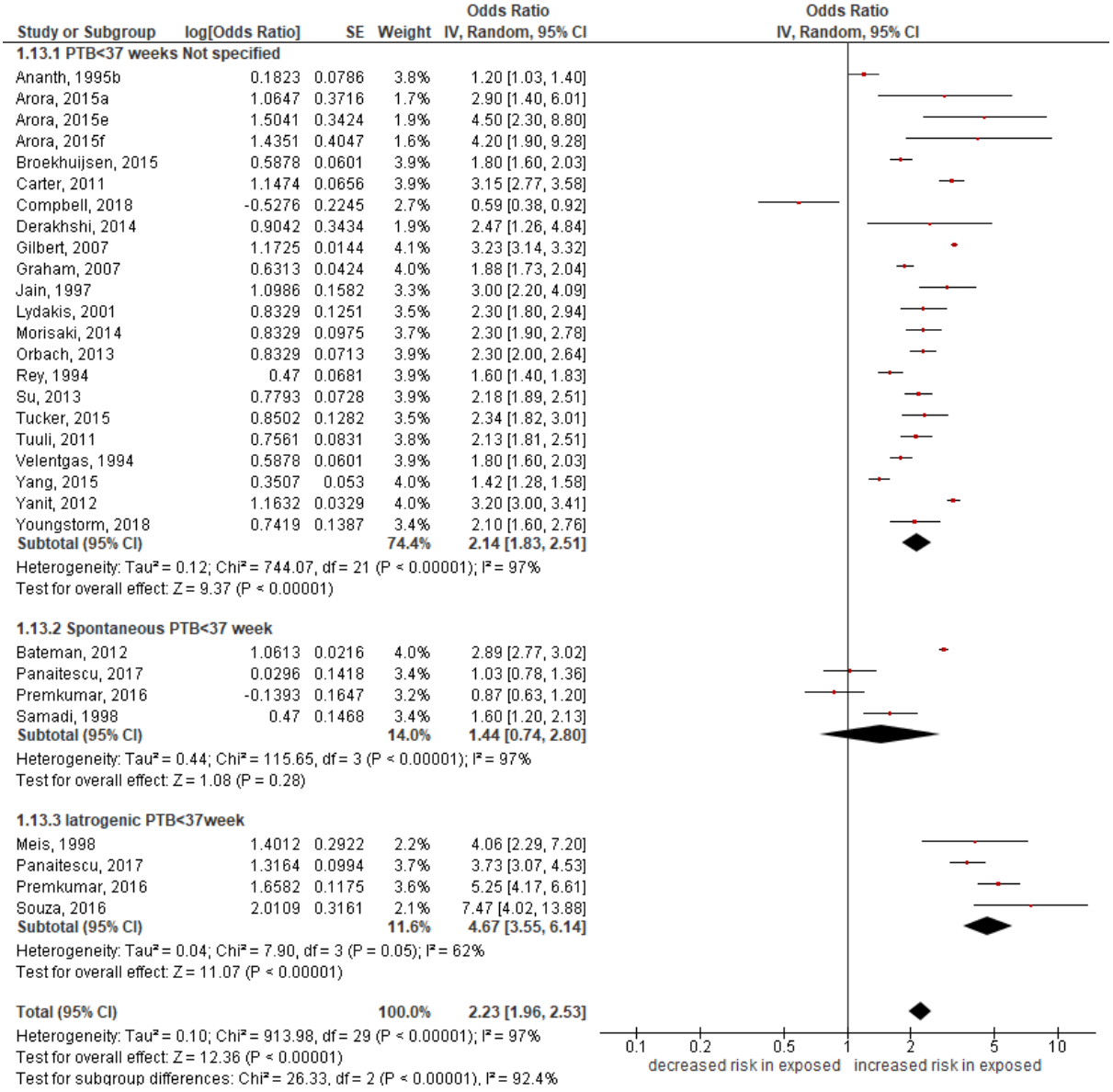


Figure S8c: Funnel plot for adjusted estimates of PTB (chronic hypertension versus normotensive)

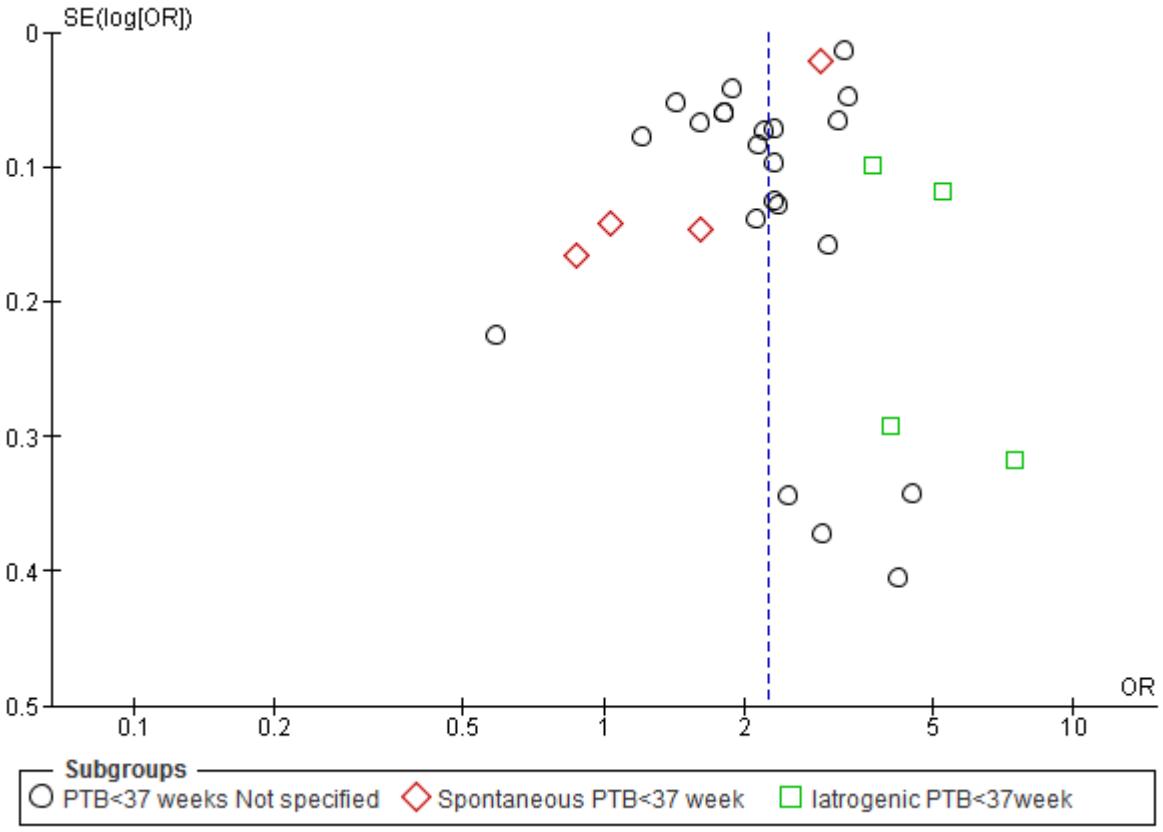


Figure S9a: Forest plot of crude estimates of the association between chronic hypertension and SGA

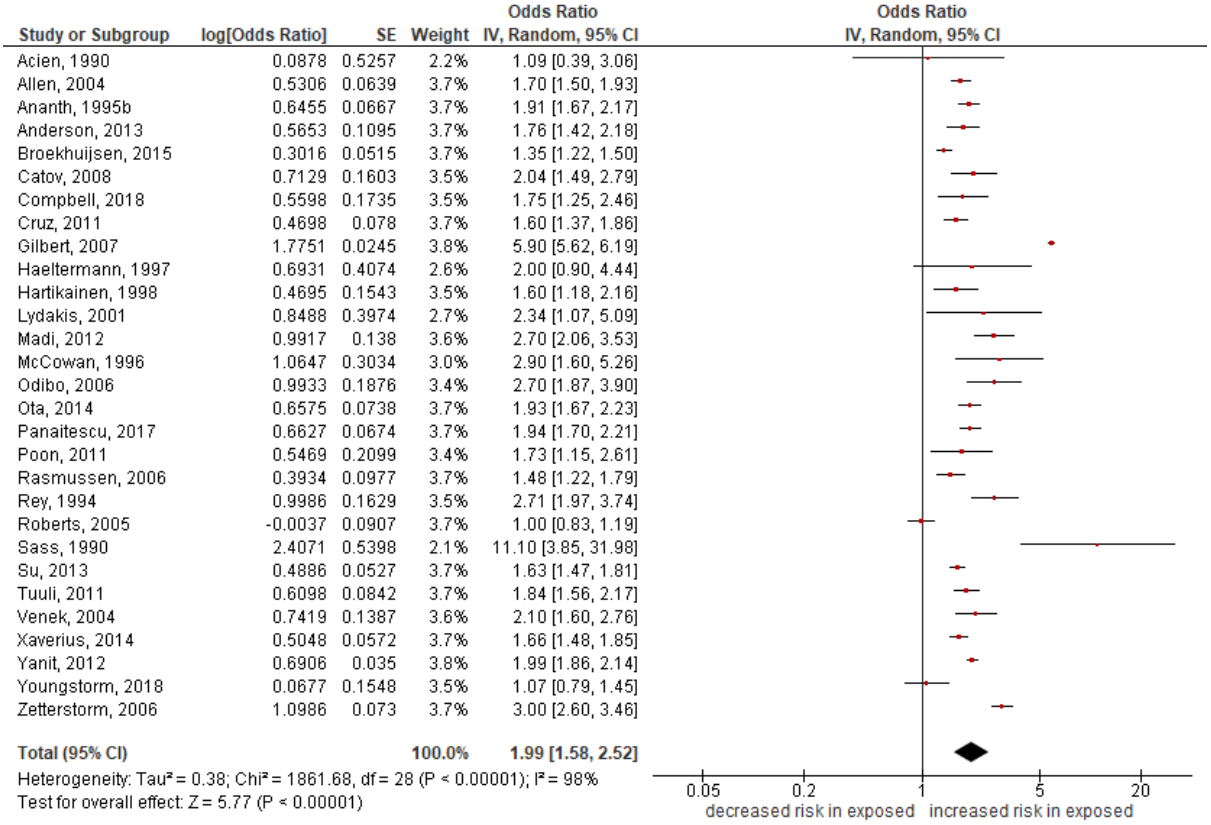


Figure S9b: Funnel plot for adjusted estimates of SGA (chronic hypertension versus normotensive)

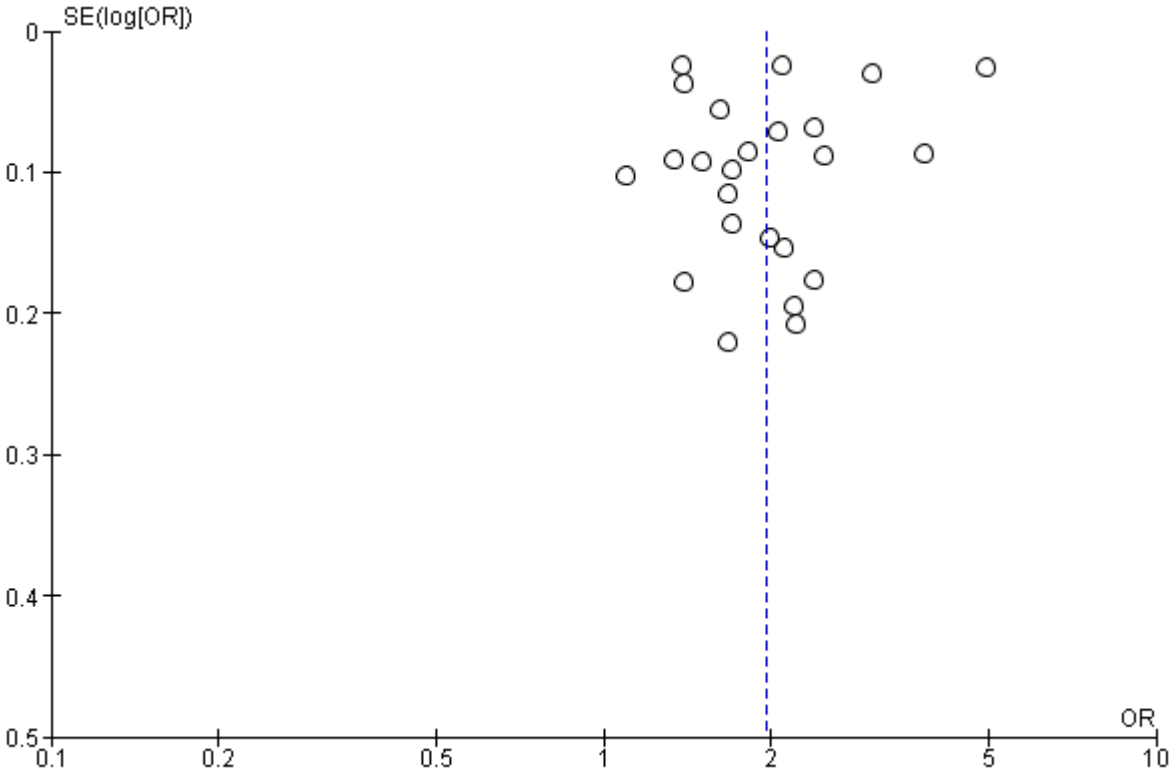


Figure S10a: Forest plot of crude estimates of the association between chronic hypertension and LBW

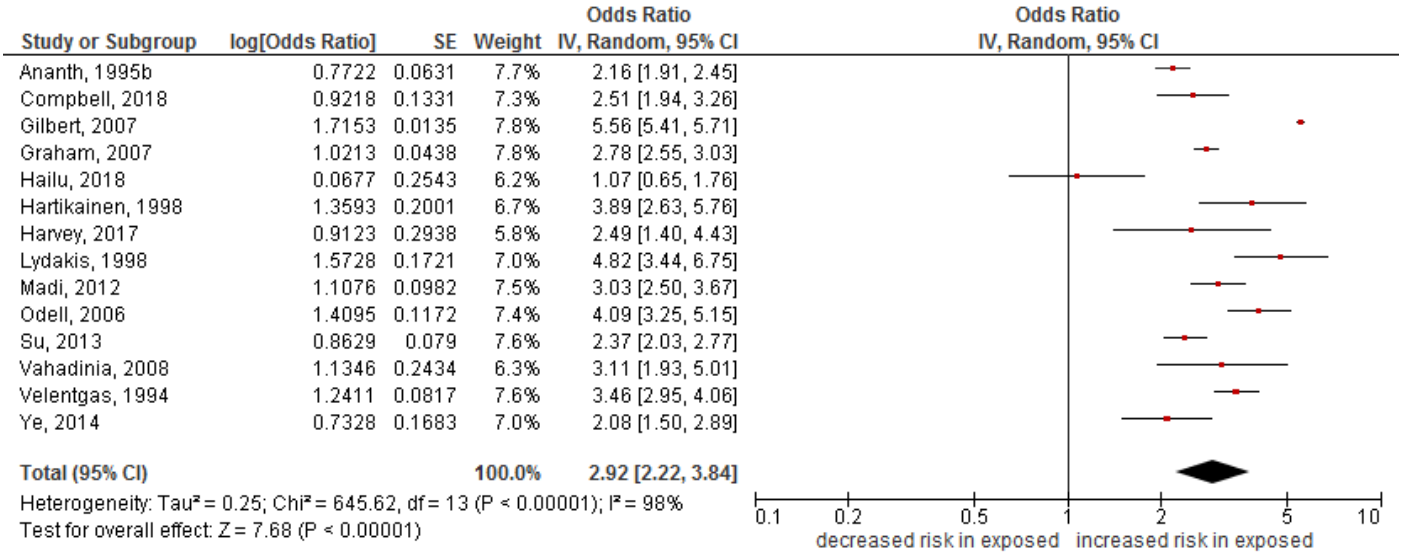


Figure S10b: Funnel plot for adjusted estimates of LBW (chronic hypertension versus normotensive)

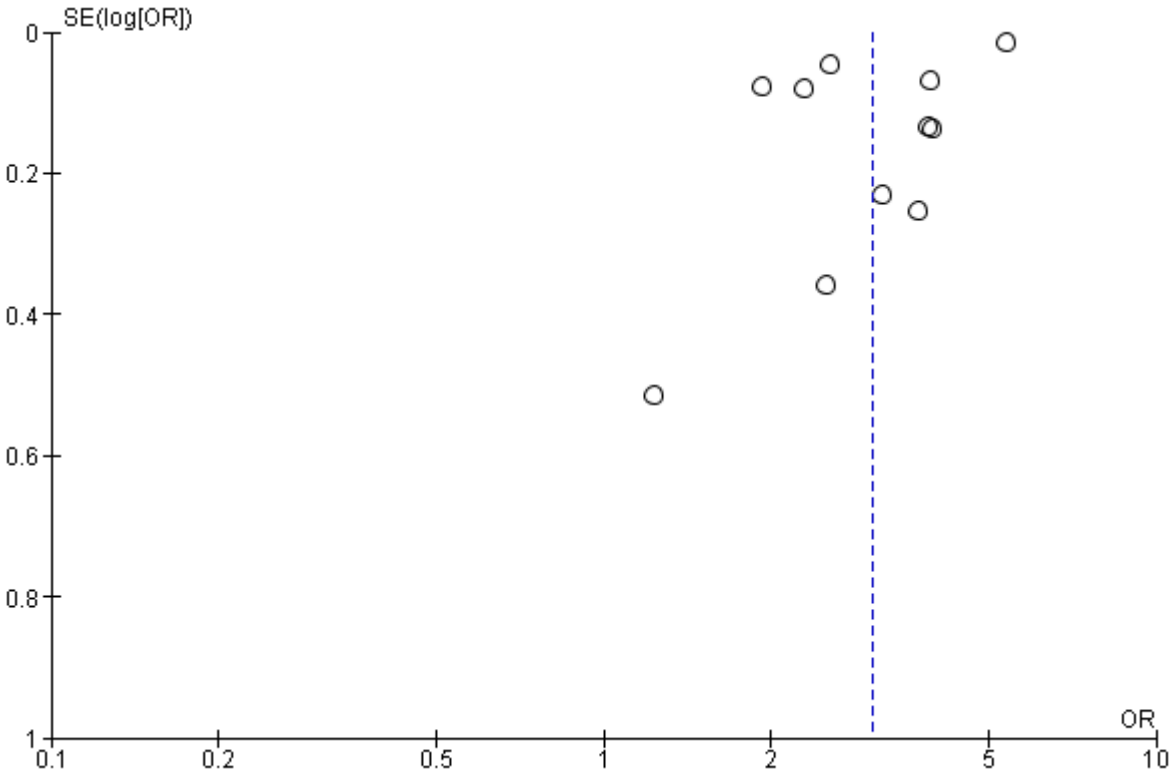


Figure S11a: Forest plot of crude estimates of the association between chronic hypertension and neonatal death

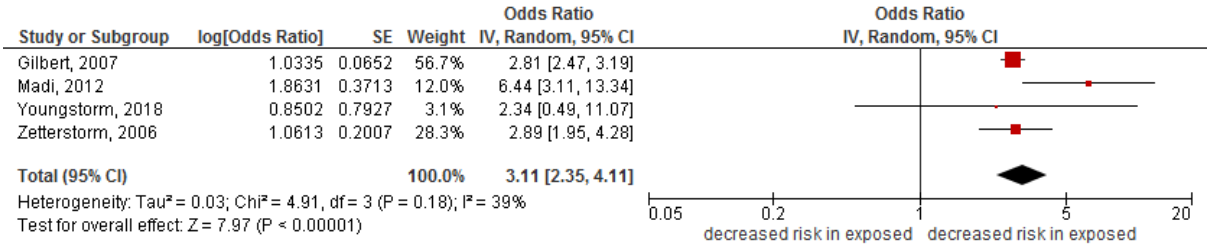


Figure S11b: Forest plot of adjusted estimates of the association between chronic hypertension and neonatal death

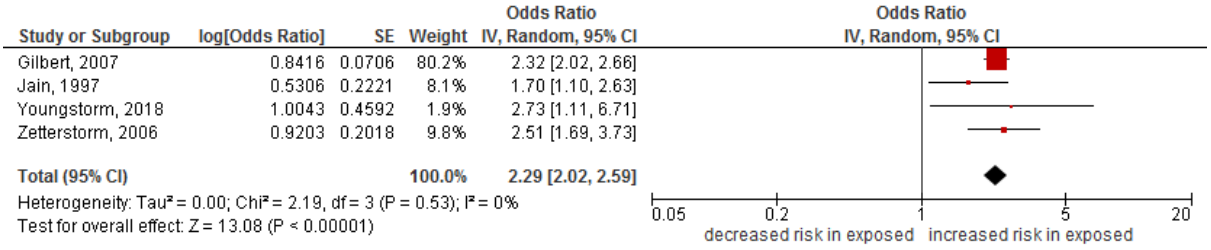


Figure S12a: Forest plot of crude estimates of the association between chronic hypertension and perinatal death

Study or Subgroup	log[Odds Ratio]	SE	Weight	Odds Ratio IV, Random, 95% CI
Acien, 1990	1.8225	0.7323	4.7%	6.19 [1.47, 25.99]
Ahmed, 2012	0.8459	0.1317	14.9%	2.33 [1.80, 3.02]
Broekhuijsen, 2015	-0.3567	0.275	12.0%	0.70 [0.41, 1.20]
Hartikainen, 1998	1.3371	0.3334	10.7%	3.81 [1.98, 7.32]
McCowan, 1996	0.5306	0.7382	4.6%	1.70 [0.40, 7.22]
Rey, 1994	1.3229	0.2909	11.6%	3.75 [2.12, 6.64]
Sass, 1990	1.6045	0.5658	6.5%	4.98 [1.64, 15.08]
Venek, 2004	0.5306	0.2221	13.1%	1.70 [1.10, 2.63]
Ye, 2014	1.296	0.2499	12.5%	3.65 [2.24, 5.96]
Youngstorm, 2018	0.7839	0.4	9.3%	2.19 [1.00, 4.80]
Total (95% CI)			**100.0%**	**2.46 [1.70, 3.55]**

Heterogeneity: Tau² = 0.22; Chi² = 33.00, df = 9 (P = 0.0001); I² = 73%
Test for overall effect: Z = 4.79 (P < 0.00001)

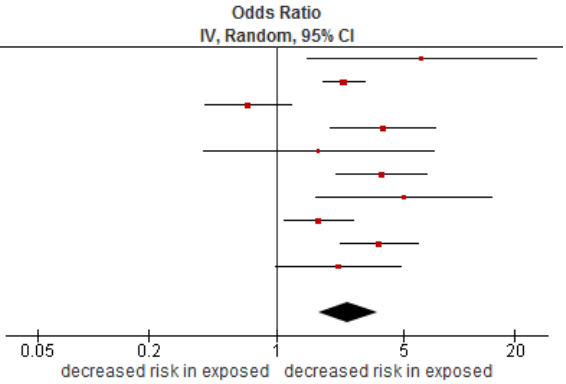


Figure S12b: Forest plot of adjusted estimates of the association between chronic hypertension and perinatal death

Study or Subgroup	log[Odds Ratio]	SE	Weight	Odds Ratio IV, Random, 95% CI
Ahmed, 2012	0.802	0.1325	20.7%	2.23 [1.72, 2.89]
Broekhuijsen, 2015	-0.2231	0.2398	16.3%	0.80 [0.50, 1.28]
Jain, 1997	0.8755	0.1468	20.2%	2.40 [1.80, 3.20]
Rey, 1994	1.1632	0.2936	14.1%	3.20 [1.80, 5.69]
Venek, 2004	0.47	0.2398	16.3%	1.60 [1.00, 2.56]
Youngstorm, 2018	0.6366	0.3403	12.4%	1.89 [0.97, 3.68]
Total (95% CI)			**100.0%**	**1.87 [1.33, 2.63]**

Heterogeneity: Tau² = 0.13; Chi² = 20.27, df = 5 (P = 0.001); I² = 75%
Test for overall effect: Z = 3.60 (P = 0.0003)

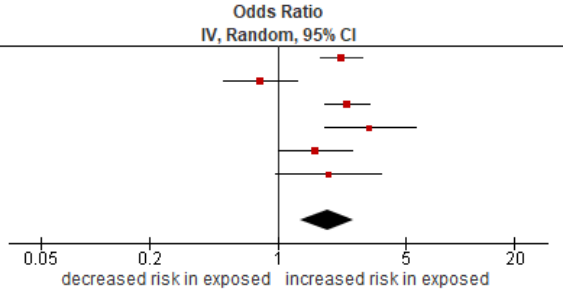


Figure S13: Forest plot of crude estimates of the association between chronic hypertension and NICU admission

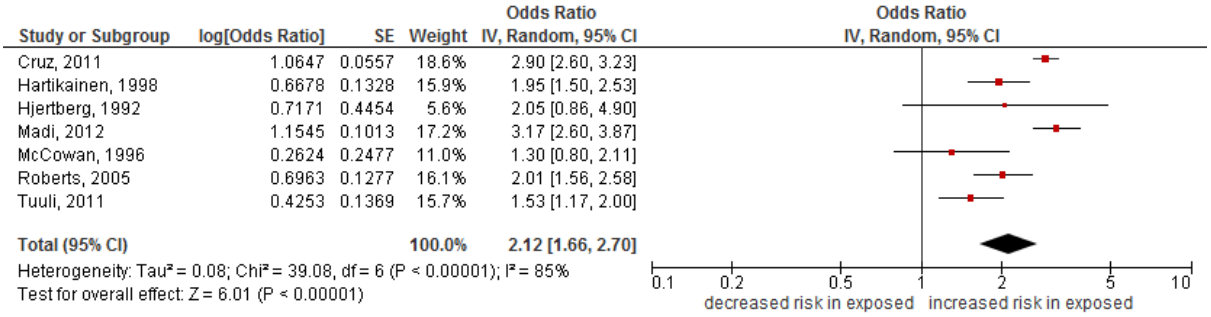


Figure S14: Forest plot of crude estimates of the association between chronic hypertension and miscarriage

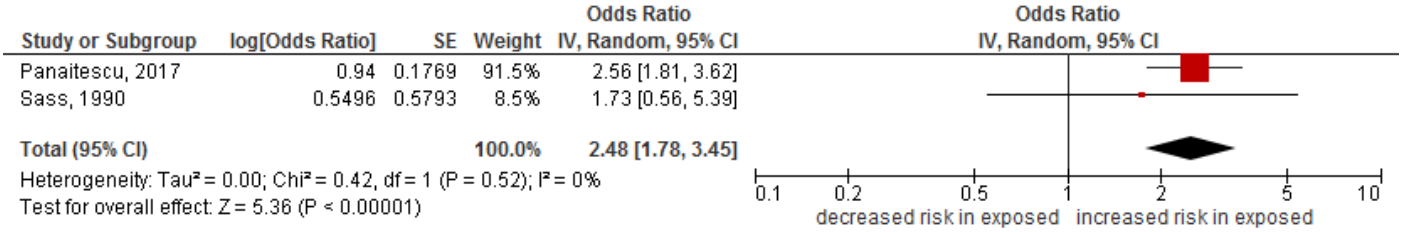


Figure S15: Forest plot of adjusted estimates of the association between chronic hypertension and stillbirth by ethnicity

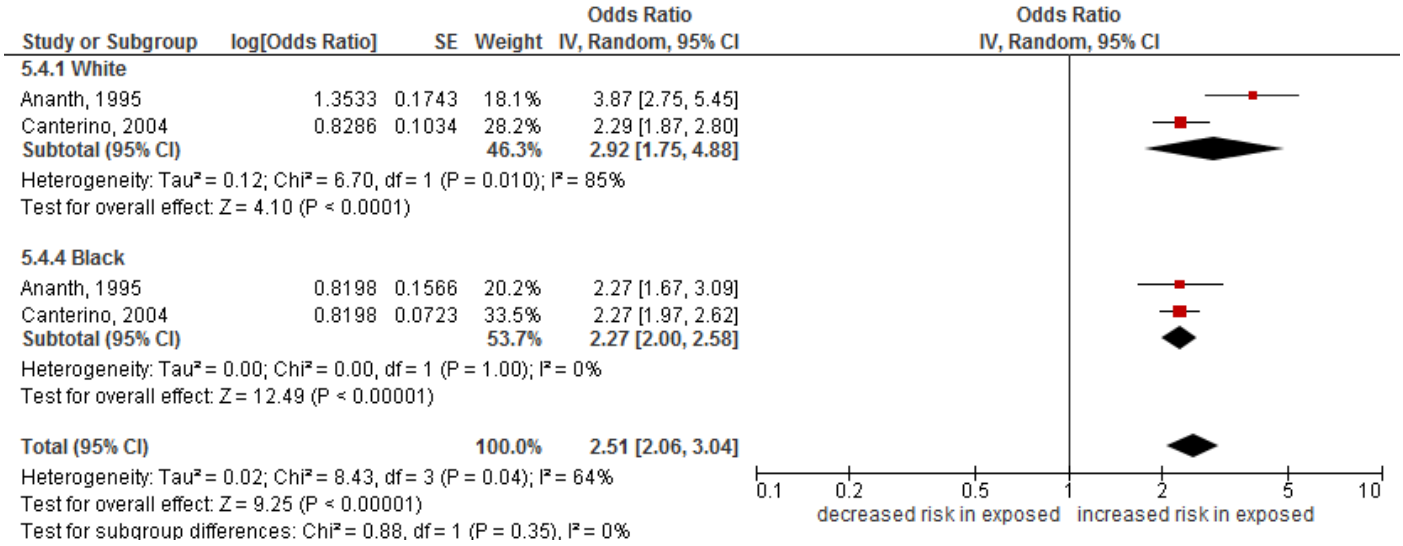


Figure S16: Forest plot of adjusted estimates of the association between chronic hypertension and PTB by ethnicity

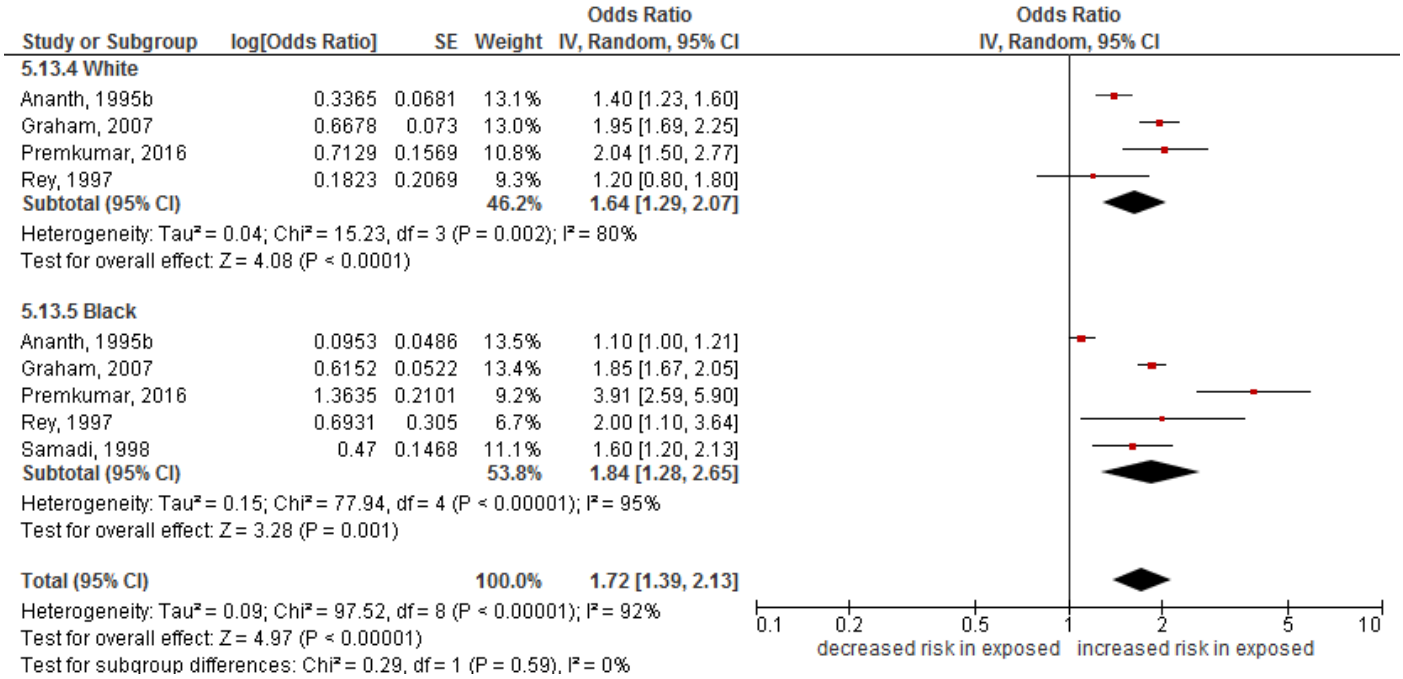


Figure S17: Forest plot of adjusted estimates of the association between chronic hypertension and SGA by ethnicity

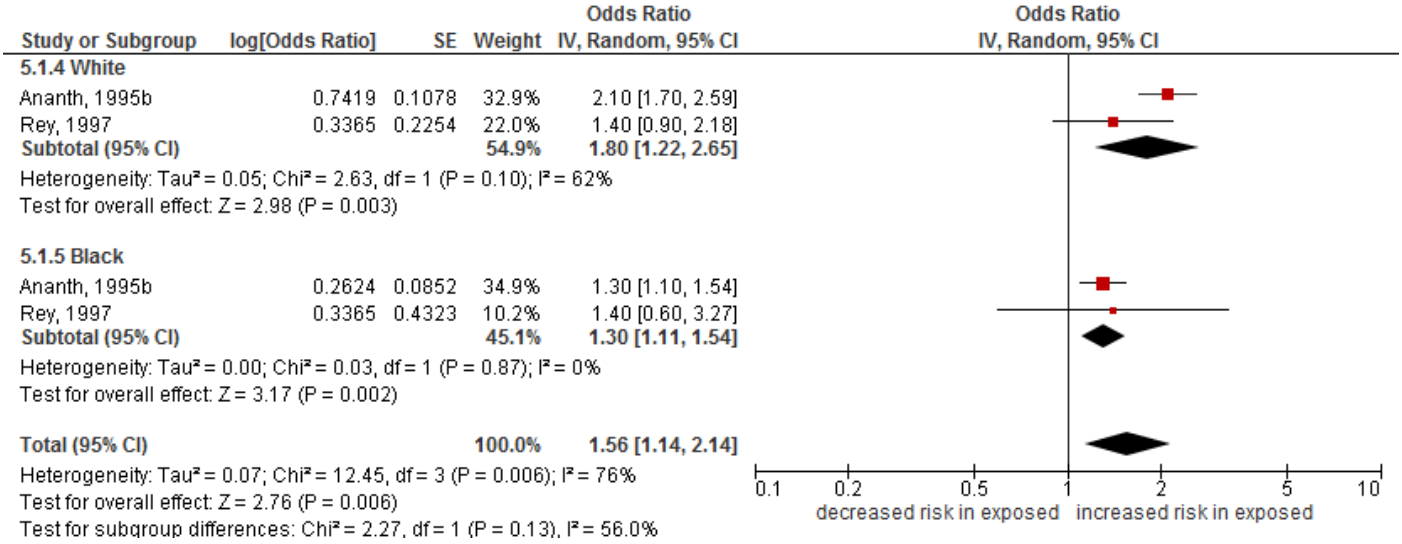
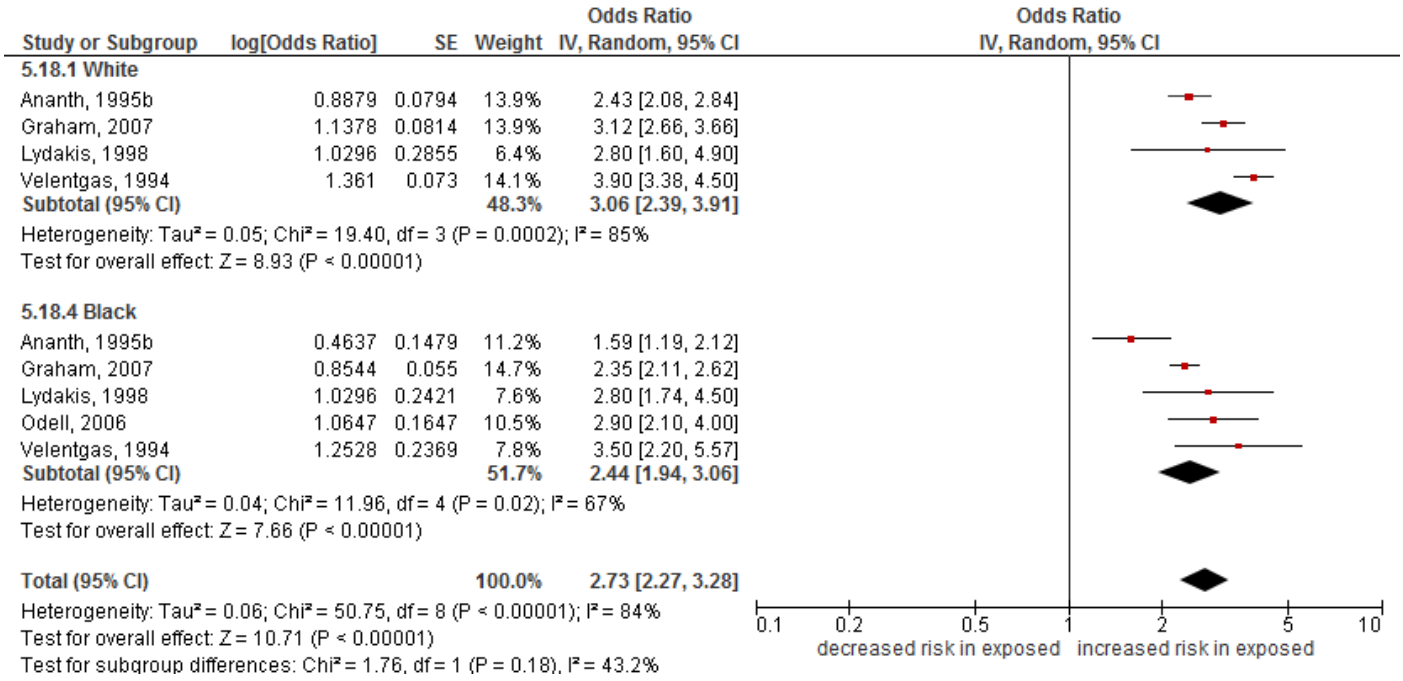


Figure S18: Forest plot of adjusted estimates of the association between chronic hypertension and LBW by ethnicity



Results of sensitivity analyses of adjusted estimates for adverse outcome by study location and year of publication (women with chronic hypertension vs. without)

Maternal Outcome	No. of studies (estimates)	Overall adjusted OR [95% CI]	I^2,%
PE			
Location			
North America	5	5.22 [2.87, 9.50]	100
Europe	4	4.83 [3.14, 7.42]	97
Other	8	6.01 [3.34, 10.79]	99
Decades			
1990-1999	1	6.50 [5.20, 8.13]	-
2000-2009	7	3.25 [1.74, 6.07]	99
2010-2019	14	7.29 [6.15, 8.65]	97
HELLP			
Location			
North America	1	2.66 [2.14, 3.31]	-
Europe	2	3.15 [1.09, 9.15]	80
Decades			
2000-2009	1	2.66 [2.14, 3.31]	-
2010-2019	2	3.15 [1.09, 9.15]	80
CS			
Location			
North America	5	2.08 [1.88, 2.31]	95
Europe	2	1.11 [1.01, 1.21]	0
Other	1	2.70 [2.40, 3.04]	-
Decades			
1990-1999	1	1.80 [1.60, 2.03]	-
2000-2009	2	2.48 [2.15, 2.86]	83
2010-2019	5	1.71 [1.28, 2.30]	98
PPH			
Location			
North America	2	1.45 [1.36, 1.54]	0
Europe	1	1.10 [0.90, 1.34]	-
Other	1	2.20 [1.40, 3.46]	-
Decades			

2000-2009	2	1.67 [1.12, 2.49]	70
2010-2019	2	1.32 [0.87, 2.02]	74
Maternal mortality			
Location			
North America	4	4.80 [3.04, 7.58]	71
Decades			
2000-2009	2	3.66 [2.00, 6.69]	55
2010-2019	2	6.04 [3.82, 9.53]	46
Fetal, neonatal outcome			
Stillbirth			
Location			
North America	12	2.37 [2.25, 2.50]	3
Europe	4	2.18 [1.87, 2.55]	0
Other	2	2.27 [2.03, 2.54]	0
Decades			
1990-1999	2	2.89 [2.39, 3.49]	0
2000-2009	7	2.34 [2.21, 2.47]	0
2010-2019	9	2.26 [2.10, 2.44]	0
VPTB<34 wks.			
Location			
North America	3	2.51 [0.91, 6.92]	99
Europe	2	1.55 [1.15, 2.09]	0
Other	1	1.25 [1.02, 1.53]	-
Decades			
1990-1999	1	1.40 [1.20, 1.63]	-
2000-2009	-	-	-
2010-2019	5	2.06 [1.06, 3.98]	97
PTB<37 wks.*			
Location			
North America	16	2.08 [1.76, 2.46]	96
Europe	5	2.56 [1.90, 3.44]	78
Other	5	2.04 [1.58, 2.63]	91
Decades			

1990-1999	5	1.93 [1.40, 2.66]	90
2000-2009	3	1.90 [1.73, 2.08]	36
2010-2019	18	2.26 [1.95, 2.61]	94
SGA			
Location			
North America	13	2.18 [1.62, 2.92]	99
Europe	6	1.81 [1.45, 2.27]	92
Australia	1	1.68 [1.34, 2.11]	-
Other	4	1.62 [1.39, 1.89]	58
Decades			
1990-1999	3	2.50 [1.44, 4.35]	95
2000-2009	7	2.30 [1.49, 3.56]	98
2010-2019	14	1.73 [1.44, 2.07]	97
LBW			
Location			
North America	7	3.21 [2.19, 4.68]	98
Europe	1	3.91 [2.99, 5.11]	-
Other	3	2.46 [1.60, 3.77]	59
Decades			
1990-1999	3	3.07 [1.85, 5.10]	97
2000-2009	4	3.74 [2.28, 6.14]	99
2010-2019	4	2.38 [1.95, 2.92]	13
Neonatal death			
Perinatal mortality			
Location			
North America	3	2.45 [1.92, 3.11]	0
Europe	2	1.36 [0.50, 3.72]	93
Other	1	1.60 [1.00, 2.56]	-
Decades			
1990-1999	2	2.54 [1.97, 3.29]	0
2000-2009	1	1.60 [1.00, 2.56]	-
2010-2019	3	1.51 [0.76, 2.99]	86

*Sub-group estimates (spontaneous, iatrogenic) for two studies were combined.

Figure S19a: Forest plot of crude estimates of the association between antihypertensive treatment and Pre-eclampsia

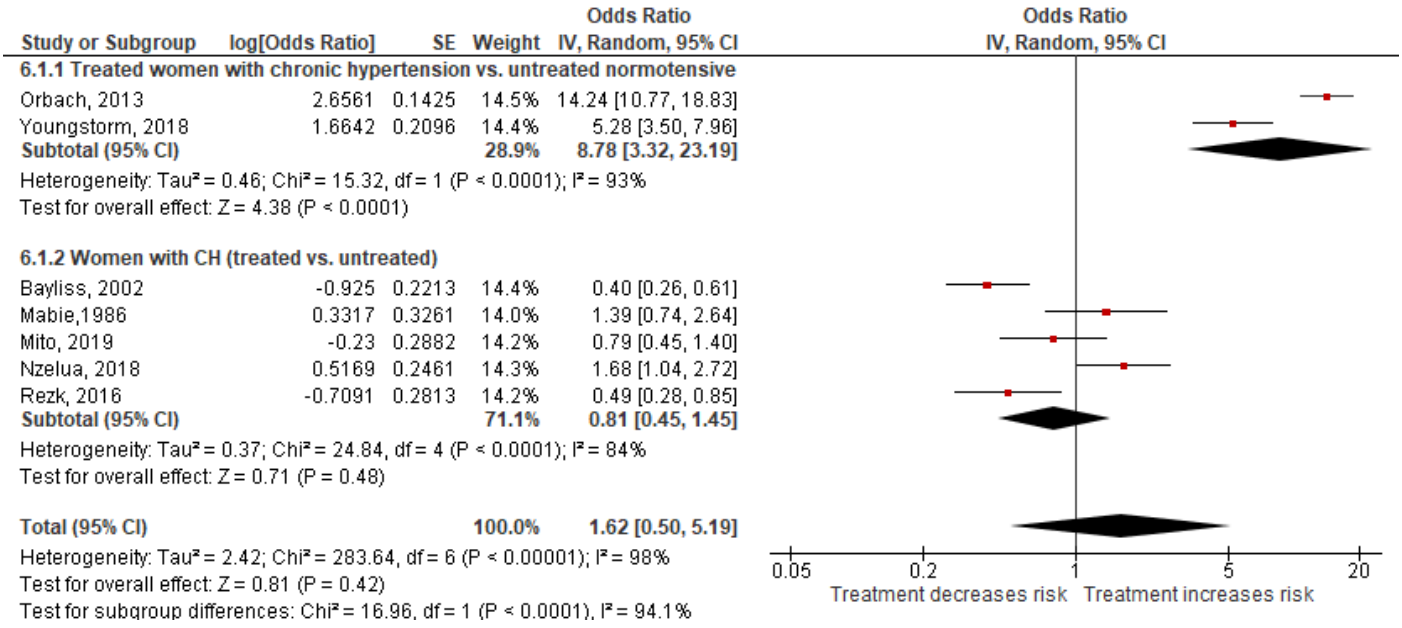


Figure S19b: Forest plot of adjusted estimates of the association between antihypertensive treatment and Pre-eclampsia

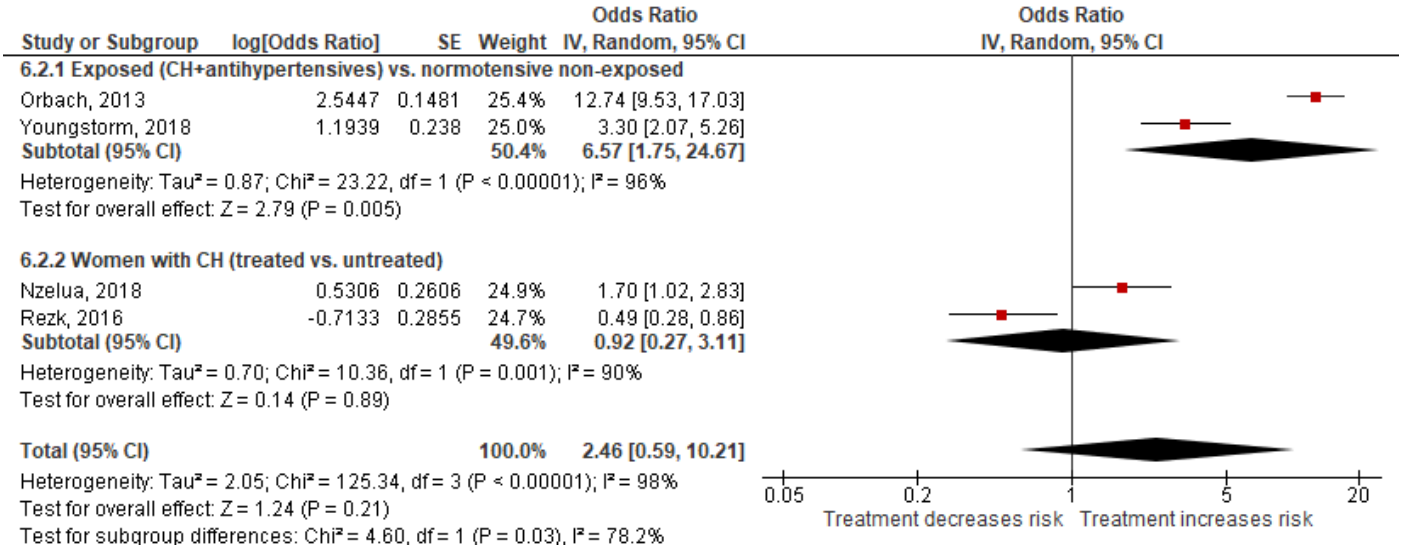


Figure S20: Forest plot of crude estimates of the association between antihypertensive treatment and caesarean section

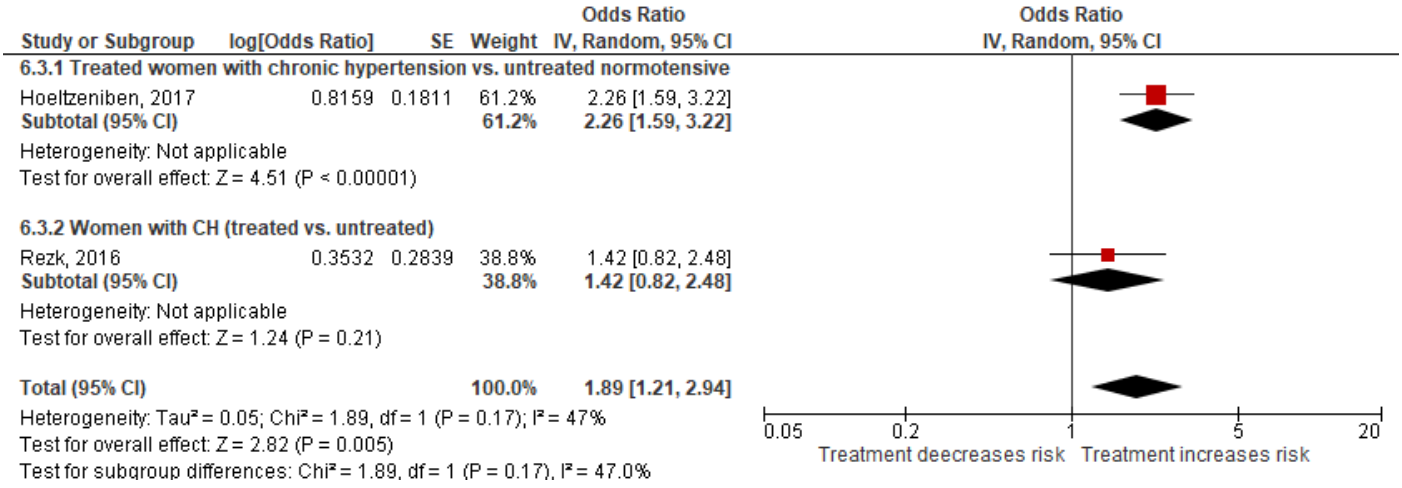


Figure S21: Forest plot of crude estimates of the association between antihypertensive treatment and miscarriage

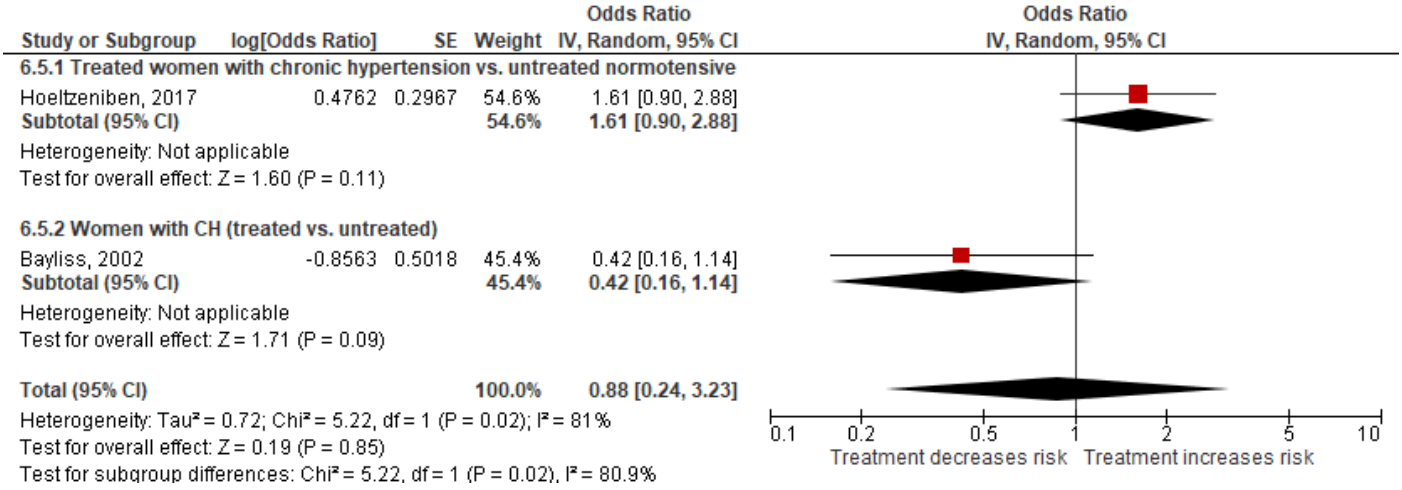


Figure S22: Forest plot of crude estimates of the association between antihypertensive treatment and stillbirth

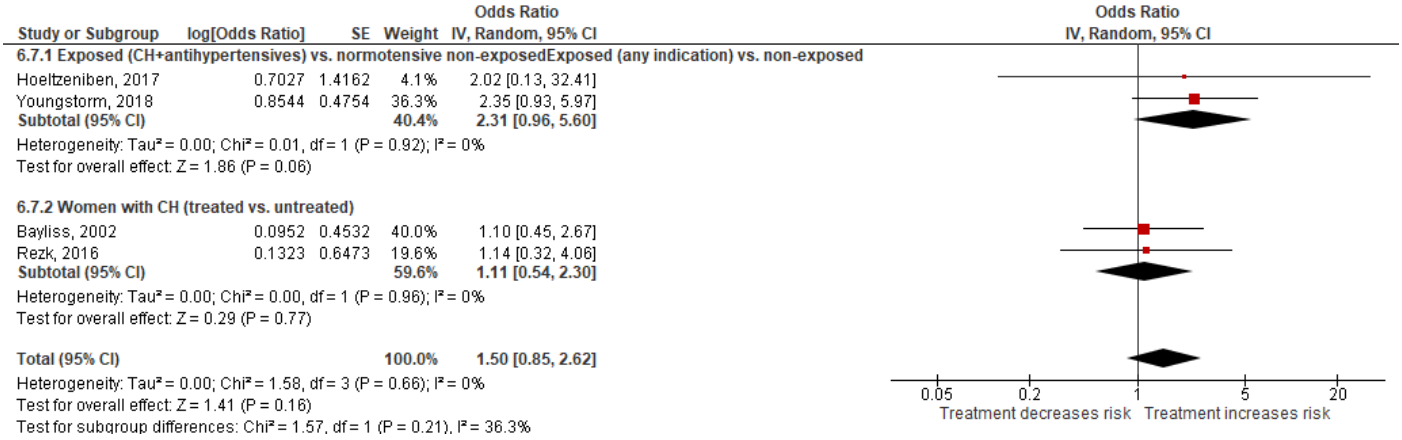


Figure S23a: Forest plot of crude estimates of the association between antihypertensive treatment and preterm birth

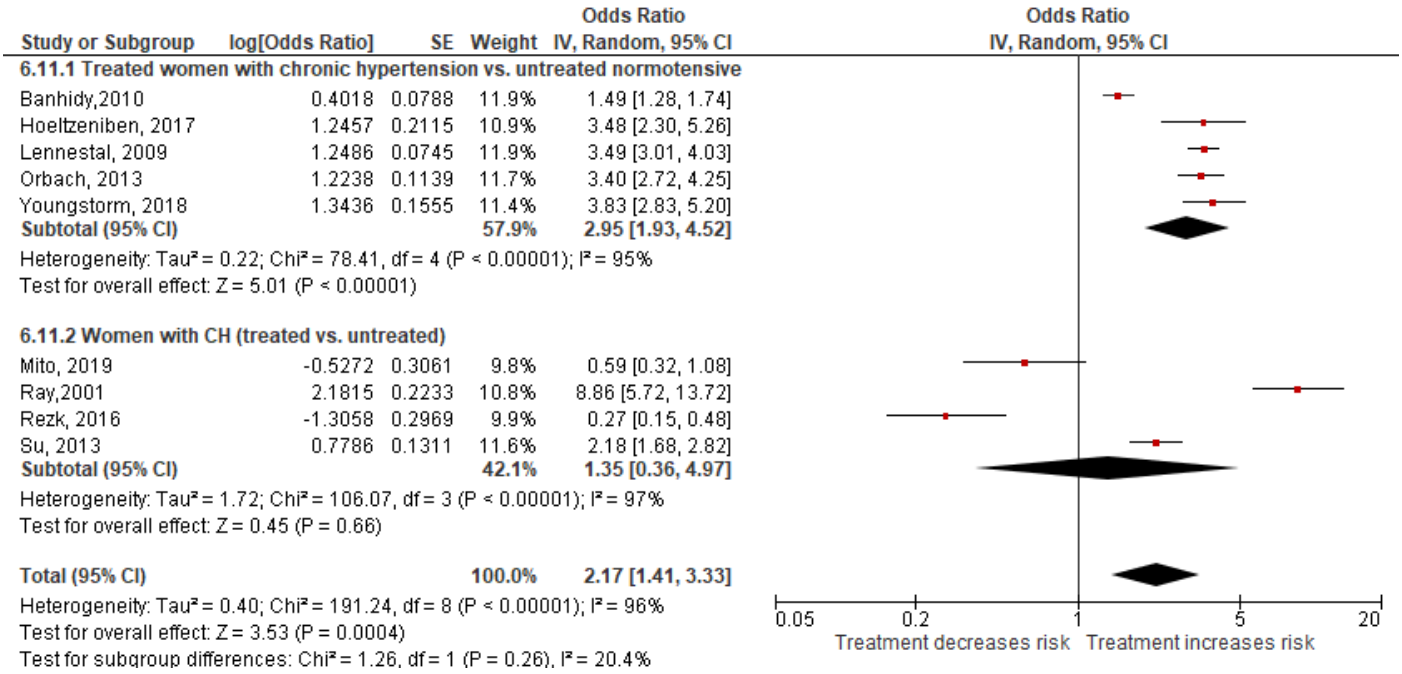


Figure S23b: Forest plot of adjusted estimates of the association between antihypertensive treatment and preterm birth

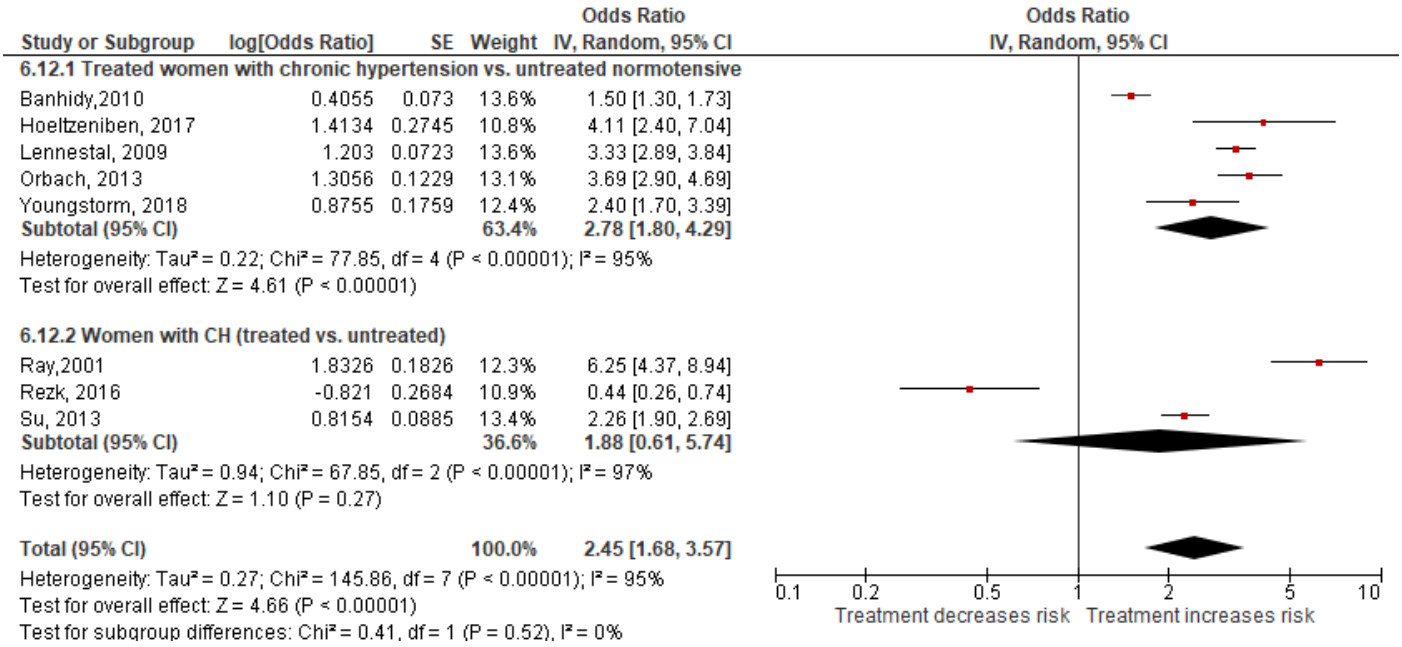


Figure S23c: Forest plot of crude estimates of the association between antihypertensive treatment and preterm birth (β-blockers vs. untreated)

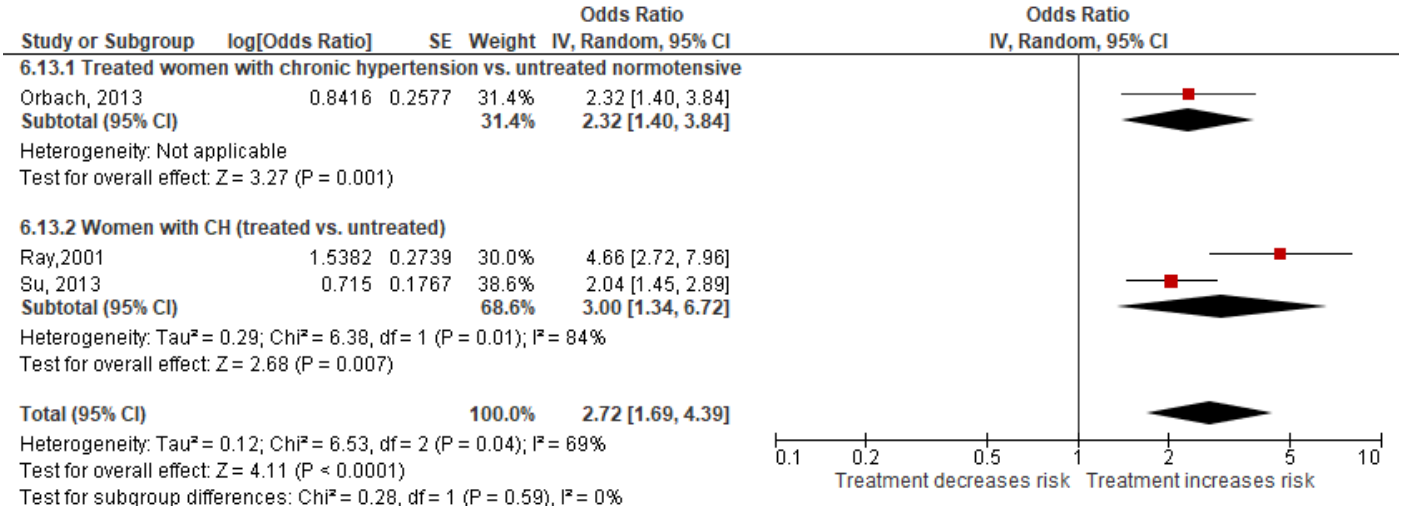


Figure S23d: Forest plot of adjusted estimates of the association between antihypertensive treatment and preterm birth (β-blockers vs. untreated)

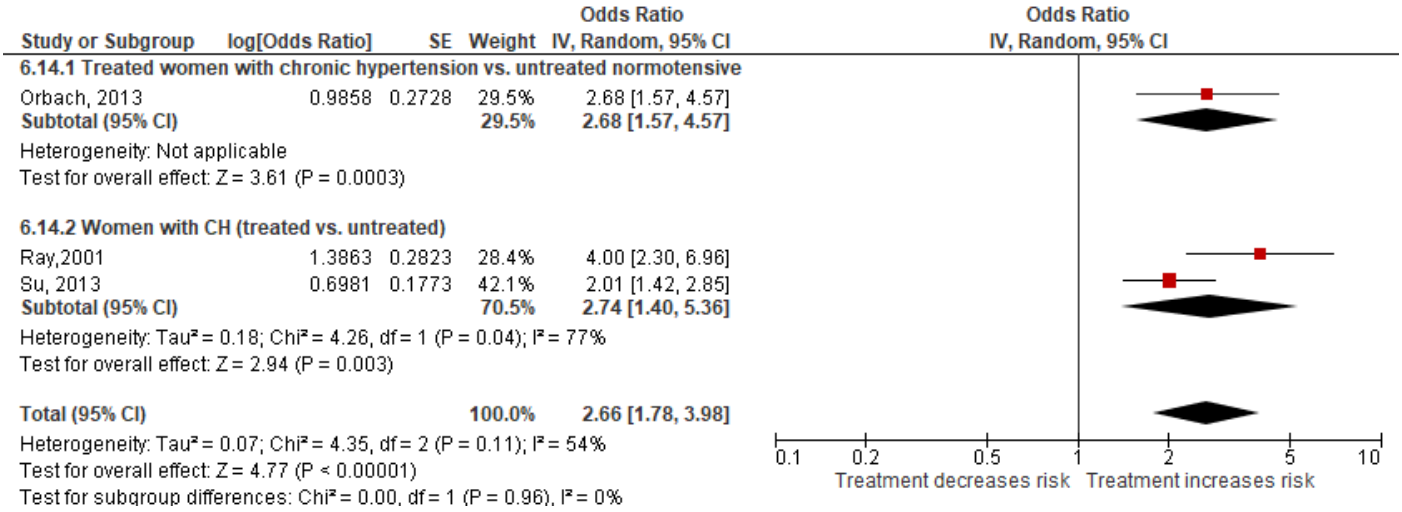


Figure S24a: Forest plot of crude estimates of the association between antihypertensive treatment and small for gestational age

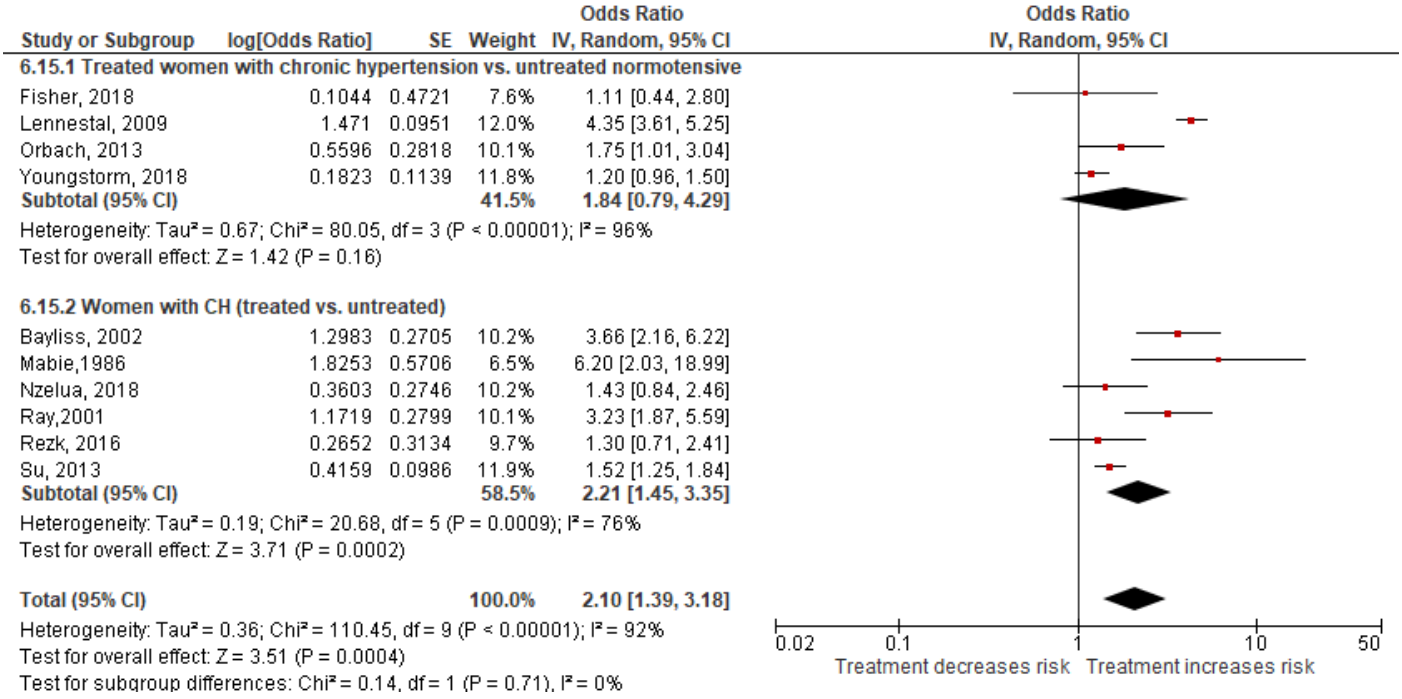


Figure S24b: Forest plot of adjusted estimates of the association between antihypertensive treatment and small for gestational age

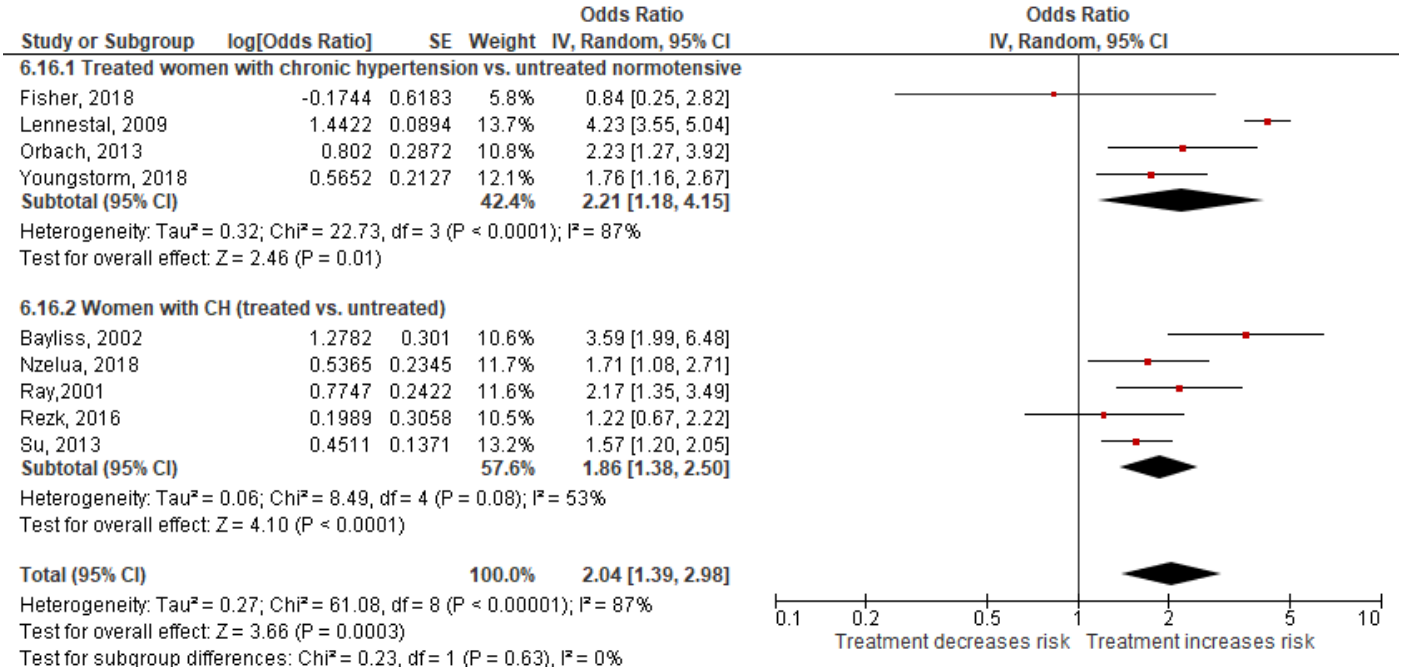


Figure S24c: Forest plot of crude estimates of small for gestational age (β-blockers only vs. untreated)

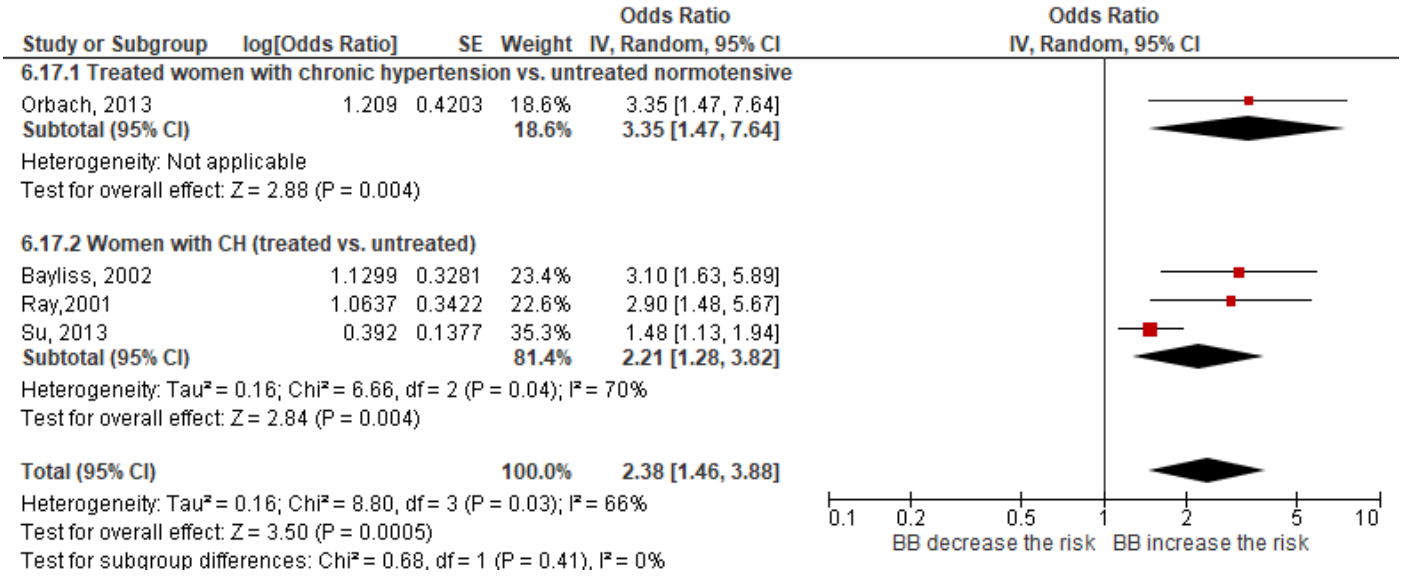


Figure S24d: Forest plot of adjusted of small for gestational age (β-blockers only vs. untreated)

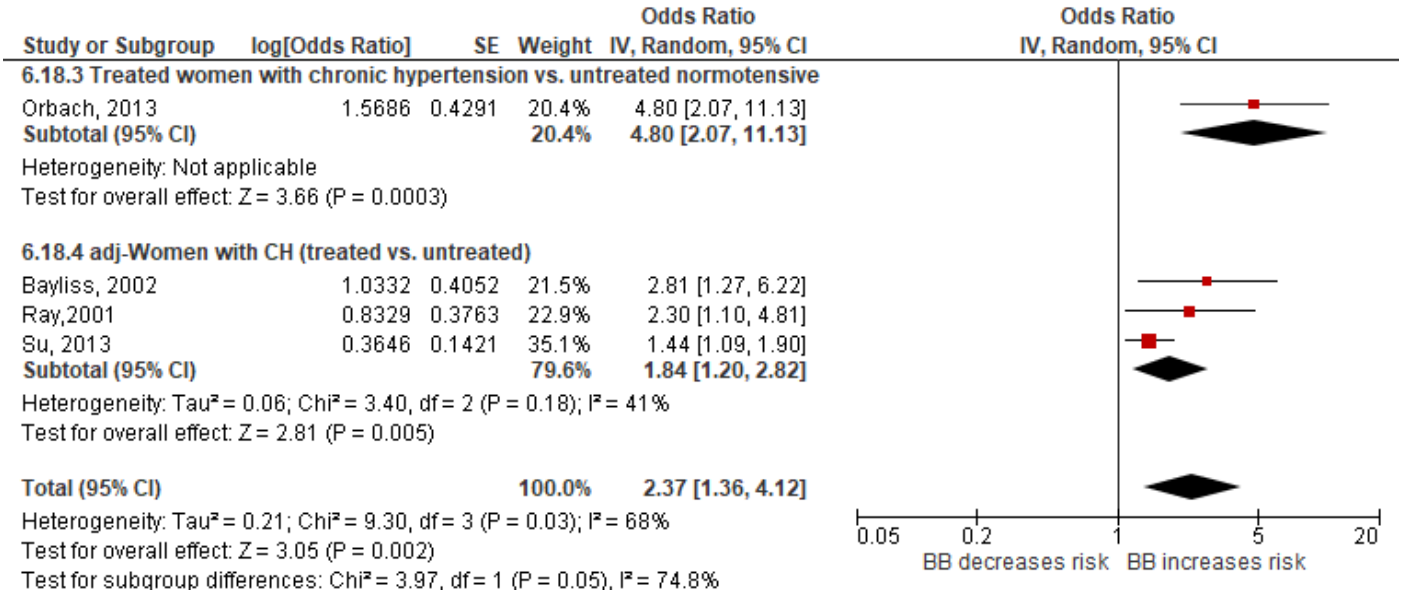


Figure S24e: Forest plot of crude estimates of small for gestational age (centrally acting antiadrenergic vs. untreated)

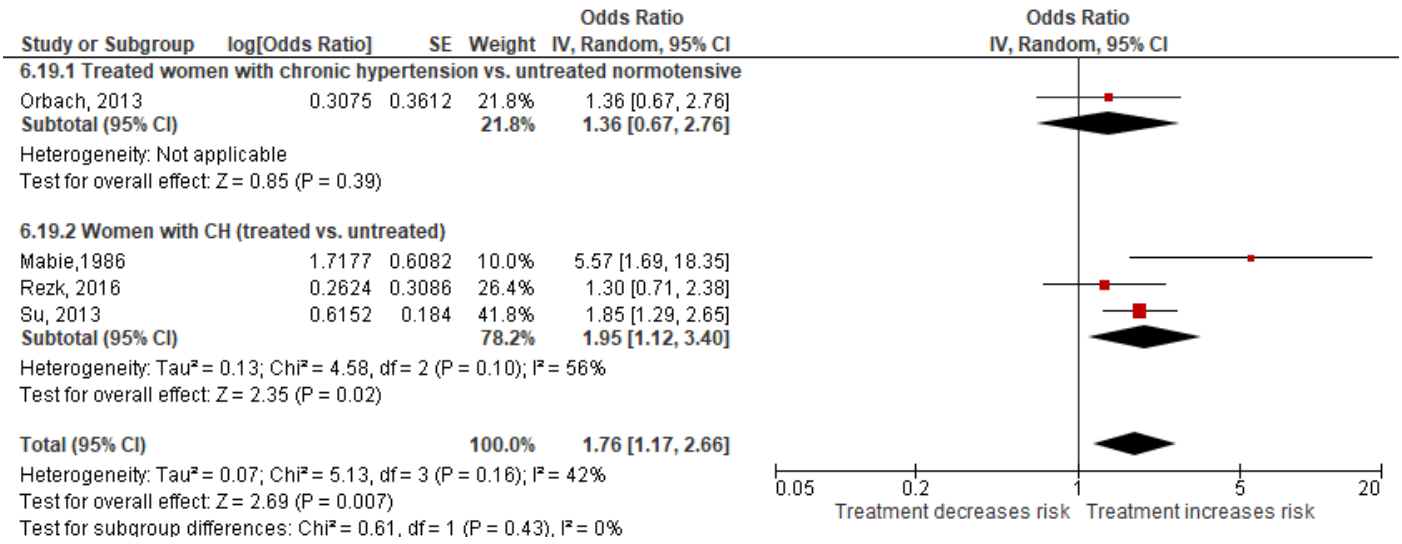


Figure S24f: Forest plot of adjusted estimates of small for gestational age (centrally acting antiadrenergic vs. untreated)

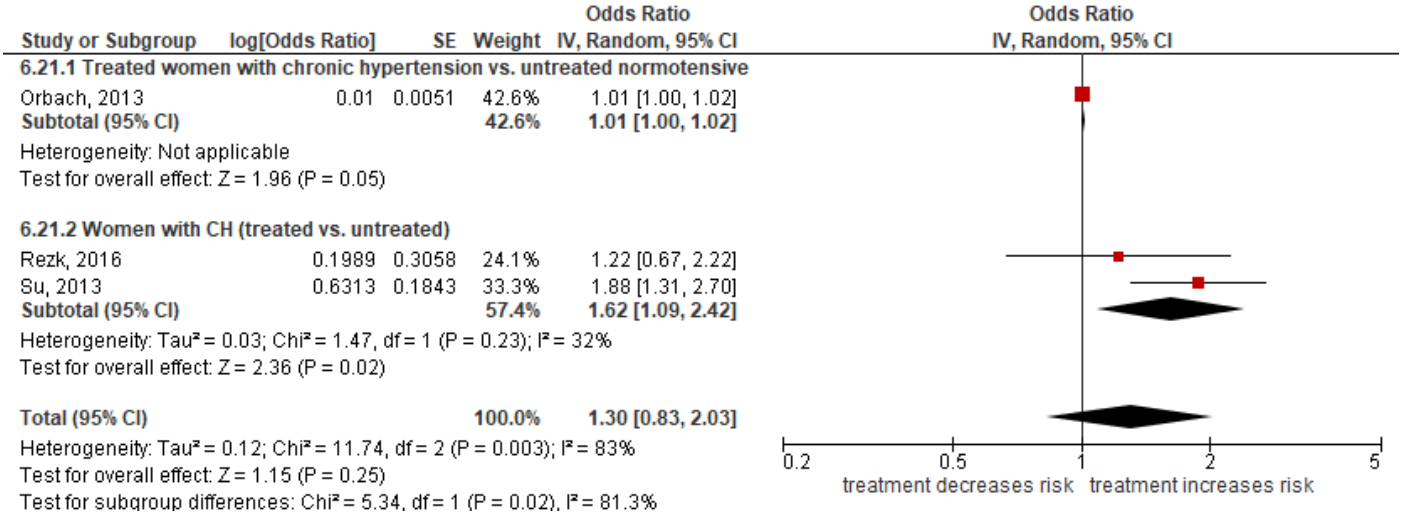


Figure S24g: Forest plot of adjusted estimates of small for gestational age (single/multiple agents vs. untreated)

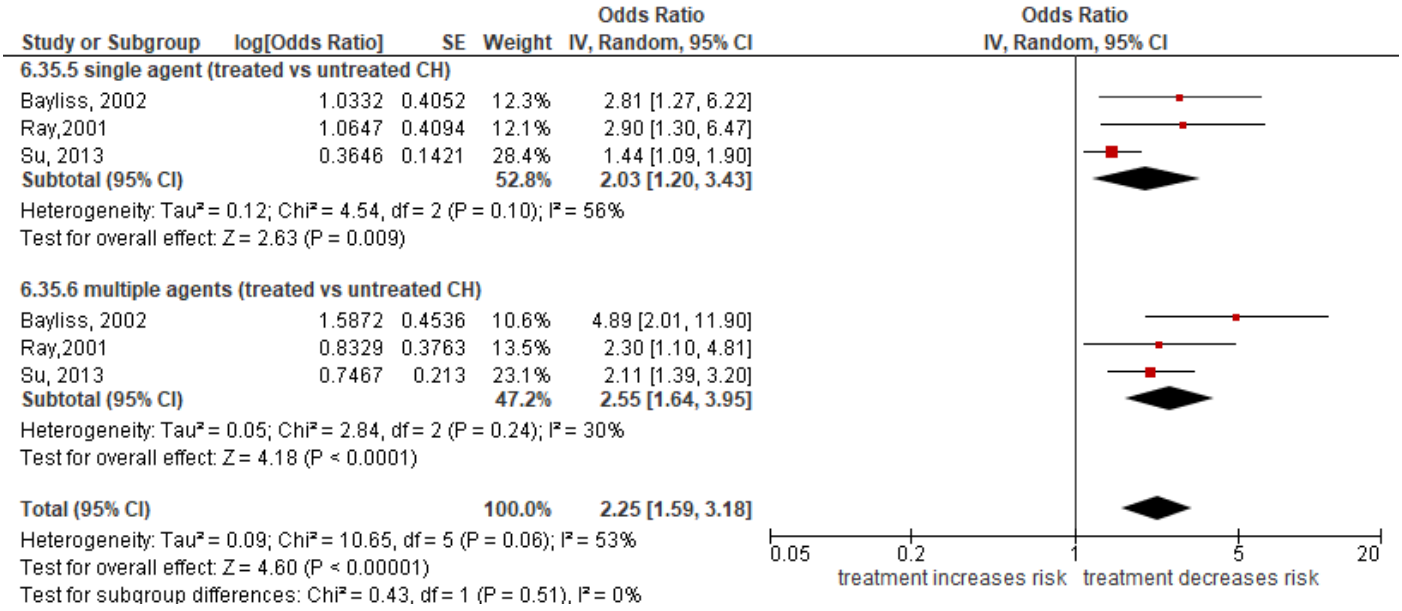


Figure S24h: Forest plot of crude estimates of small for gestational age (β-blockers vs. methyldopa)

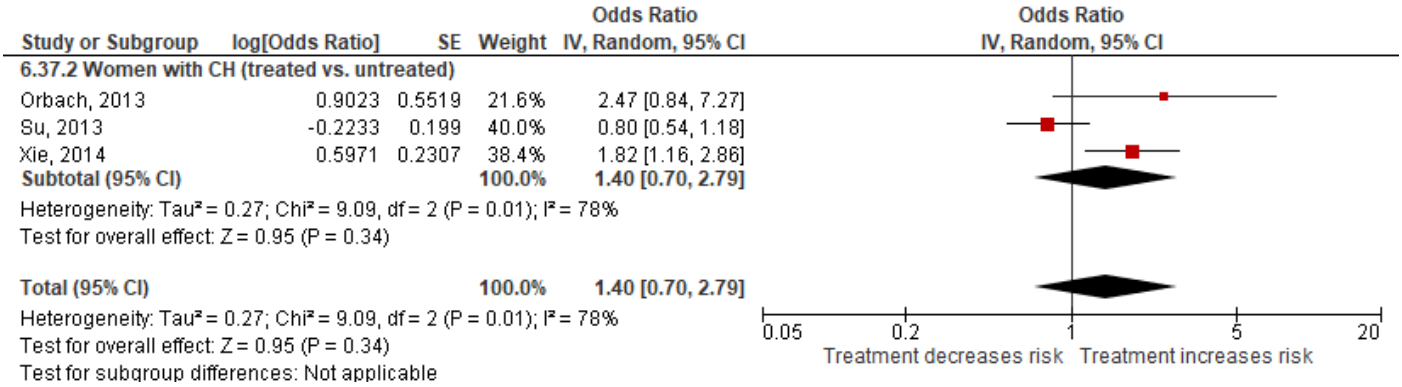


Figure S24i: Forest plot of crude estimates of SGA (methyldopa vs other agents)

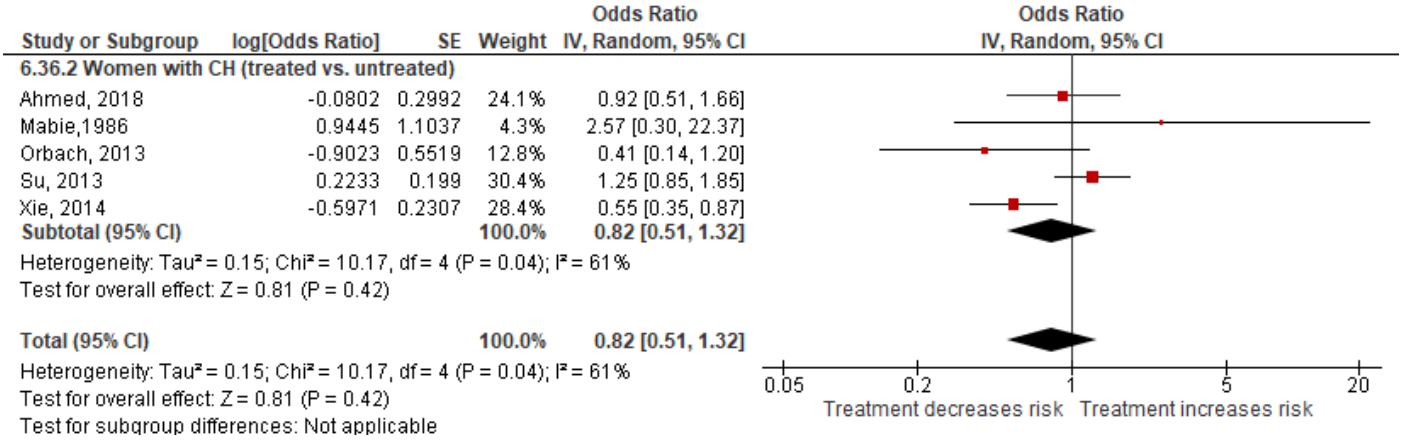


Figure S25a: Forest plot of crude estimates of the association between antihypertensive treatment and low birthweight

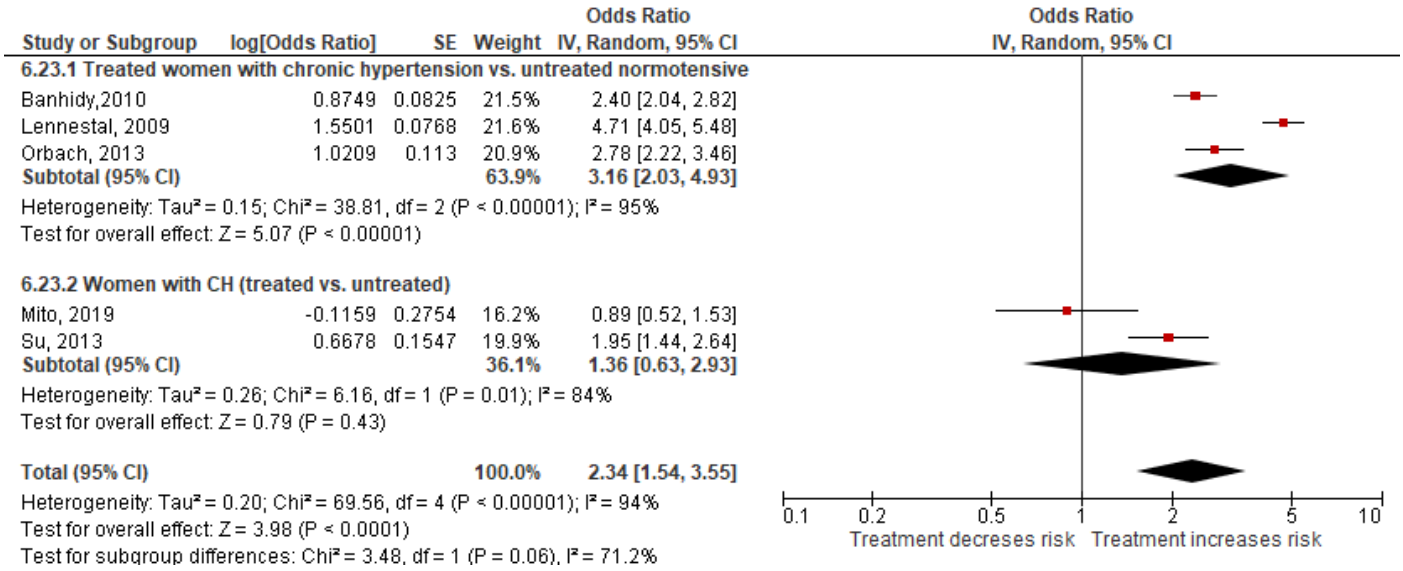


Figure S25b: Forest plot of adjusted estimates of the association between antihypertensive treatment and low birthweight

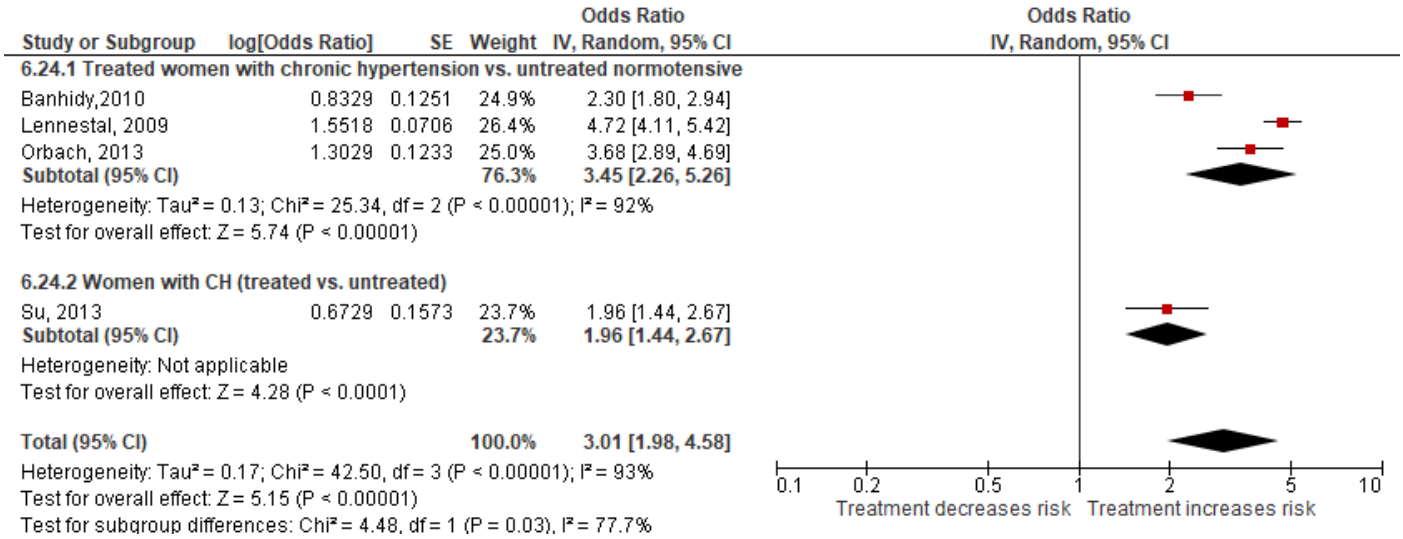


Figure S26: Forest plot of crude estimates of the association between antihypertensive treatment and neonatal intensive care unit admission

